14th ISPA 2025
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29–31 October 2025 · Coimbra · Portugal
F 2.2

Automating Electrical Grid Asset Inspections:

Evaluating the Impact of Annotation Detail Level

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1290 Faculdade de Ciências e Tecnologia Universidade de Coimbra
isec Engenharia
deec.uc Departamento de Engenharia Eletrotécnica e de Computadores

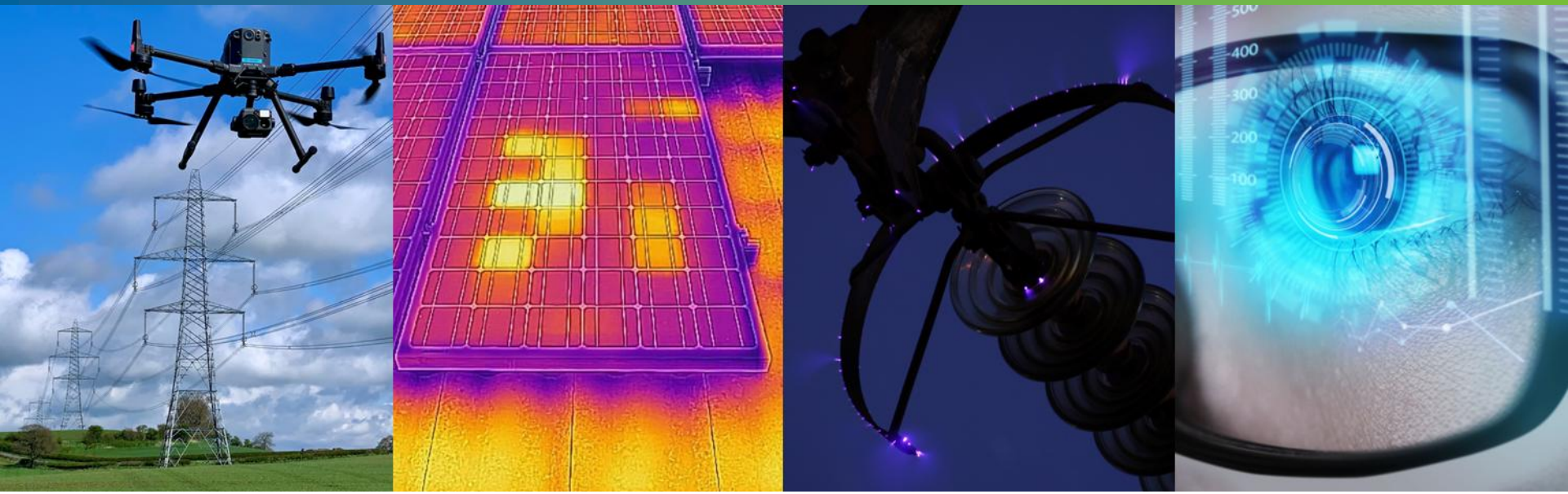
INTRODUCTION

- Context
- Motivation
- Objectives
- Overview



PROBLEM

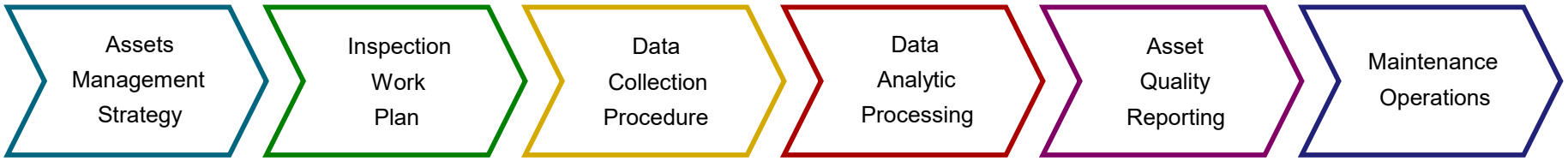
- Electrical Power Growing Demand
- Grid Reliability and Generation Efficiency
- Maintenance Periodicity and Inspection Magnitude



SOLUTION

- Remote Visual Inspections Outsourcing
- Invisible Electromagnetic Spectrum Technologies
- Image Processing Effective Methods

Electrical Grid Asset Inspection Current Workflow



Electrical Grid Asset Inspection Current Workflow



Assets Management Strategy

Power Lines
- Medium Voltage
- High Voltage
- Very High Voltage

Substations
Solar Plants
Wind Turbines
Hydroelectric Dams

Electrical Grid Asset Inspection Current Workflow





Assets Management Strategy

Inspection Work Plan

Power Lines
- Medium Voltage
- High Voltage
- Very High Voltage

Substations
Solar Plants
Wind Turbines
Hydroelectric Dams

Employed Resources
- Foot Patrol
- Helicopter
- UAV

Assets Location
Work Schedule

Electrical Grid Asset Inspection Current Workflow







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Assets Location
Work Schedule

Data Collection Procedure

Technologies
- RGB
- Infra-Red
- Ultra-Violet
- LiDAR + GNSS

Data Homogeneity
Traceability

Electrical Grid Asset Inspection Current Workflow









Assets Management Strategy	Inspection Work Plan	Data Collection Procedure	Data Analytic Processing
Power Lines - Medium Voltage - High Voltage - Very High Voltage Substations Solar Plants Wind Turbines Hydroelectric Dams	Employed Resources - Foot Patrol - Helicopter - UAV Assets Location Work Schedule	Technologies - RGB - Infra-Red - Ultra-Violet - LiDAR + GNSS Data Homogeneity Traceability	Defects - Missing Parts - Broken Parts - Flashed Insulators - Loose Connections - Corrosion - Dirtiness - Vegetation - Bird Nests

Electrical Grid Asset Inspection Current Workflow



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Data Analytic Processing

Defects
- Missing Parts
- Broken Parts
- Flashed Insulators
- Loose Connections
- Corrosion
- Dirtiness
- Vegetation
- Bird Nests

Asset Quality Reporting

Assets
- Structures
- Supports
- Components

Severity Degree
Repair Suggestion

Electrical Grid Asset Inspection Current Workflow








Assets Management Strategy	Inspection Work Plan	Data Collection Procedure	Data Analytic Processing	Asset Quality Reporting	Maintenance Operations
Power Lines - Medium Voltage - High Voltage - Very High Voltage Substations Solar Plants Wind Turbines Hydroelectric Dams	Employed Resources - Foot Patrol - Helicopter - UAV Assets Location Work Schedule	Technologies - RGB - Infra-Red - Ultra-Violet - LiDAR + GNSS Data Homogeneity Traceability	Defects - Missing Parts - Broken Parts - Flashed Insulators - Loose Connections - Corrosion - Dirtiness - Vegetation - Bird Nests	Assets - Structures - Supports - Components Severity Degree Repair Suggestion	Repair Replacement Reinstall Cleaning Clearing

Electrical Grid Asset Inspection Current Workflow





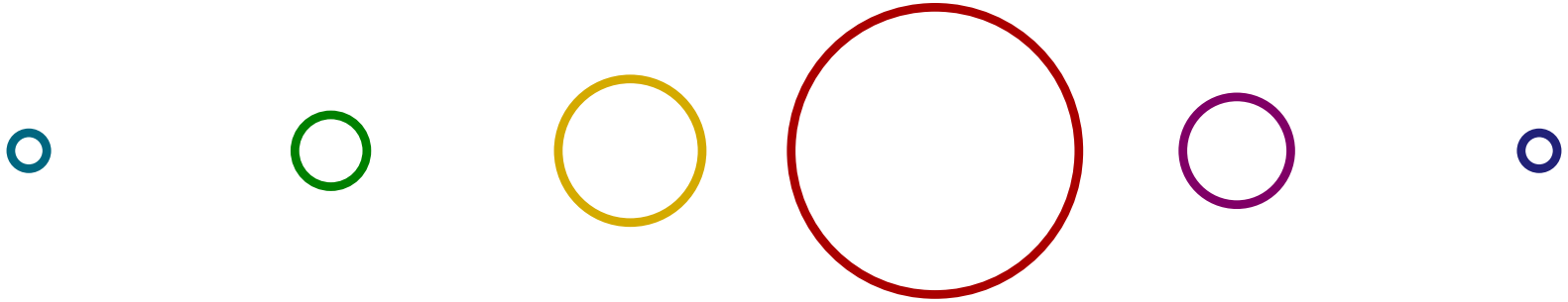








Assets Management Strategy → Inspection Work Plan → Data Collection Procedure → Data Analytic Processing → Asset Quality Reporting → Maintenance Operations



Electrical Grid Asset Inspection Current Workflow







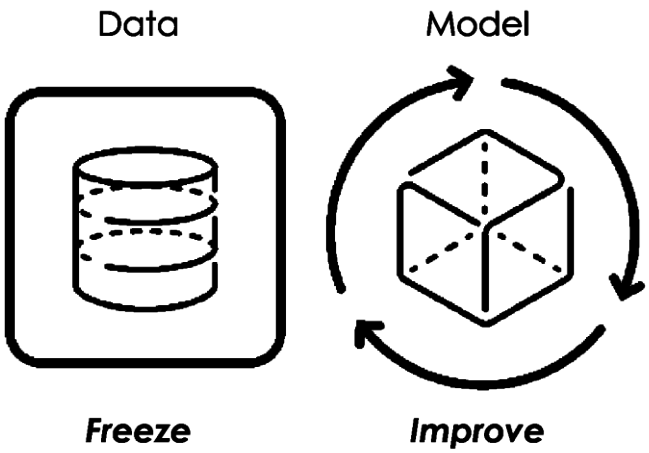






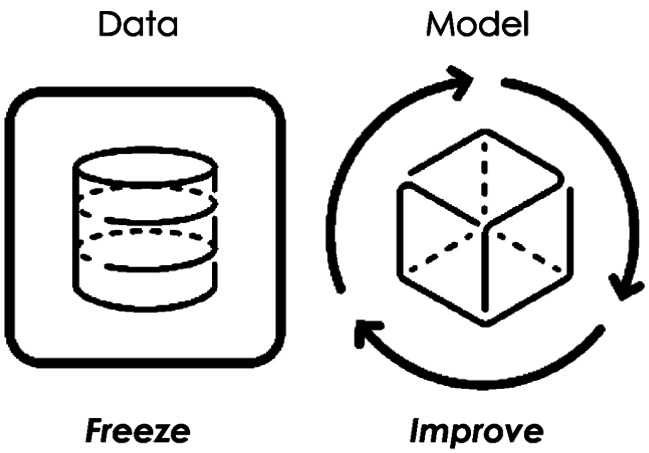
Assets Management Strategy → Inspection Work Plan → Data Collection Procedure → Data Analytic Processing → Asset Quality Reporting → Maintenance Operations

Model-centric ML Development



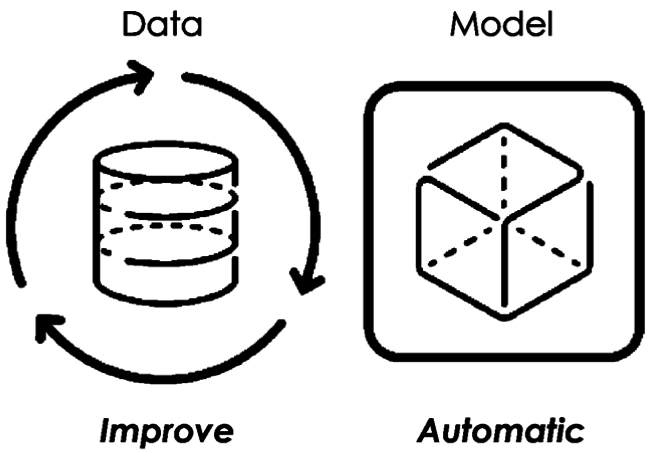
Traditionally, the main focus was on model improvement

Model-centric ML Development



Traditionally, the main focus was on model improvement

Data-centric ML Development



However, it is important to focus on data as well

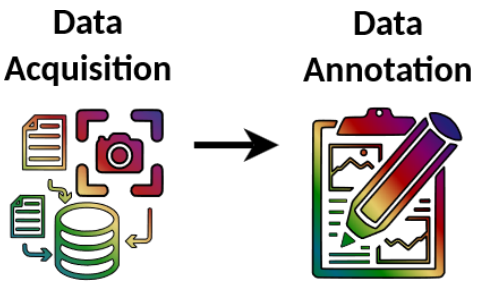
Model-Centric Approach

Model-Centric Approach

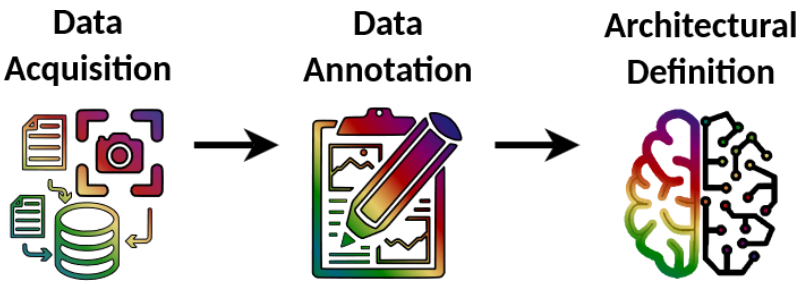
Data Acquisition



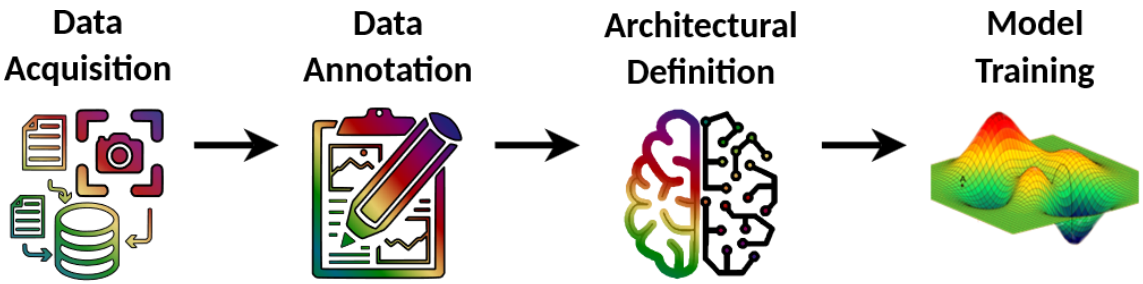
Model-Centric Approach



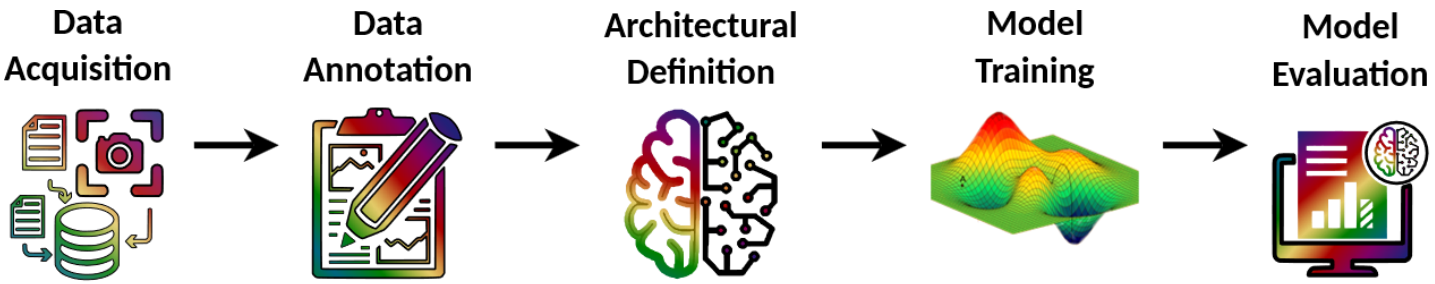
Model-Centric Approach



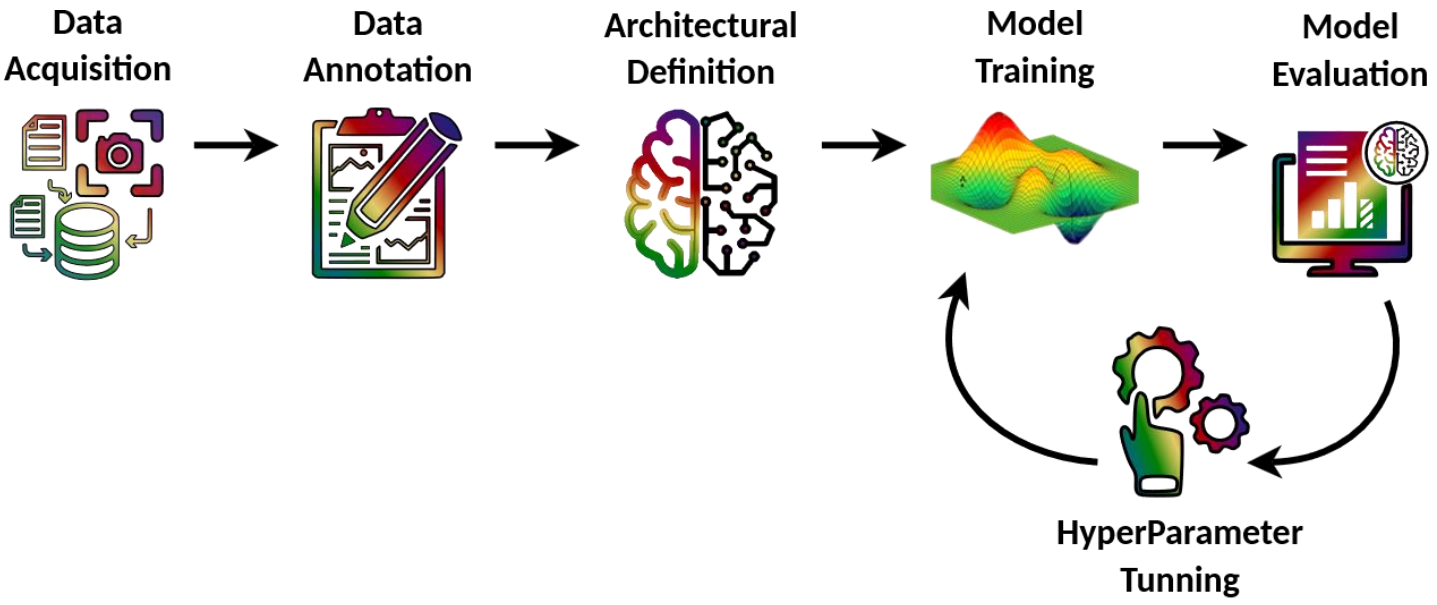
Model-Centric Approach



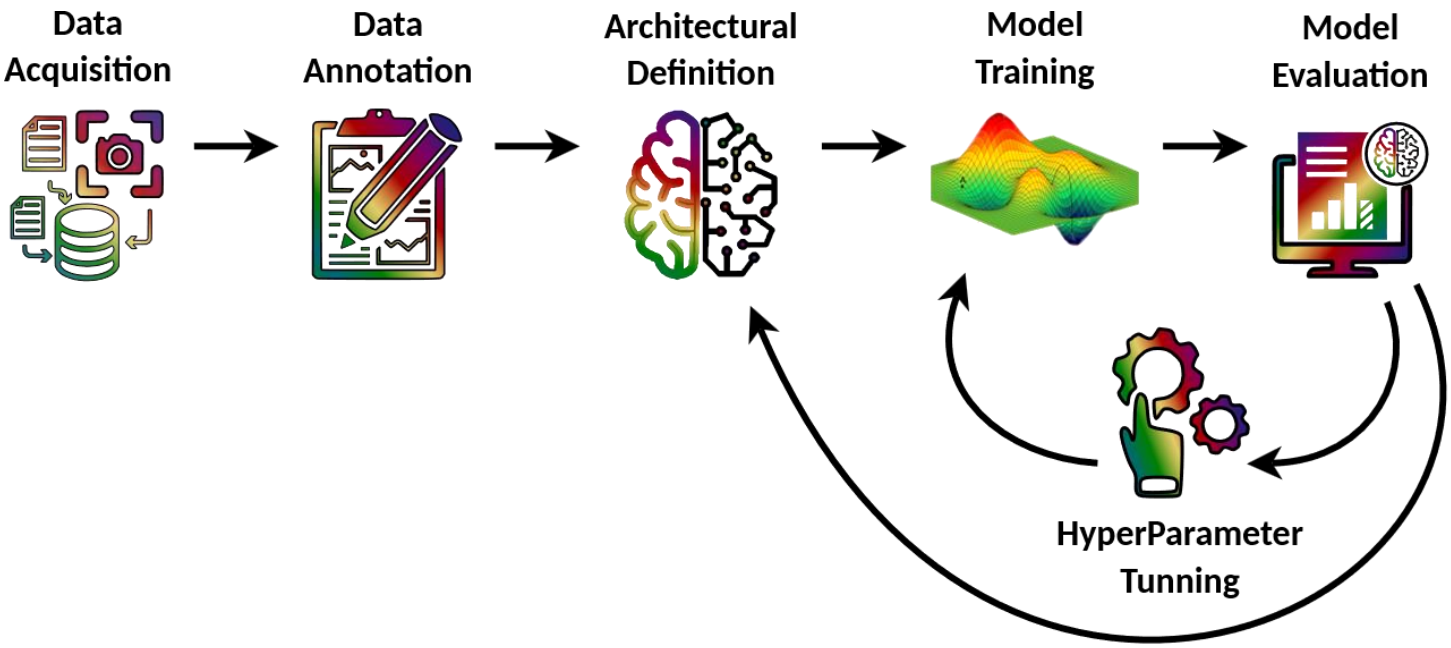
Model-Centric Approach



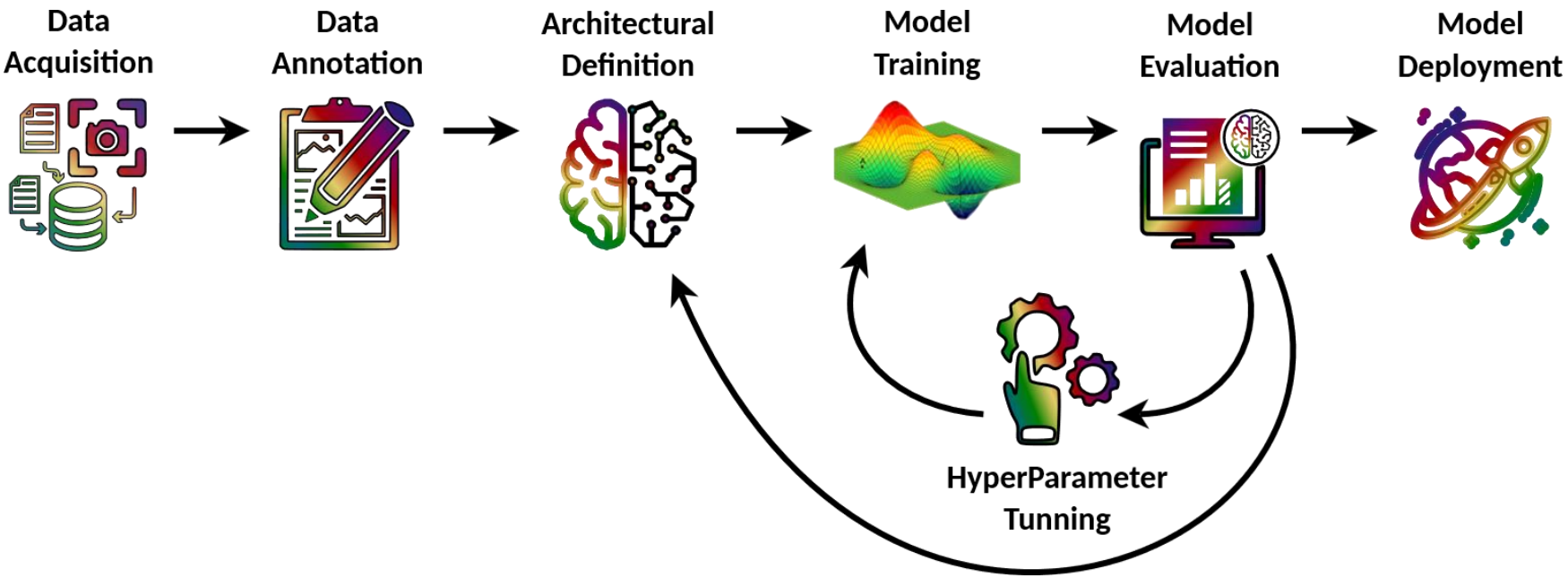
Model-Centric Approach



Model-Centric Approach

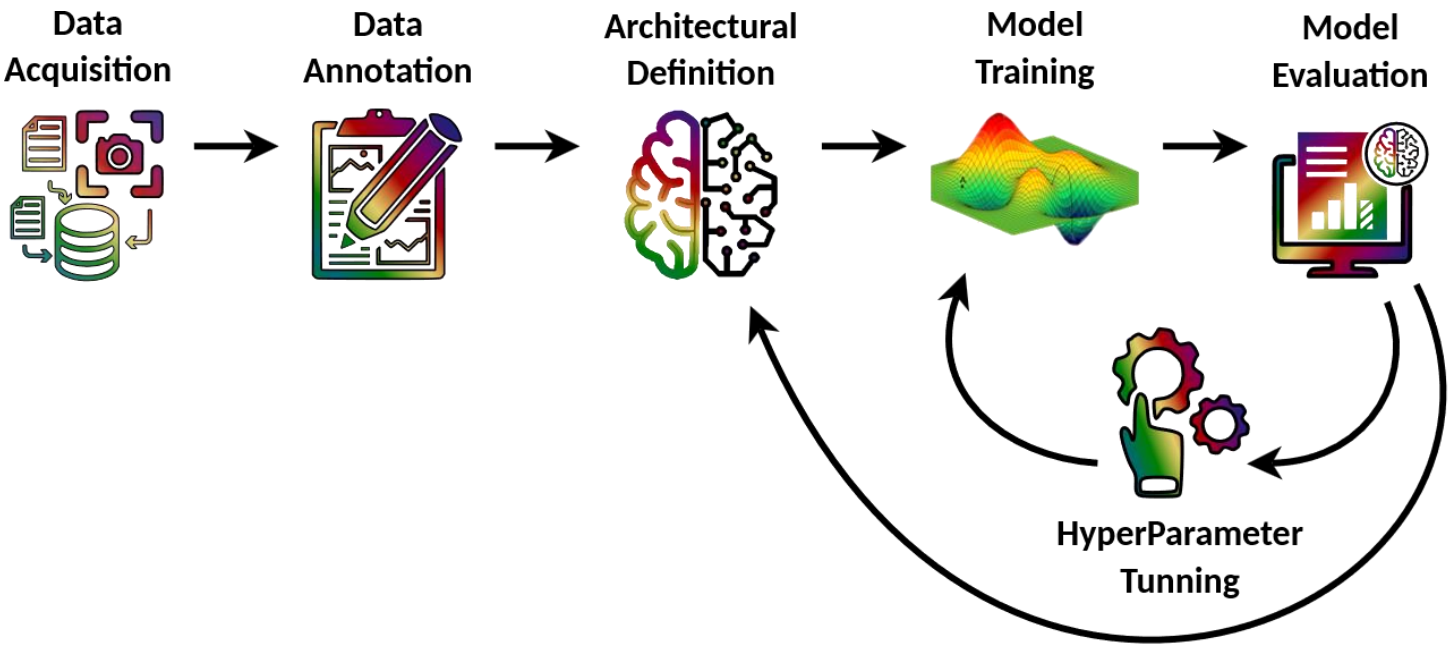


Model-Centric Approach

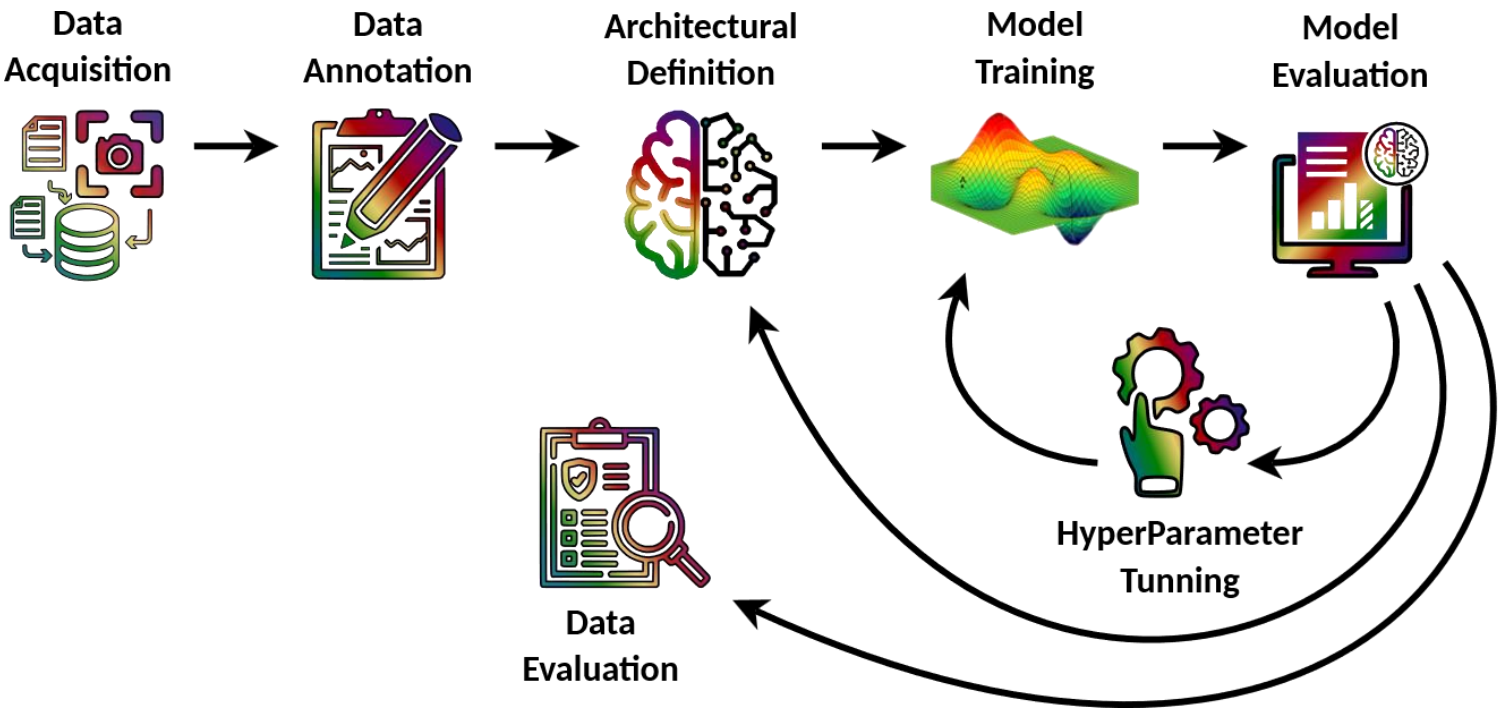


Data-Centric Approach

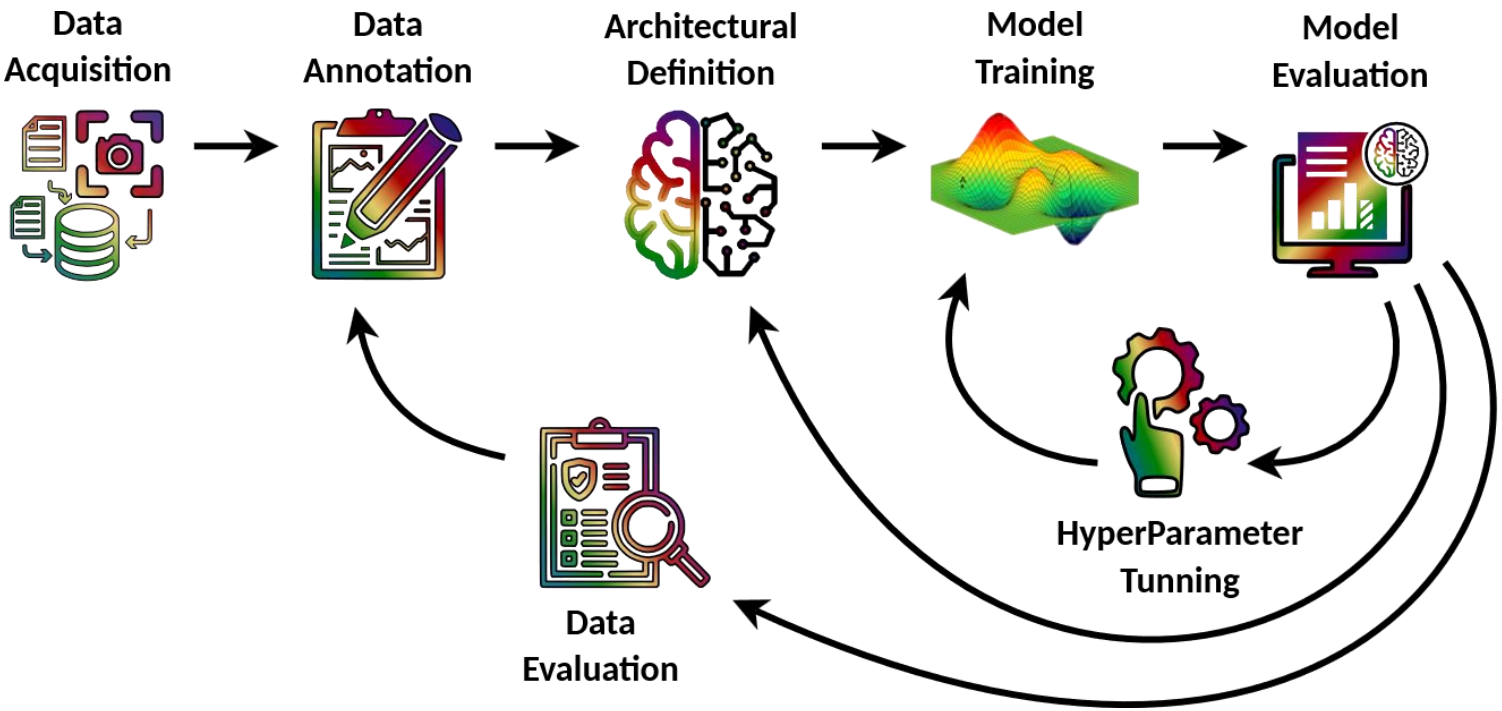
Data-Centric Approach



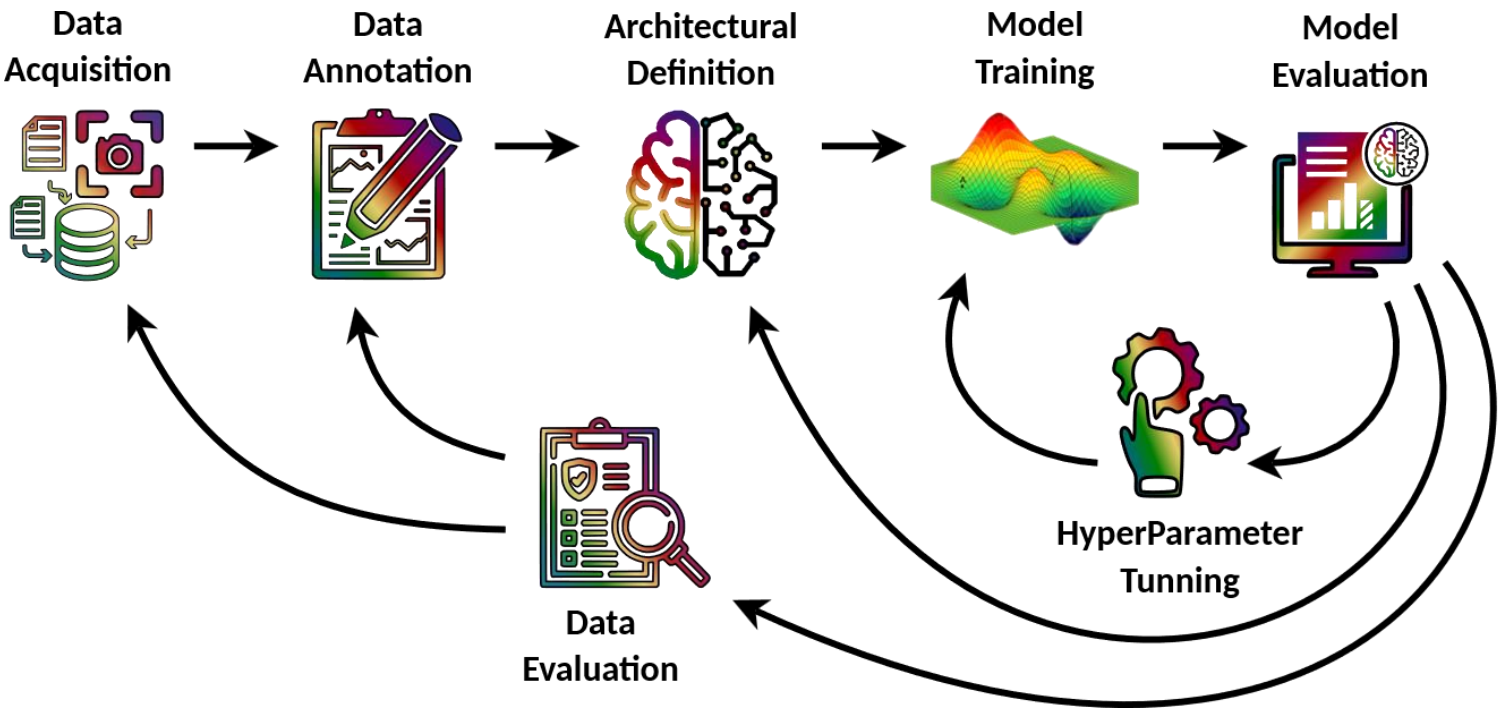
Data-Centric Approach



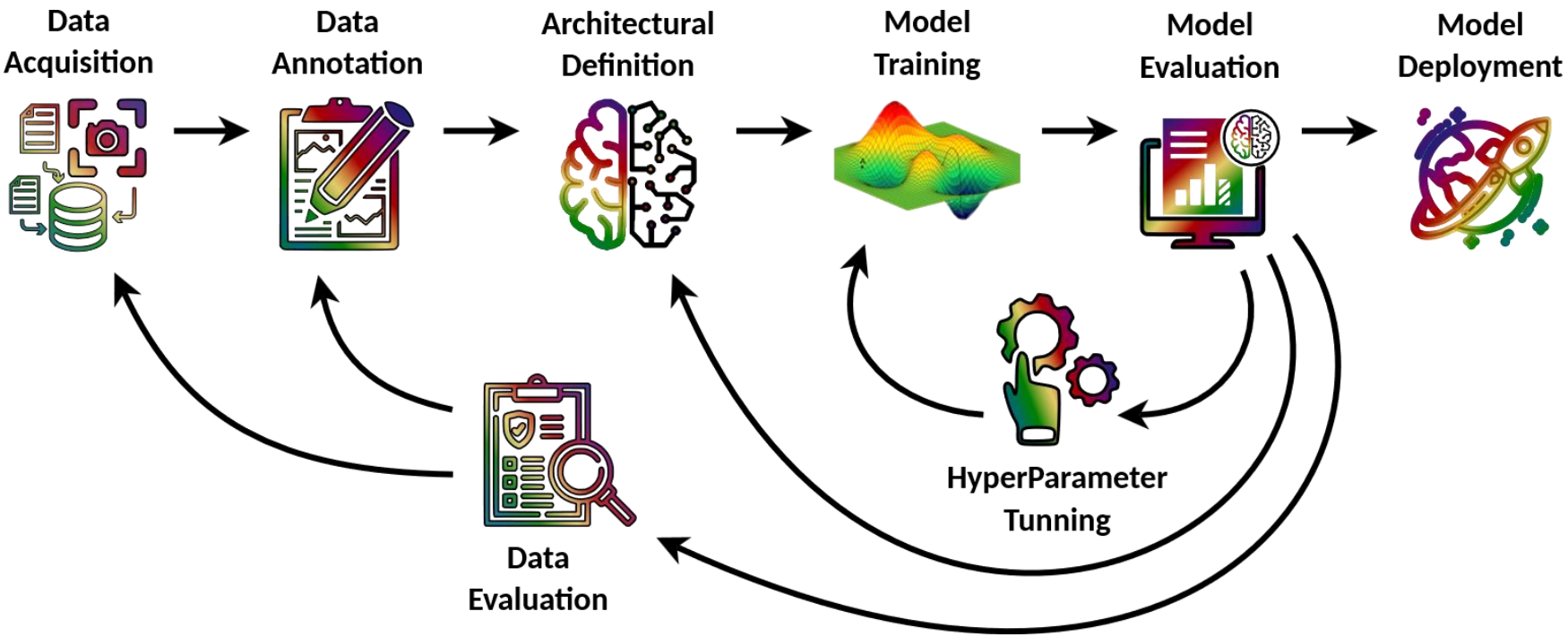
Data-Centric Approach



Data-Centric Approach



Data-Centric Approach



How Can Data Quality be Measured?

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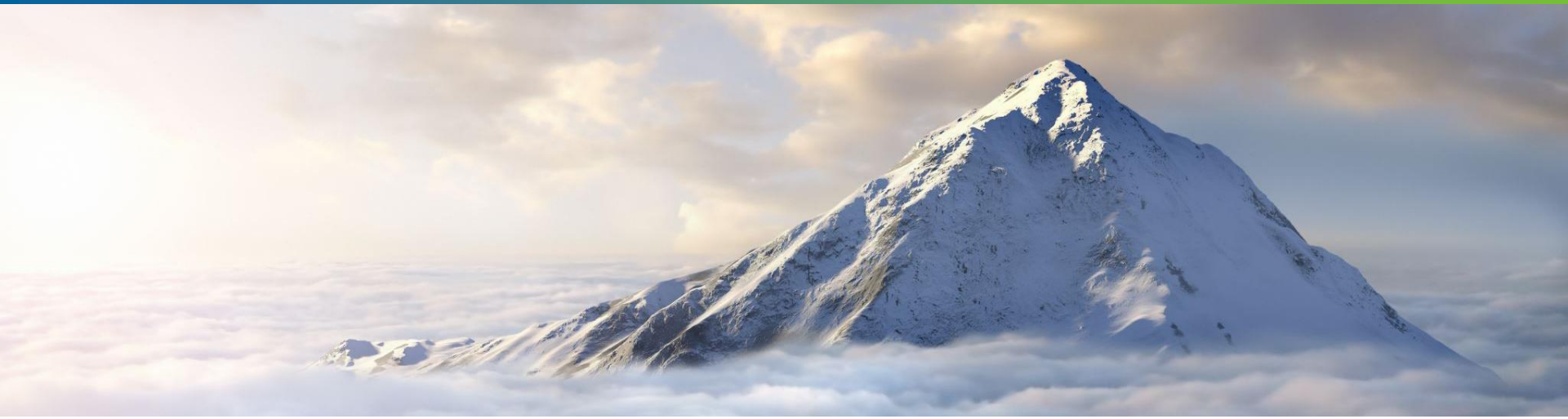
- Labelling Consistency

How Can Data Quality be Measured?

- Labelling Consistency
- Spatial Granularity

How Can Data Quality be Measured?

- Labelling Consistency
- Spatial Granularity
- Categorical Definition



OBJECTIVES

- Define Annotations Quality
- Propose Annotations Procedures
- Benchmark SotA CV Models
- Analyze Annotations Impact



Overview

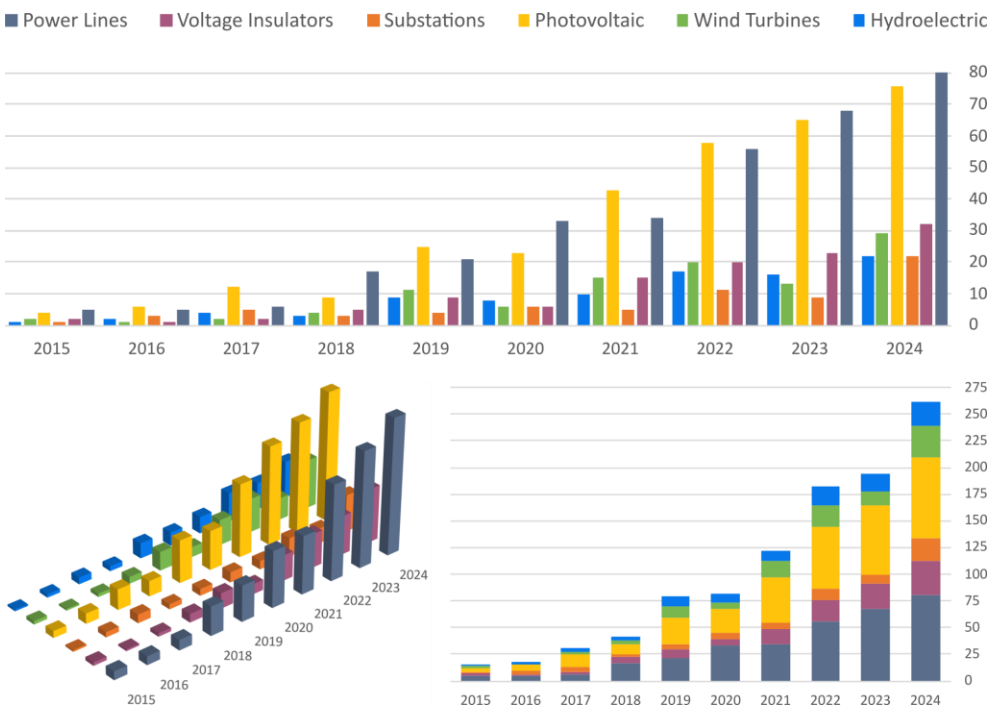
- Related Work
- Experimental Methodology
- Results and Discussion
- Summary and Challenges

Related Work

- Literature Survey
- IDID Dataset
- Annotations Versions
- Key Labeling Recommendations

Automating Electrical Grid Asset Inspection

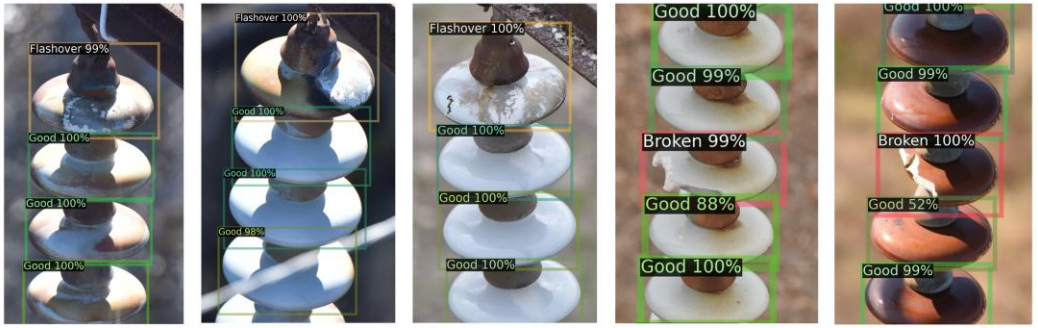
- Growing Field of Study
- Model Centric Approaches
- Inconsistent Validation Practices
- Dataset Limitations
- Metrics Optimization
- Operational Testing

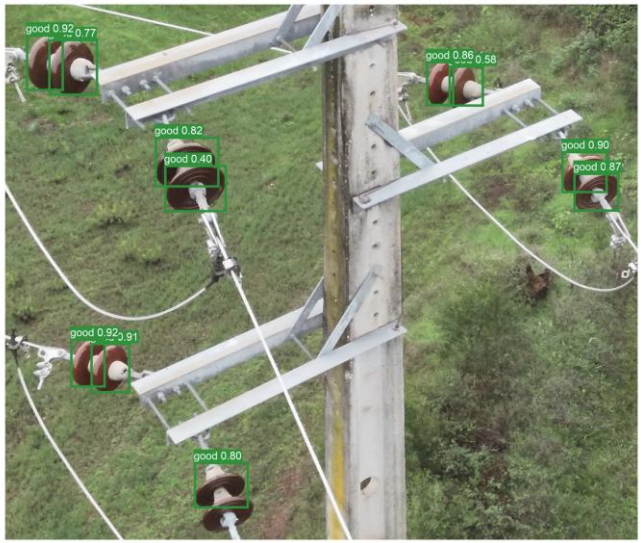


P. D. Rocha, F. Lopes, and L. A. Da Silva Cruz, "Automating Electrical Grid Asset Inspection: From Current Challenges to Future Directions," IEEE Access, pp. 1–45, 2025, Submitted.

Fine-Tune SotA Models using Original Annotations



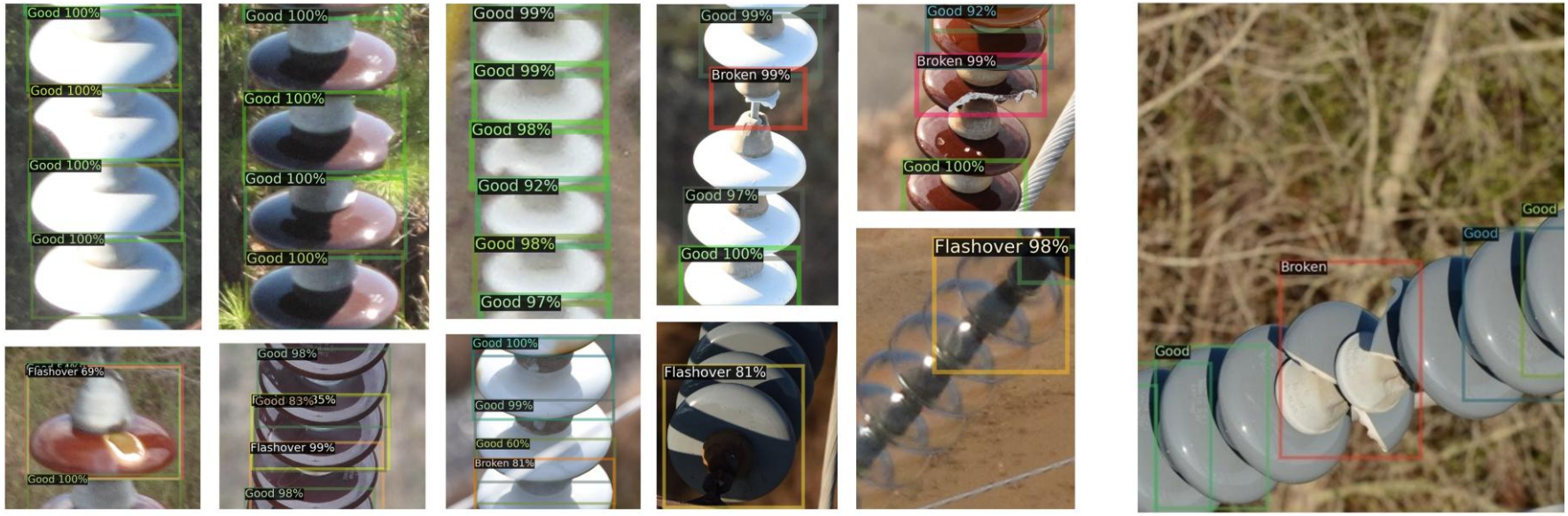




- YOLOv8s mAP@50 = 87.9%
- Faster R-CNN mAP@50 = 87.2%

F. Lopes, P. Rocha, and A. Coelho, ‘‘Towards Automated Visual Inspection of Electrical Grid Assets for the Smart Grid - An Application to HV Insulators,’’ in 2024 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm). Oslo, Norway: IEEE, Sep. 2024, pp. 347–352.

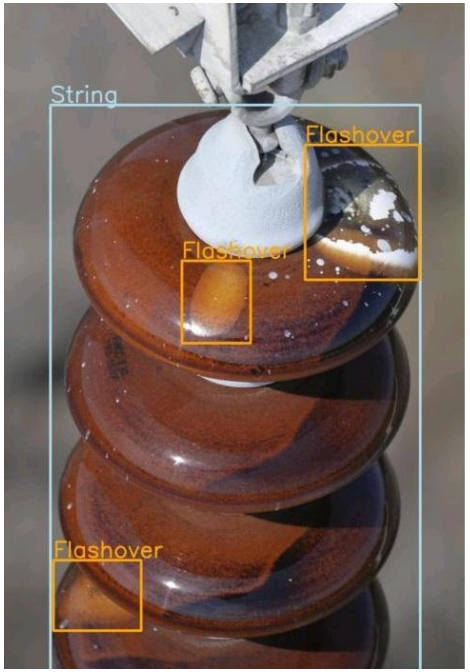
Performance Influenced by Dataset Limitations



- Insufficient Data Diversity and Objects Labels Missing

F. Lopes, P. Rocha, and A. Coelho, ‘‘Towards Automated Visual Inspection of Electrical Grid Assets for the Smart Grid - An Application to HV Insulators,’’ in 2024 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm). Oslo, Norway: IEEE, Sep. 2024, pp. 347–352.

Different Annotation Procedure

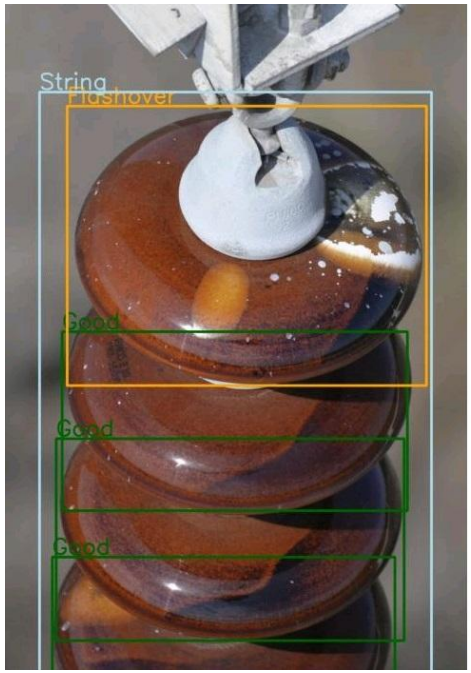


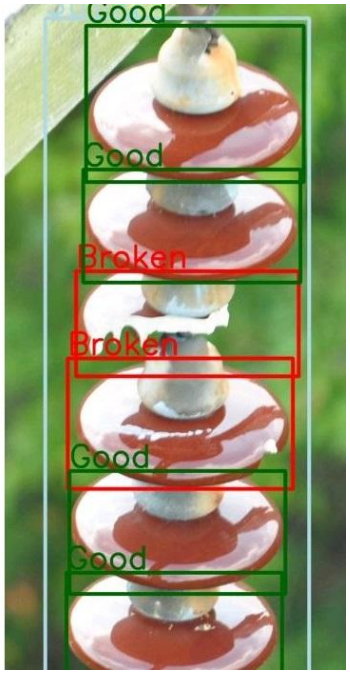


- Box Bounding the Defects

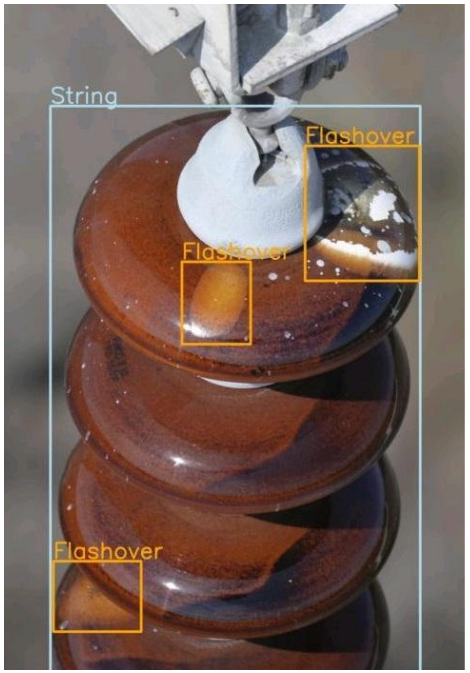
Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

Different Annotation Procedure





- Box Bounding the Discs

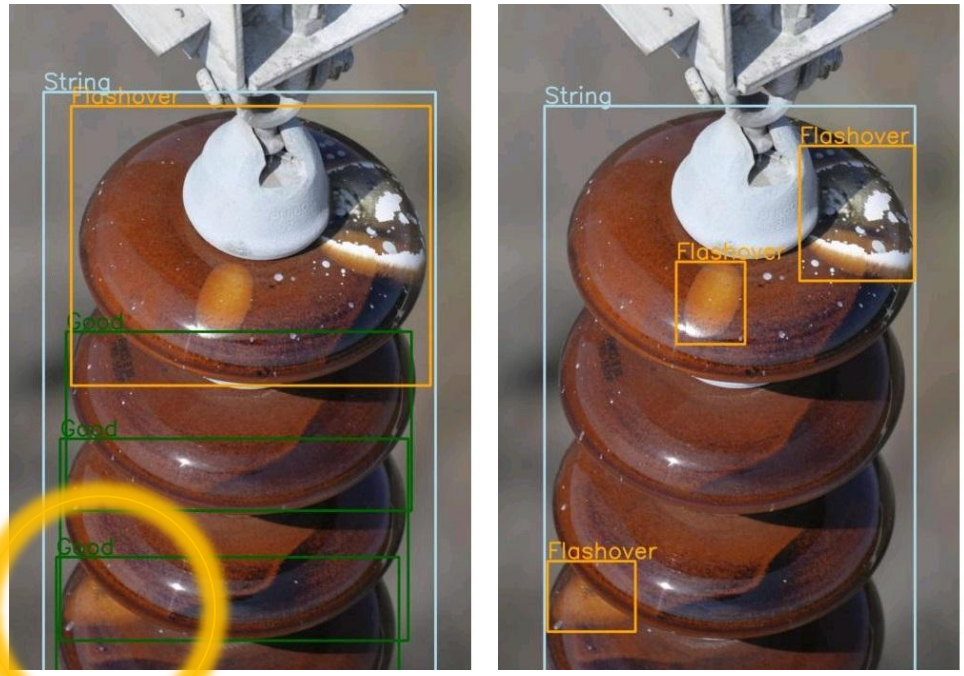




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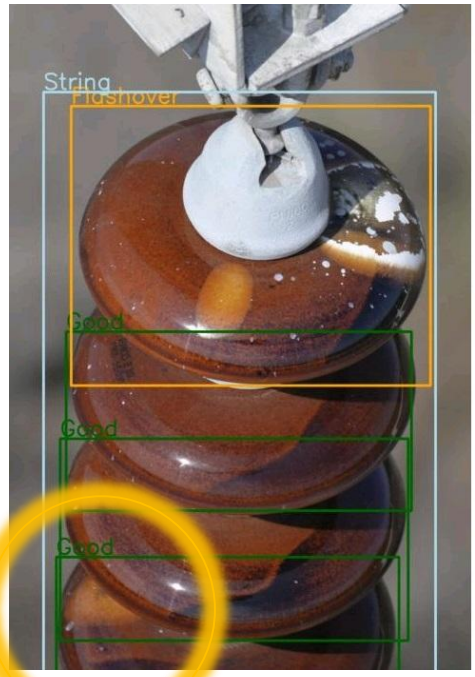
D. Lewis and P. Kulkarni, “IDID,” Aug. 2021. [Online]. Available: https://ieee-dataport.org/competitions/insulator-defect-detection

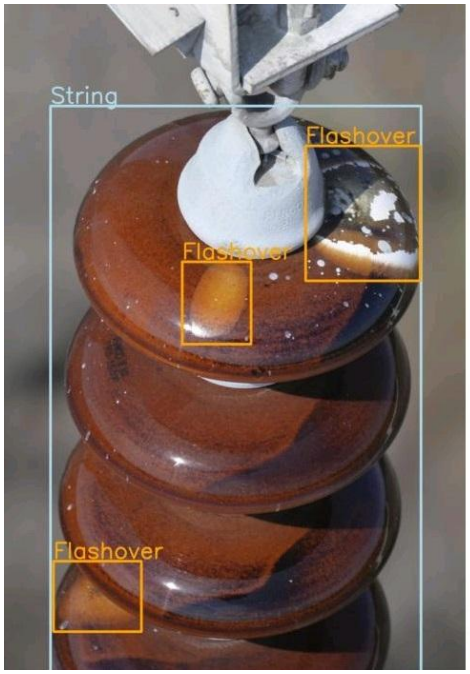
Inconsistencies in Annotations



- Spotted Missing Annotations

Inconsistencies in Annotations





- Spotted Missing Annotations





- Introduced Noise

Insulator Defect Image Dataset 1600 Images

- Redundant Data obtained with Augmented Images
- Data Leakage Risk using Radom Validation Set Splitting


160538.JPG


160538d.JPG


160538h.JPG


160538v.JPG


160605.JPG

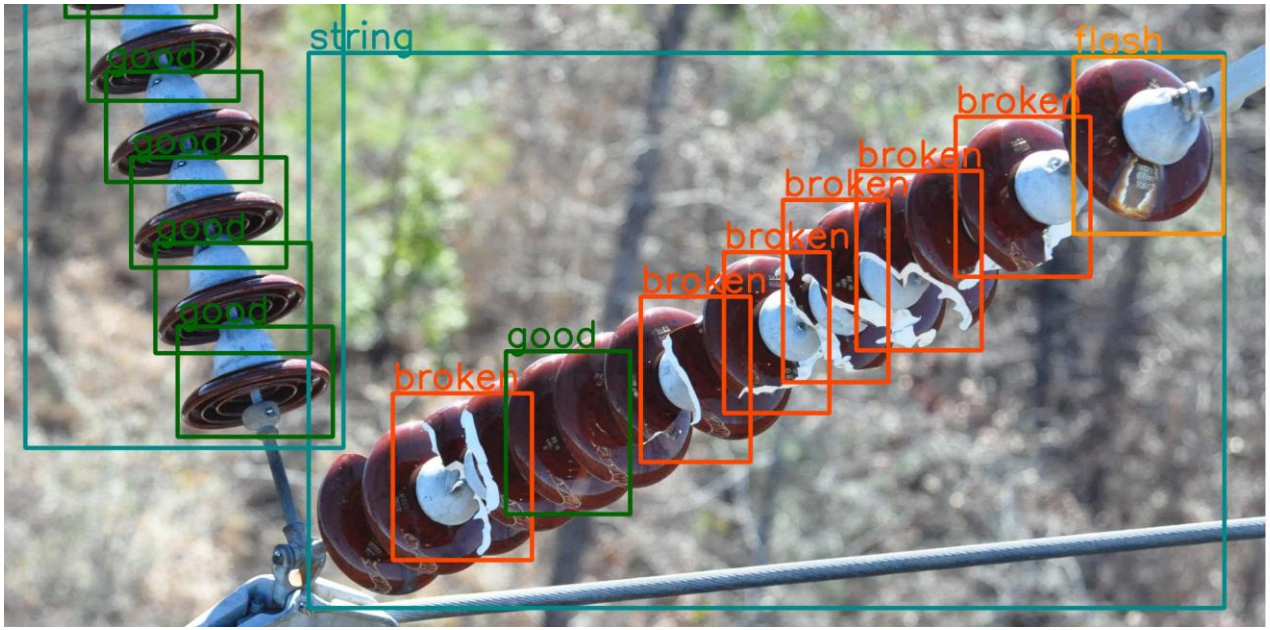

160605d.JPG


160605h.JPG


160605v.JPG

Curated Dataset

488 Images



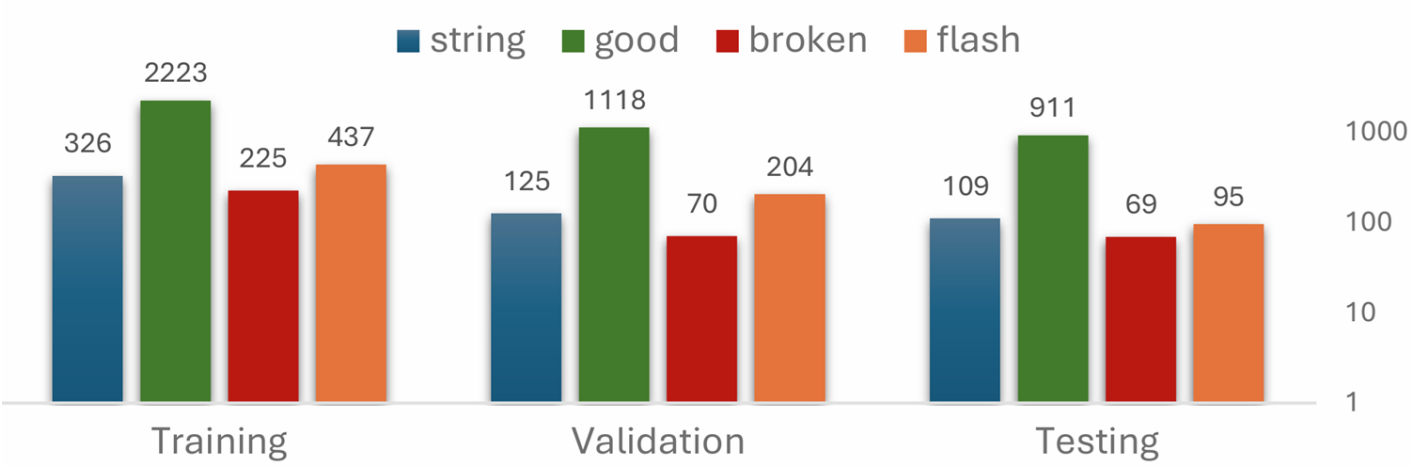


560 String **4252 Good** **364 Broken** **736 Flash**

https://universe.roboflow.com/academic-it/epri-insulators/analytics

Dataset Split

300 + 100 + 88 Images



Sub-Set	Training	Validation	Testing
Images	61.5 %	20.5 %	18.0 %
Objects	54.3 %	25.7 %	20.0 %

https://universe.roboflow.com/academic-it/epri-insulators/analytics

KEY LABELLING RECOMMENDATIONS

Key Labelling Recommendations



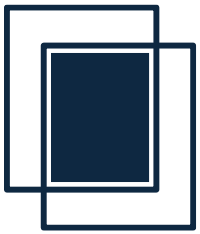
Consistency

Ultralytics, “Training tips.” [Online]. Available: https://docs.ultralytics.com/yolov5/tutorials/tips for best training results

Key Labelling Recommendations



Consistency



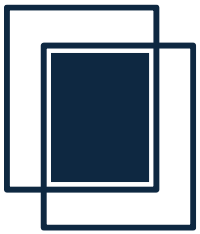
Accuracy

Ultralytics, “Training tips.” [Online]. Available: https://docs.ultralytics.com/yolov5/tutorials/tips for best training results

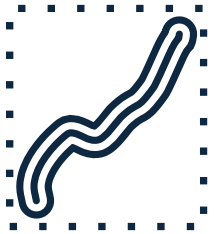
Key Labelling Recommendations



Consistency



Accuracy



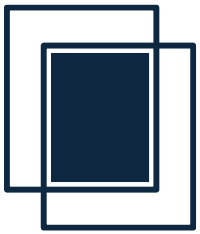
Granularity

Ultralytics, “Training tips.” [Online]. Available: https://docs.ultralytics.com/yolov5/tutorials/tips for best training results

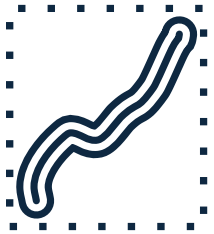
Key Labelling Recommendations



Consistency



Accuracy



Granularity



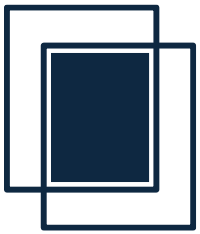
Variety

Ultralytics, “Training tips.” [Online]. Available: https://docs.ultralytics.com/yolov5/tutorials/tips for best training results

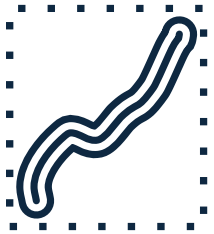
Key Labelling Recommendations



Consistency



Accuracy



Granularity



Variety



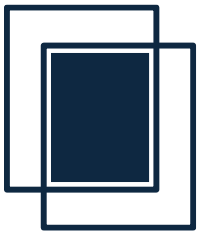
Balance

Ultralytics, “Training tips.” [Online]. Available: https://docs.ultralytics.com/yolov5/tutorials/tips for best training results

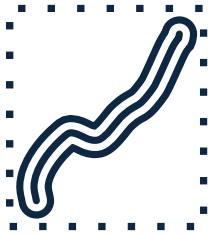
Key Labelling Recommendations



Consistency



Accuracy



Granularity



Variety



Balance



Verification

Ultralytics, “Training tips.” [Online]. Available: https://docs.ultralytics.com/yolov5/tutorials/tips for best training results

Methodology

- Experiment Design
- Imagery Diversity
- Model Complexity
- Labelling Rules

Dimensions that Impact the Performance

Dimensions that Impact the Performance

Metric

Dimensions that Impact the Performance

Metric

Error $\mathcal{L}$

Dimensions that Impact the Performance

Metric

$$\mathcal{L}\,(y, \hat{y})$$

Error $\mathcal{L}$

Dimensions that Impact the Performance

Target y **Metric**

$\mathcal{L}(y, \hat{y})$ **Error $\mathcal{L}$**

Dimensions that Impact the Performance

Target y

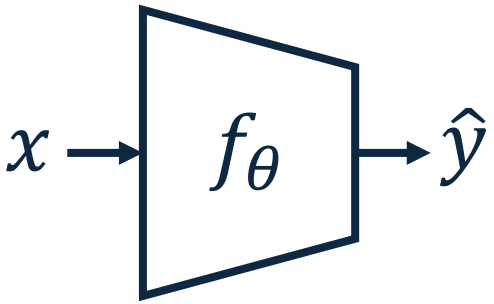
Metric

$\mathcal{L}\ (y, \hat{y})$

Error $\mathcal{L}$

Prediction $\hat{y}$

Dimensions that Impact the Performance



Target y

Metric

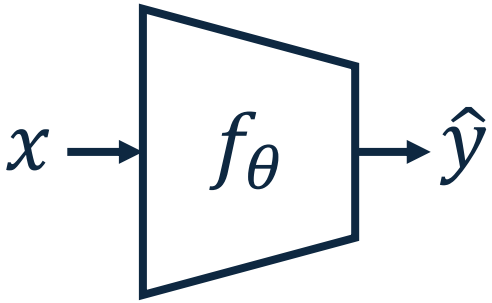
$\mathcal{L}(y, \hat{y})$

Error $\mathcal{L}$

Prediction $\hat{y}$

Dimensions that Impact the Performance

Model f



Target y

$\mathcal{L}\,(y, \hat{y})$

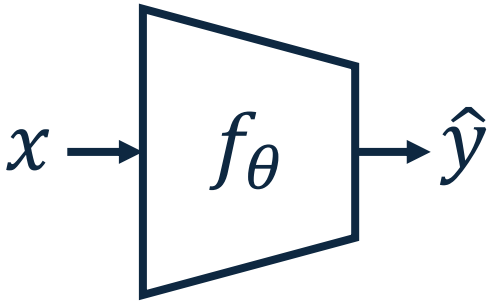
Prediction $\hat{y}$

Metric

Error $\mathcal{L}$

Dimensions that Impact the Performance

Model f



Parameters θ

Target y

$\mathcal{L}\,(y, \hat{y})$

Prediction $\hat{y}$

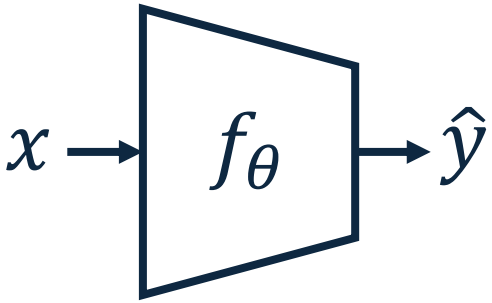
Metric

Error $\mathcal{L}$

Dimensions that Impact the Performance

Input x

Model f



Parameters θ

Target y

$$\mathcal{L}\,(y, \hat{y})$$

Prediction $\hat{y}$

Metric

Error $\mathcal{L}$

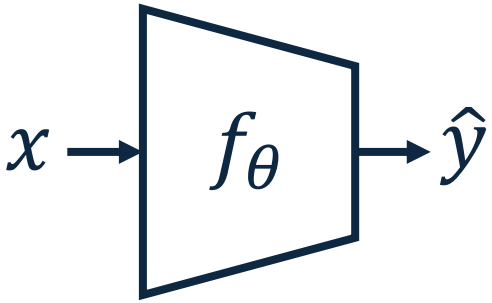
Dimensions that Impact the Performance

Input x

Model f

Target y

Metric



$\mathcal{L}(y, \hat{y})$

Error $\mathcal{L}$

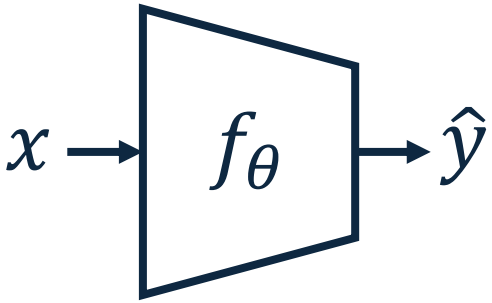
Parameters θ

Prediction $\hat{y}$

Defined

Dimensions that Impact the Performance

Input x | Model f | Target y | Metric



$\mathcal{L}\,(y, \hat{y})$

Error $\mathcal{L}$

Parameters θ

Prediction $\hat{y}$

Dependent

Defined

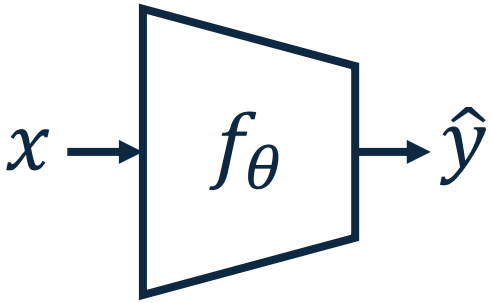
Dimensions that Impact the Performance

Input x

Model f

Target y

Metric



$\mathcal{L}\,(y, \hat{y})$

Error $\mathcal{L}$

Parameters θ

Prediction $\hat{y}$

Independent **Dependent** **Defined**

Evaluate Correlation of Independent Variables

Dimension	Symbol	Strategy
Input	x	Cross Fold Dataset Splitting
Model	f	Artificial Neural Network Size
Target	y	Varying Labelling Rules

Cross Fold Dataset Splitting

Input X

Validation Set	Training Set			Test Set
100	100	100	100	88
100	100	100	100	88
100	100	100	100	88
100	100	100	100	88
Fold 1	Fold 2	Fold 3	Fold 4	

Image Resolution

Mega Pixels

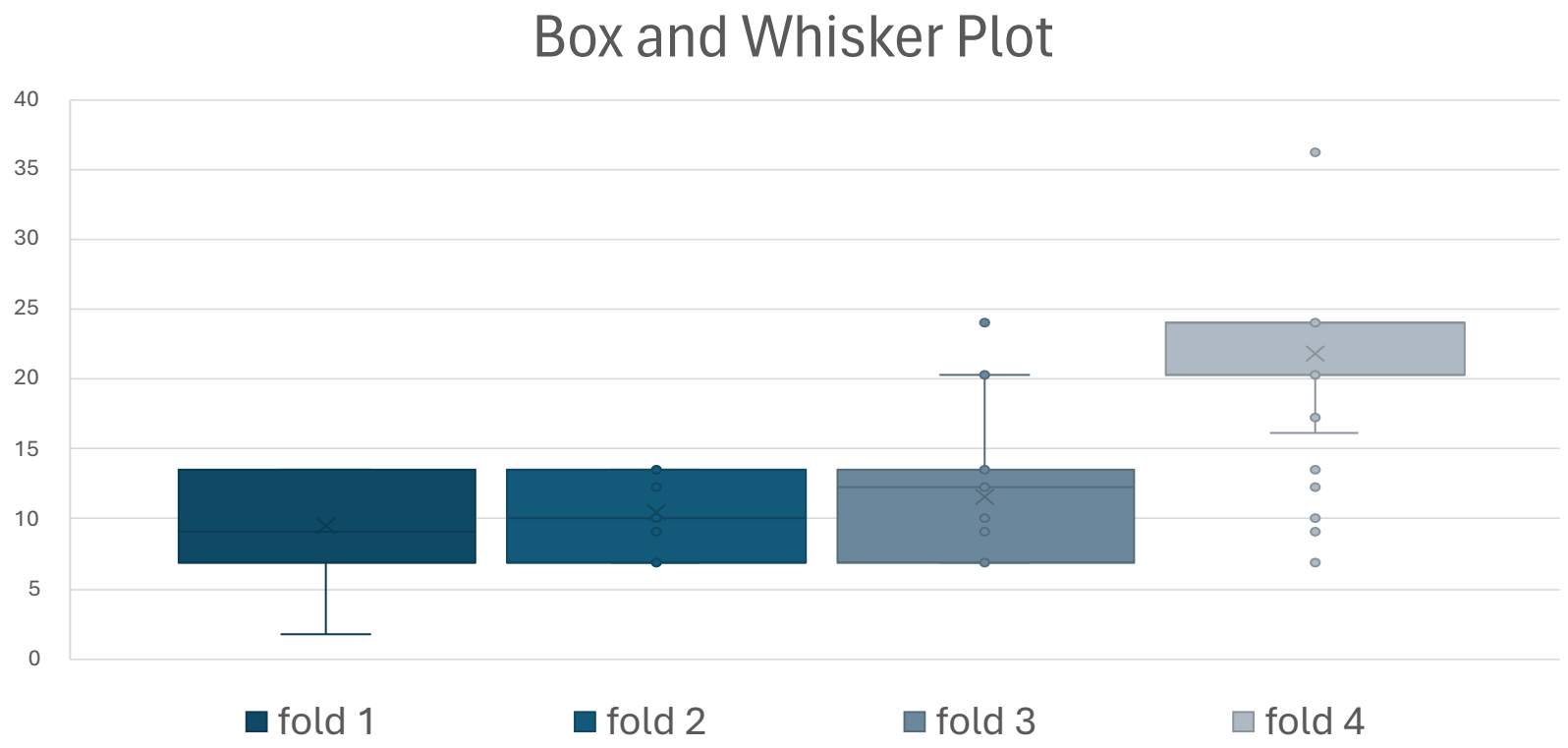
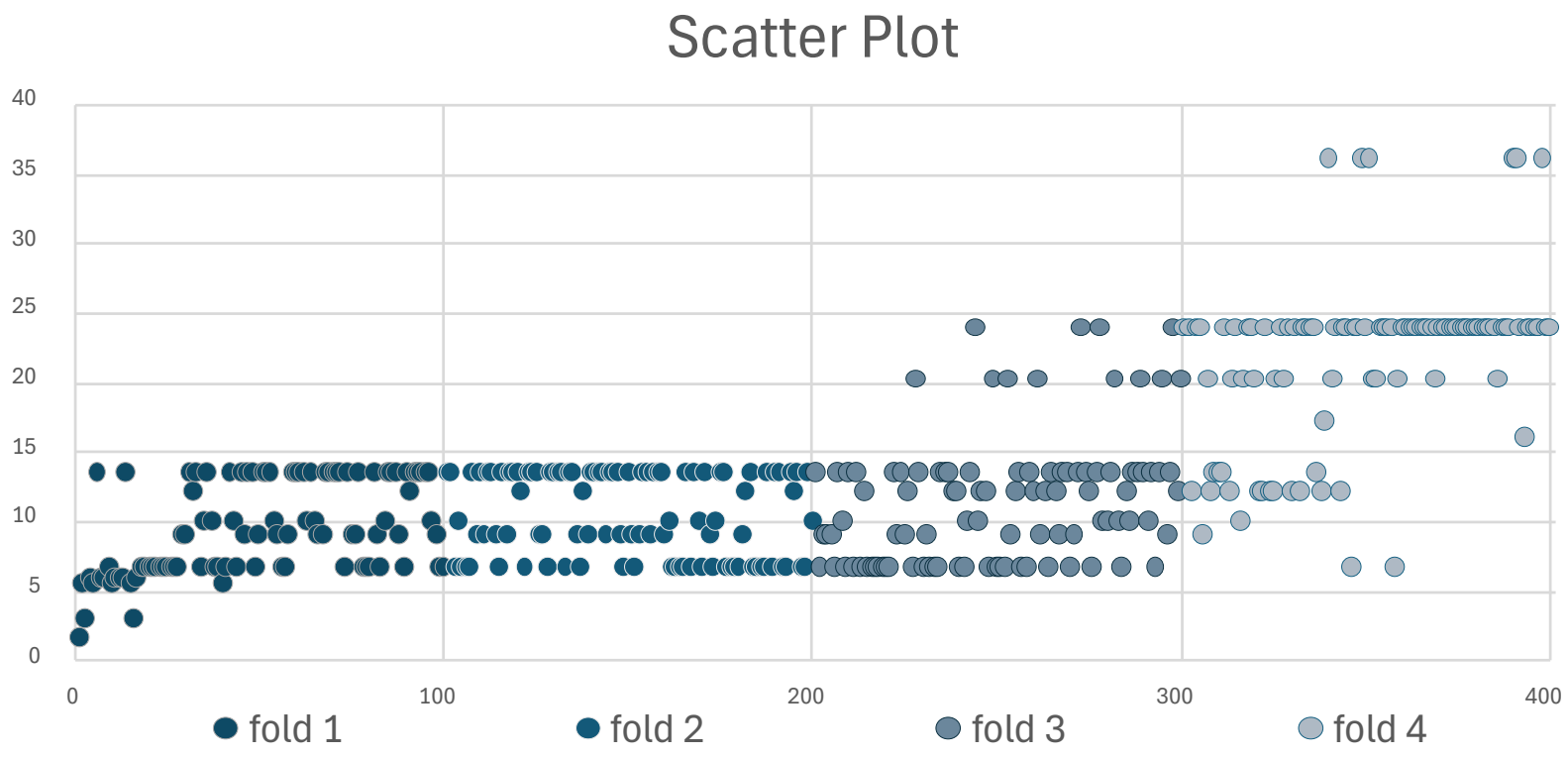


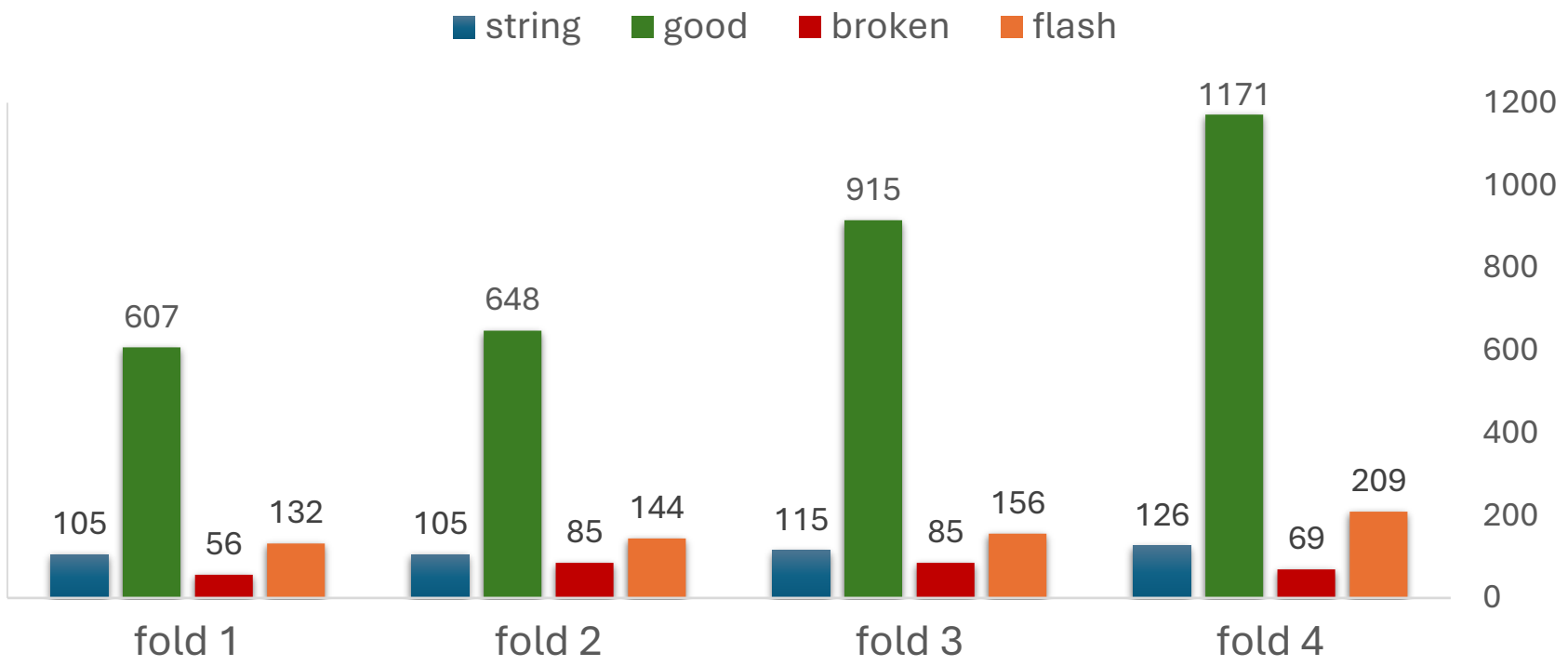
Image Resolution

Mega Pixels



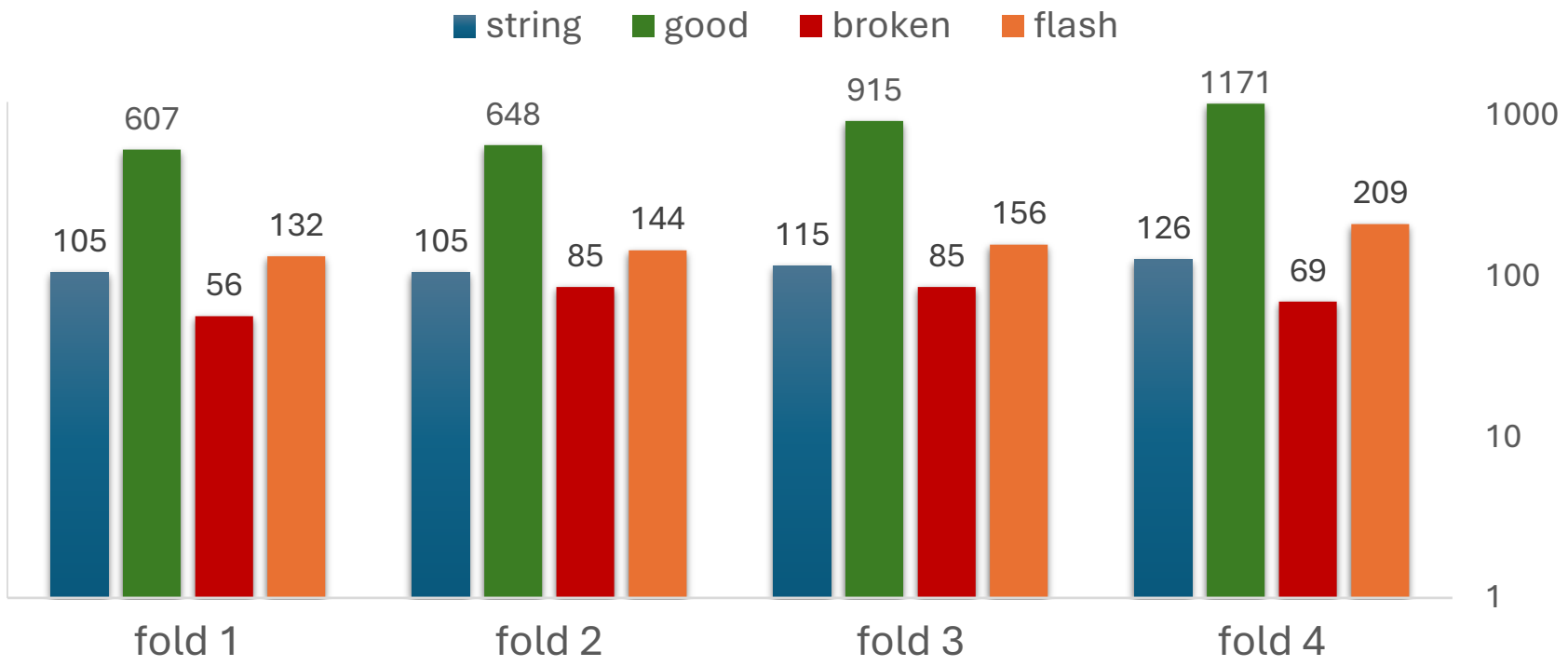
Annotated Objects

4 Classes



Annotated Objects

Logarithmic Scale



Model Complexity and Size

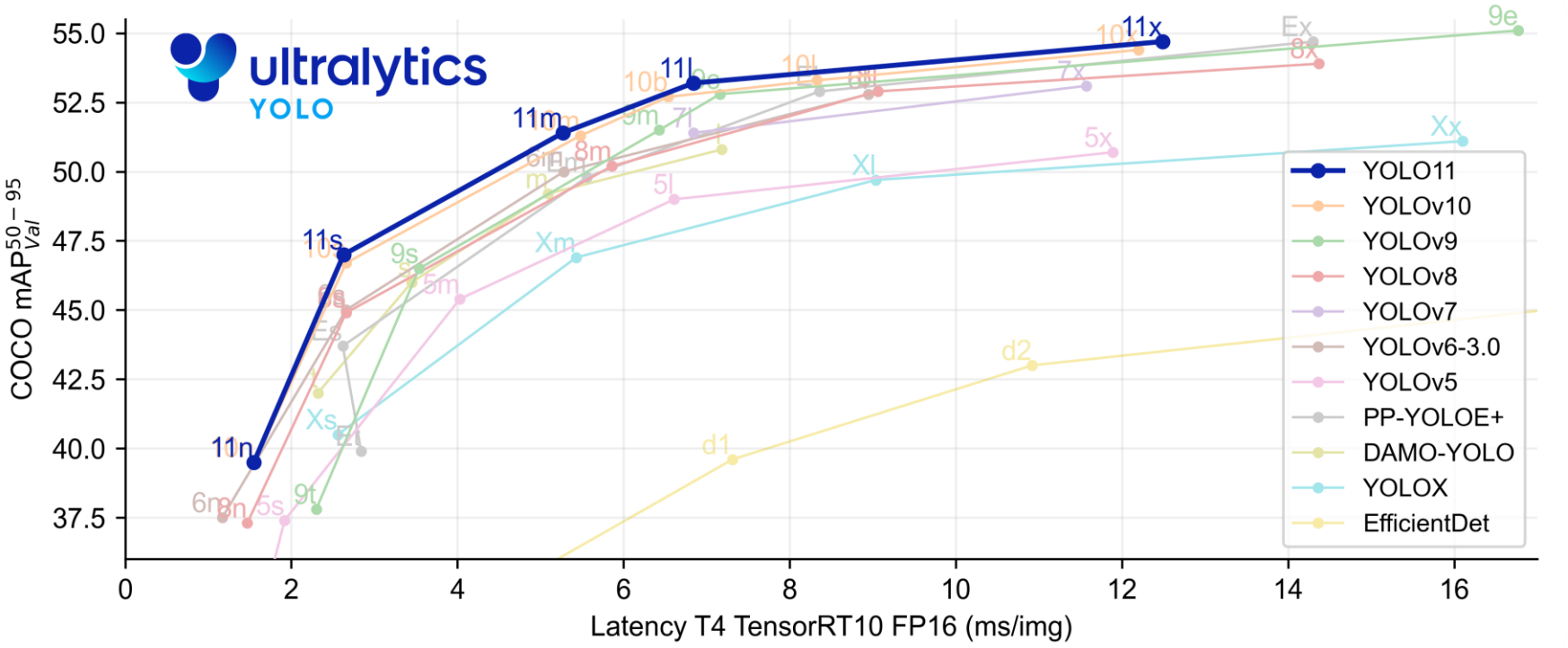
YOLO11 Variants

Variant	Depth	Width	Max Channels	Layers	Parameters	FLOPs
Nano	0.50	0.25	1024	181	2.62 M	6.6 G
Small	**0.50**	**0.50**	**1024**	**181**	**9.46 M**	**21.7 G**
Medium	**0.50**	**1.00**	**512**	**231**	**20.11 M**	**68.5 G**
Large	**1.00**	**1.00**	**512**	**357**	**25.37 M**	**87.6 G**
eXtra-L	1.00	1.00	512	357	56.97 M	196.0 G

https://github.com/ultralytics/ultralytics/blob/main/ultralytics/cfg/models/11/yolo11.yaml

MS-COCO Performance

YOLO11 Variants



https://docs.ultralytics.com/models/yolo11/

Varying Labelling Rules

Target Y

Dimension	Strategy	Domain
Consistency	Quality Control	[*basic, reference, careful*]
Spatial	Object Shape	[box_{LARGE}, box_{SMALL}, *polygon*]
Categorical	Classes Discarded	[*none, string, good*]

Annotation Consistency

- Common Labelling Errors

Annotation Consistency

- **Common Labelling Errors**
  - Duplicated Instances

Annotation Consistency

- **Common Labelling Errors**
  - Duplicated Instances
  - Bounding Box Offset

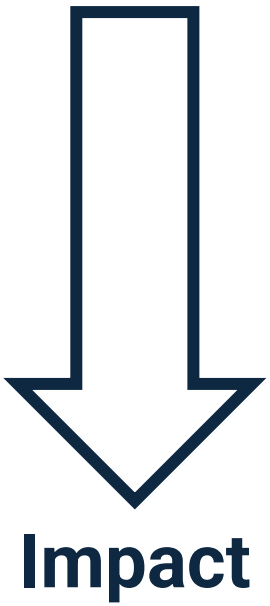
Annotation Consistency

- **Common Labelling Errors**
  - Duplicated Instances
  - Bounding Box Offset
  - Missing Objects

Annotation Consistency

- **Common Labelling Errors**
  - Duplicated Instances
  - Bounding Box Offset
  - Missing Objects
  - Wrong Class

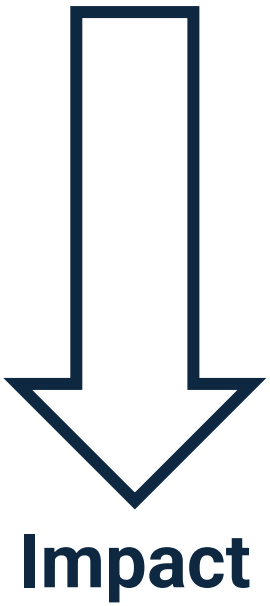
Annotation Consistency



- **Common Labelling Errors**
  - Duplicated Instances
  - Bounding Box Offset
  - Missing Objects
  - Wrong Class

Annotation Consistency

Noise Correction

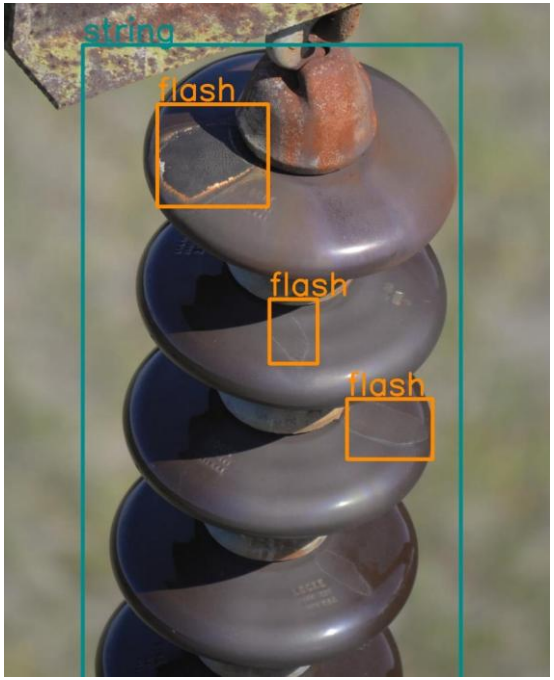


- **Common Labelling Errors**
  - Duplicated Instances
  - Bounding Box Offset
  - **Missing Objects**
  - Wrong Class

Annotation Consistency

Noise Correction

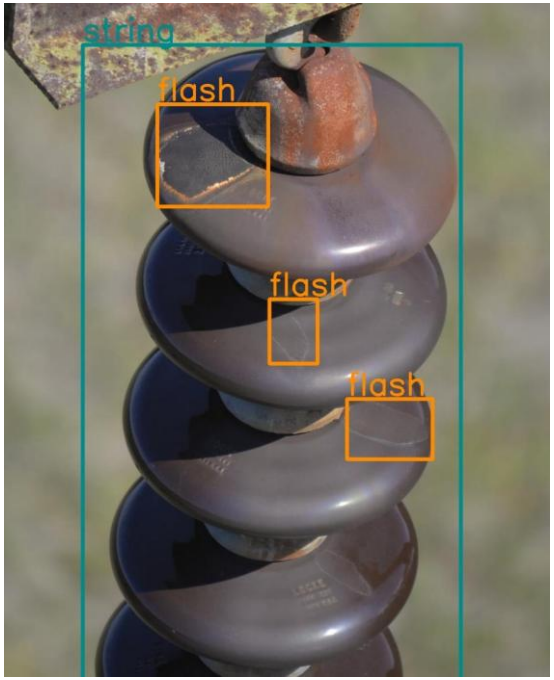
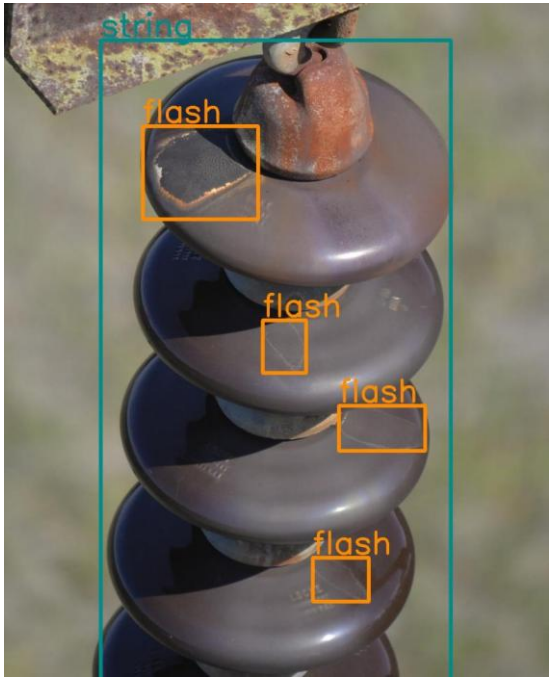
Reference [*]



[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

Annotation Consistency

Noise Correction

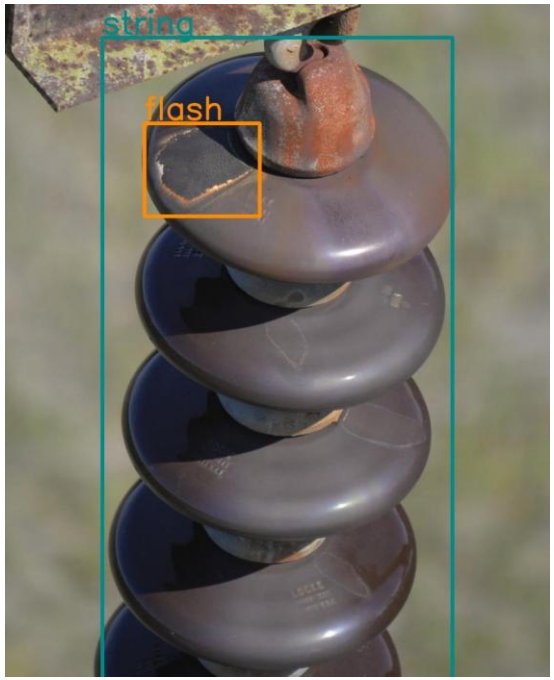
	Reference [*]	Careful
		

[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

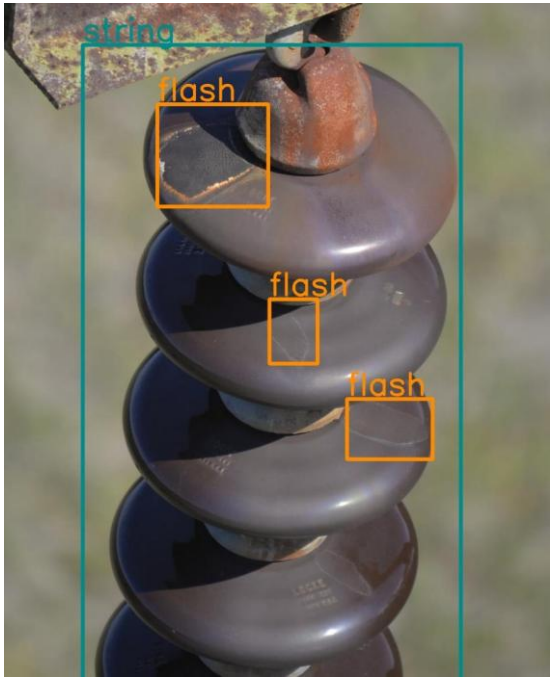
Annotation Consistency

Noise Correction

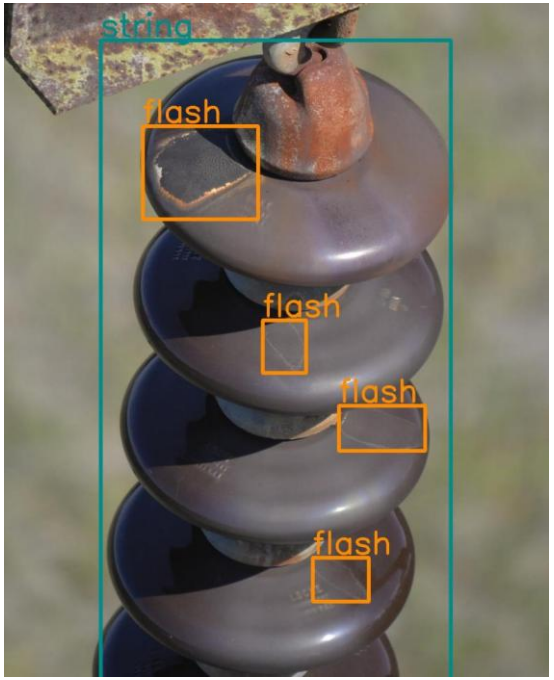
Basic



Reference [*]



Careful



[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

Spatial Granularity

Region Tightness

Spatial Granularity

Region Tightness

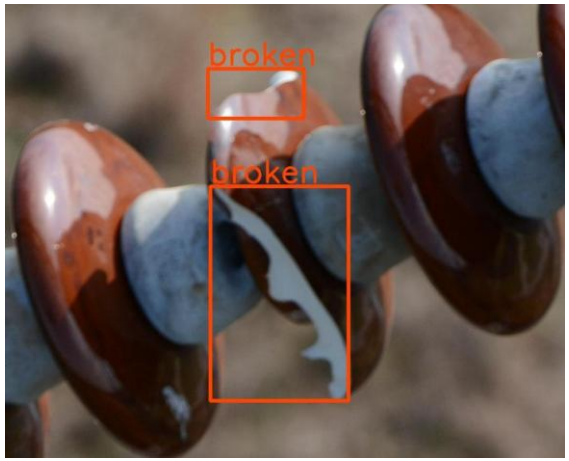
		Polygon
		Proposed Annotation



Spatial Granularity

Region Tightness

	Box_{Small}	Polygon
	Polygon Max and Min Points	Proposed Annotation

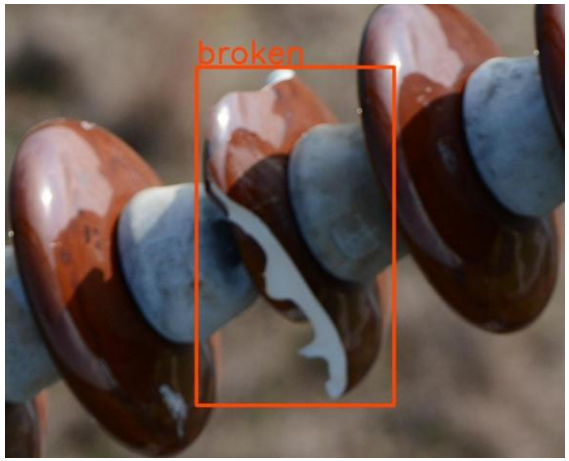


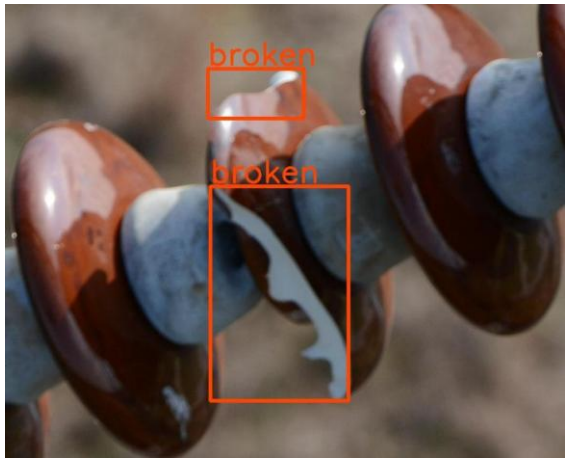


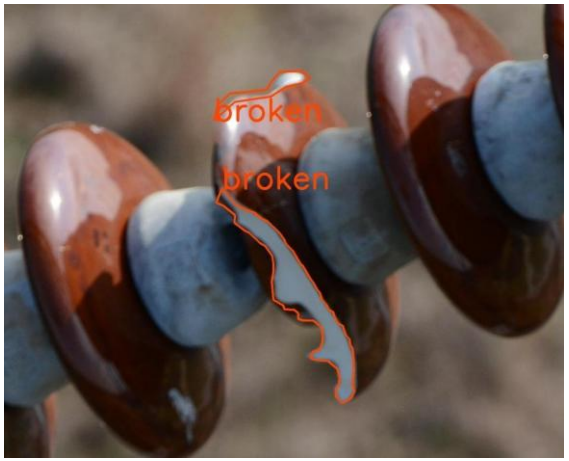
Spatial Granularity

Region Tightness

Box_{Large}	Box_{Small}	Polygon
Use Disc Points IF Defect Overlap	Polygon Max and Min Points	Proposed Annotation







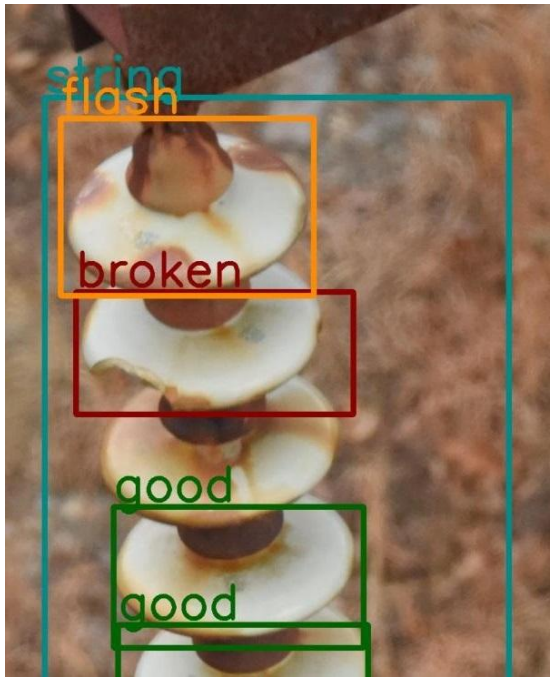
Categorical Definition

Number of Classes

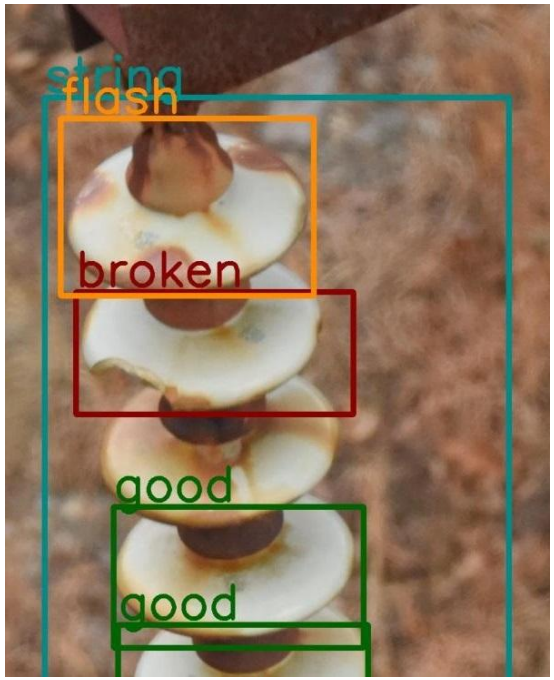
Categorical Definition

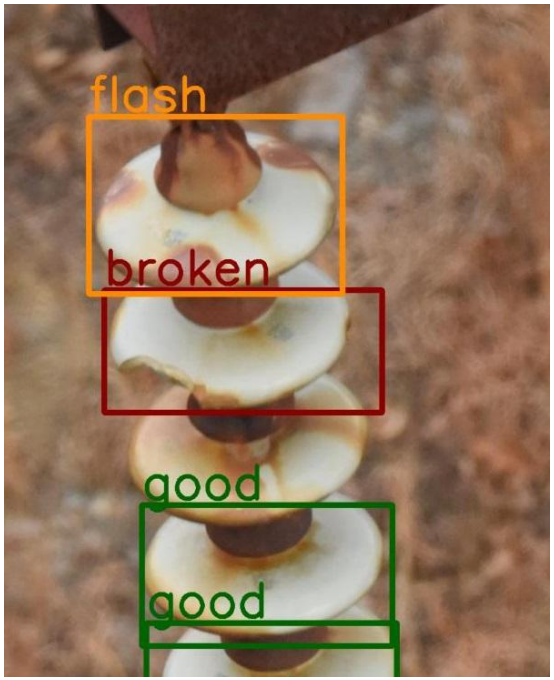
Number of Classes

4



Categorical Definition	Number of Classes
4	3

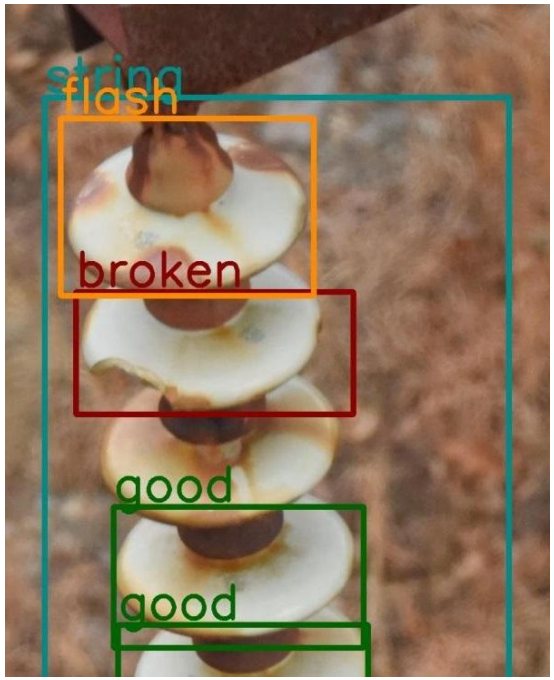


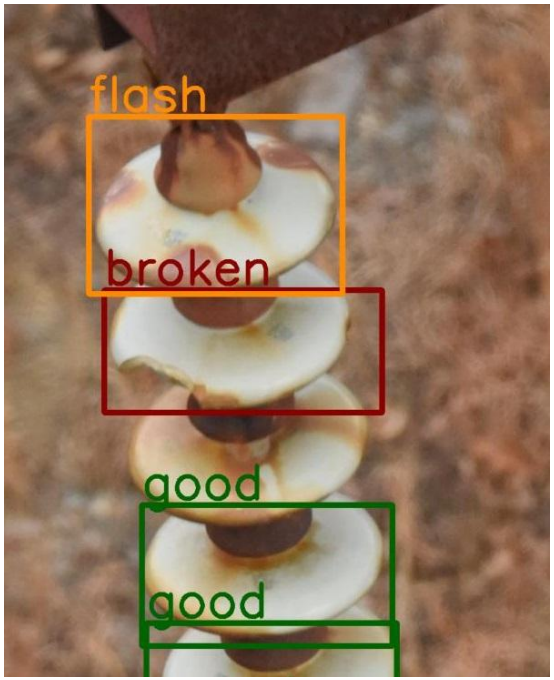


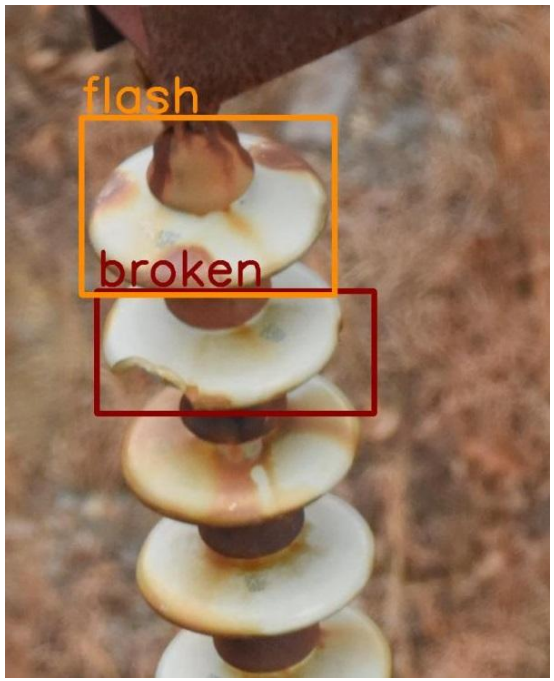
Categorical Definition

Number of Classes

4	3	2





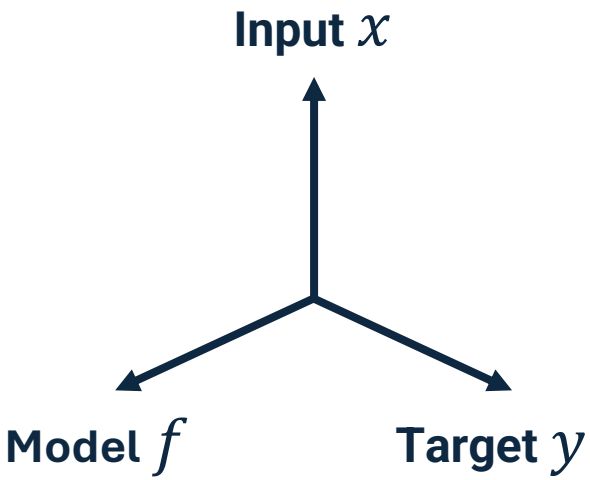


Experiments Configuration

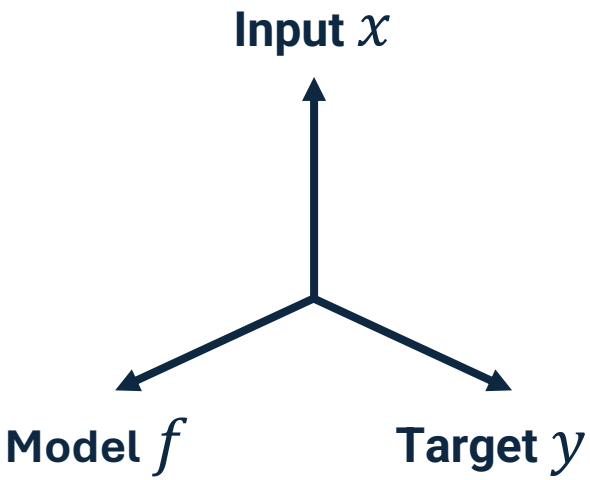
Dimensions and Domains

Experiments Configuration

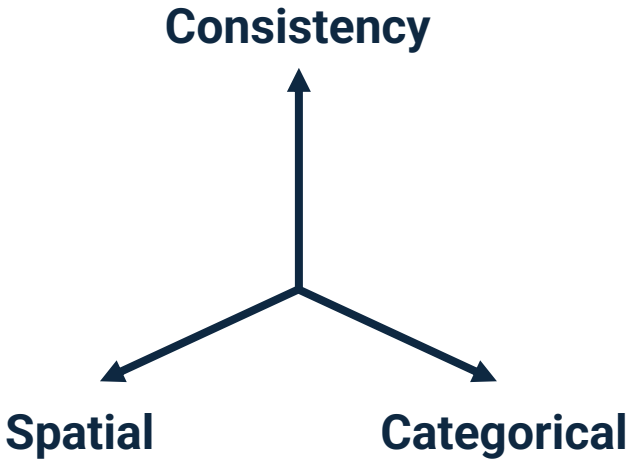
Dimensions



Experiments Configuration



Dimensions



Experiments Configuration

Domains and Range

Experiment Impact		Consistency	Spatial	Categorical
Dimension	**Domain**	**Range / Count**		
Quality Control	$[basic, ref, careful]$	[1 : 3] **3**	[3] **1**	[2] **1**
Object Shape	$[box_L, box_S, polygon]$	[2] **1**	[1 : 3] **3**	[1] **1**
Classes Discarded	$[none, string, good]$	[2] **1**	[1] **1**	[1 : 3] **3**
X-Val Fold	$[1, 2, 3, 4]$	[1 : 4] **4**	[1 : 4] **4**	[1 : 4] **4**
Model Size	$[small, medium, large]$	[1 : 3] **3**	[1 : 3] **3**	[1 : 3] **3**
Total Count	**108**	**36**	**36**	**36**

Results

- Experiments Setup
- Annotations Consistency
- Spatial Granularity
- Categorical Definition

Training Setup

Hyperparameters

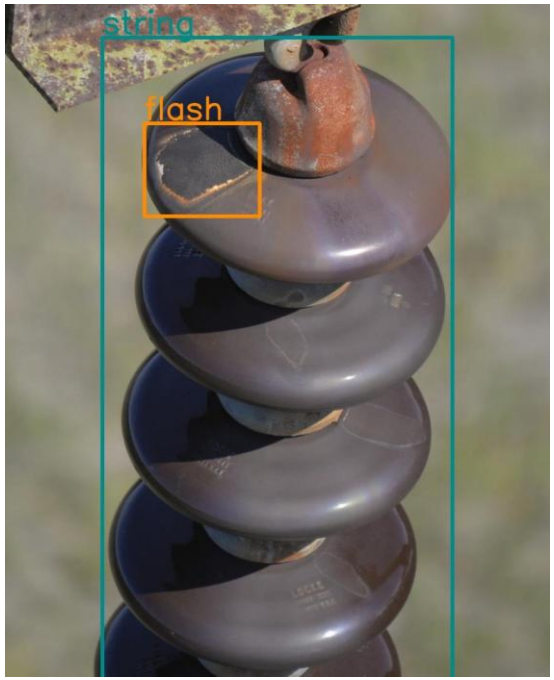
Variable	Value	Default	Description
epochs	250	100	Total number of forward- and back-prop iterations
batch	-1 (auto)	16	Number of images used to compute the mean error
imgsz	1280	640	Number of pixels along each side of the square input layer
freeze	2	None	Freezes the weights in the first N layers of the model
dropout	0.1	0.0	Fraction of neurons randomly disabled during training
lr0	0.1	0.01	Initial learning rate
lrf	0.001	0.01	Final learning rate as a fraction of the initial rate
warmup_epochs	10.0	3.0	Number of epochs using warmup learning rate

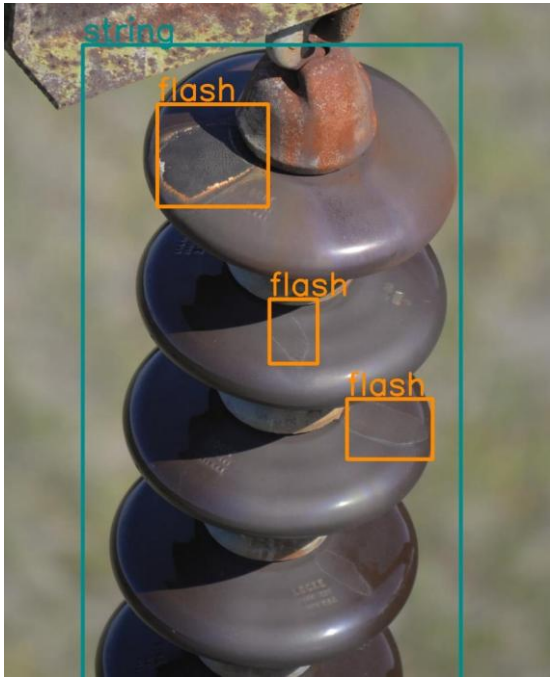
Annotation Consistency

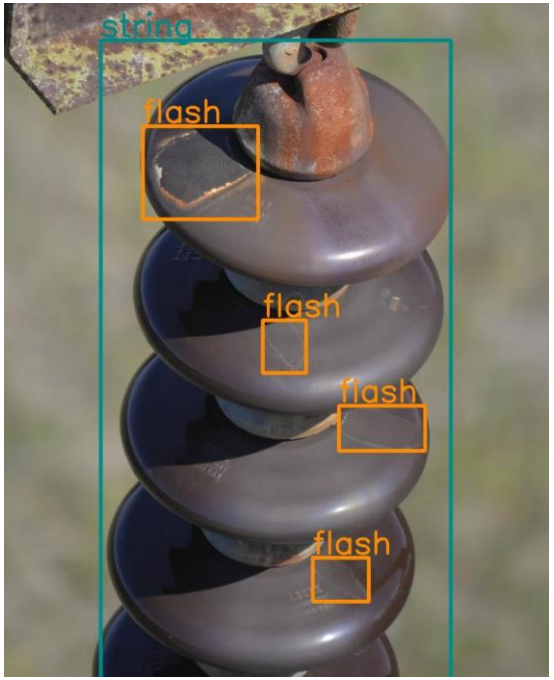
Annotation Consistency

Labeling Rule

Basic	Reference [*]	Careful





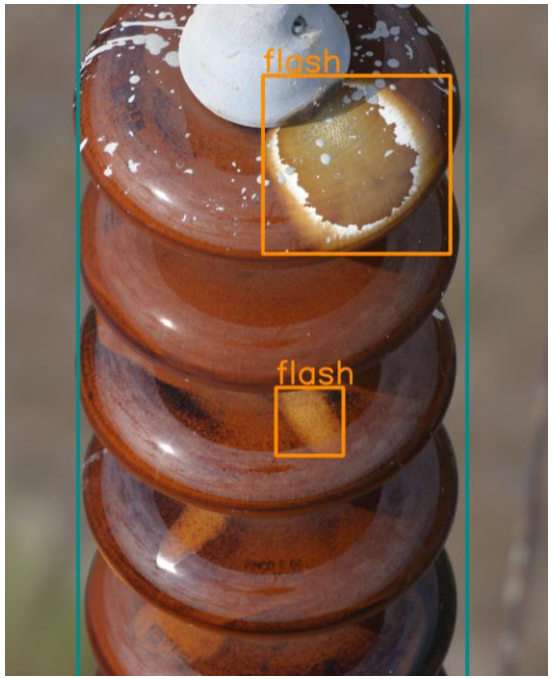


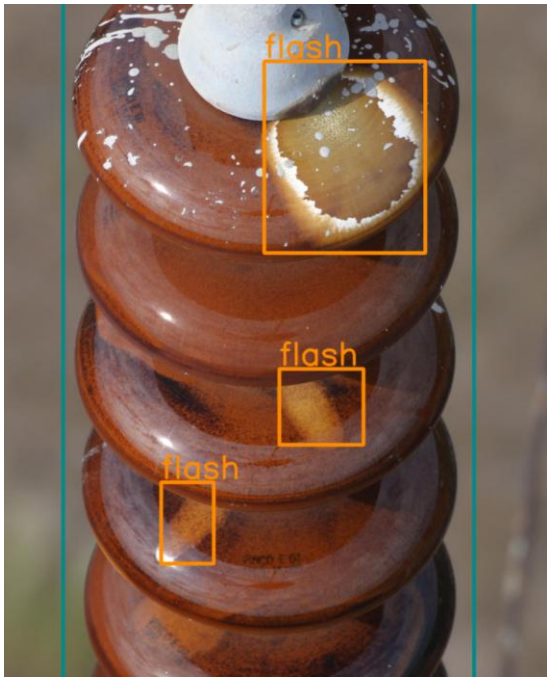
[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

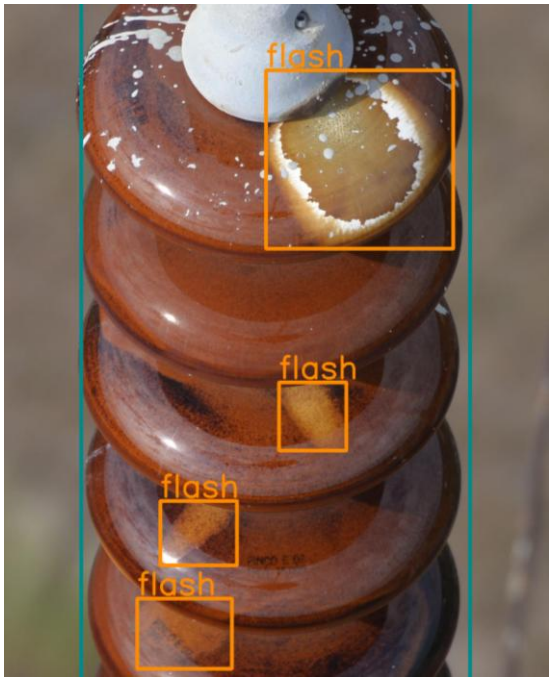
Annotation Consistency

More Samples

Basic	Reference [*]	Careful





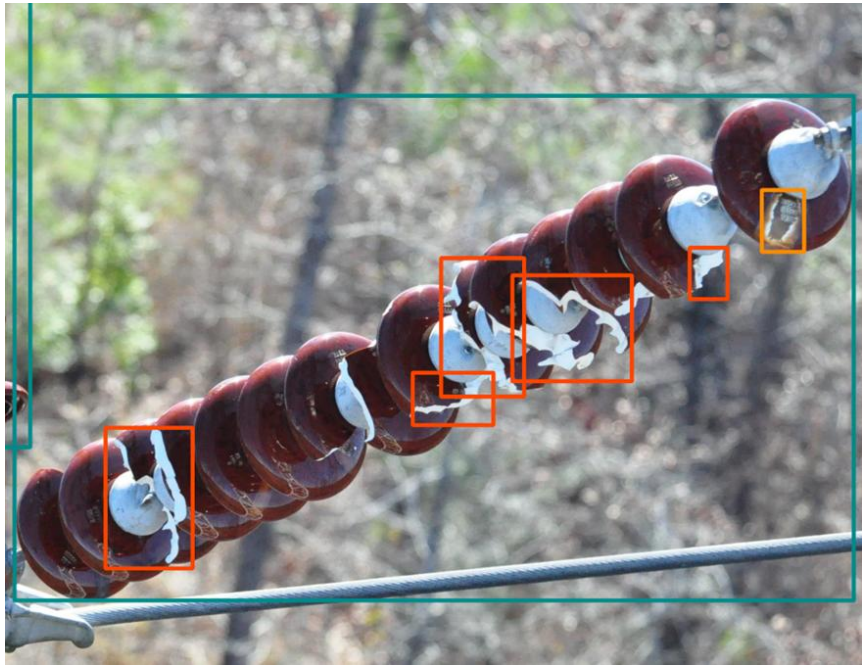


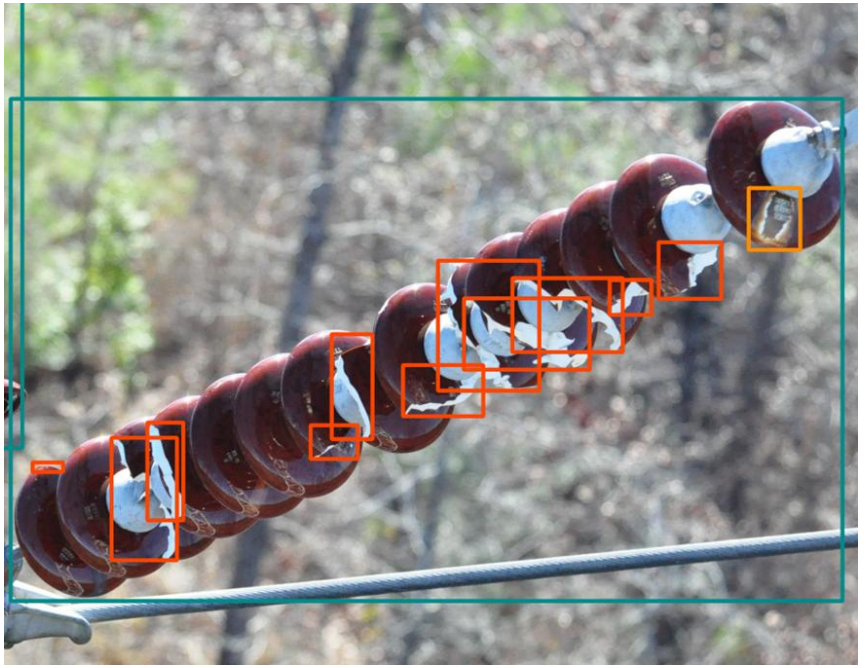
[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

Annotation Consistency

More Samples

Reference [*]	Careful



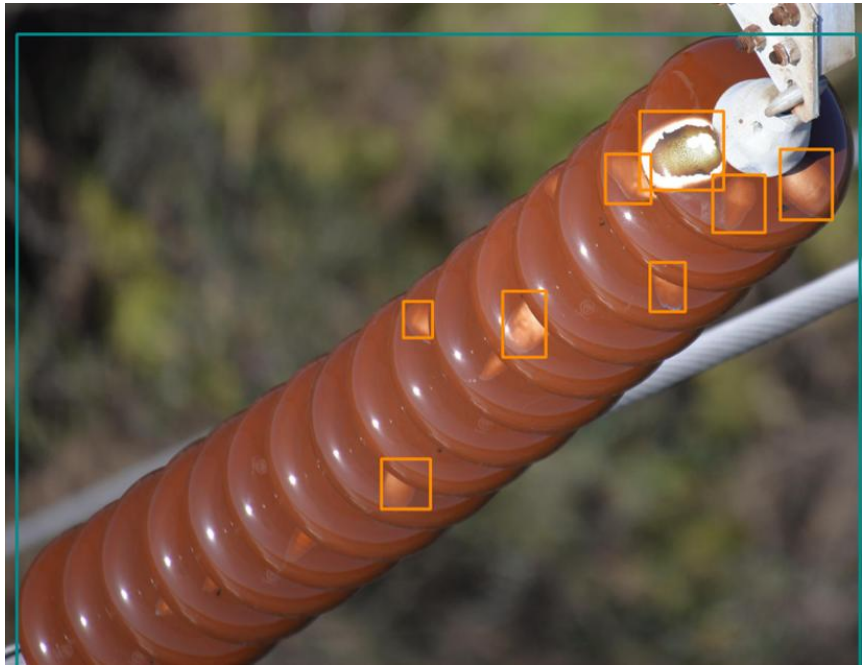


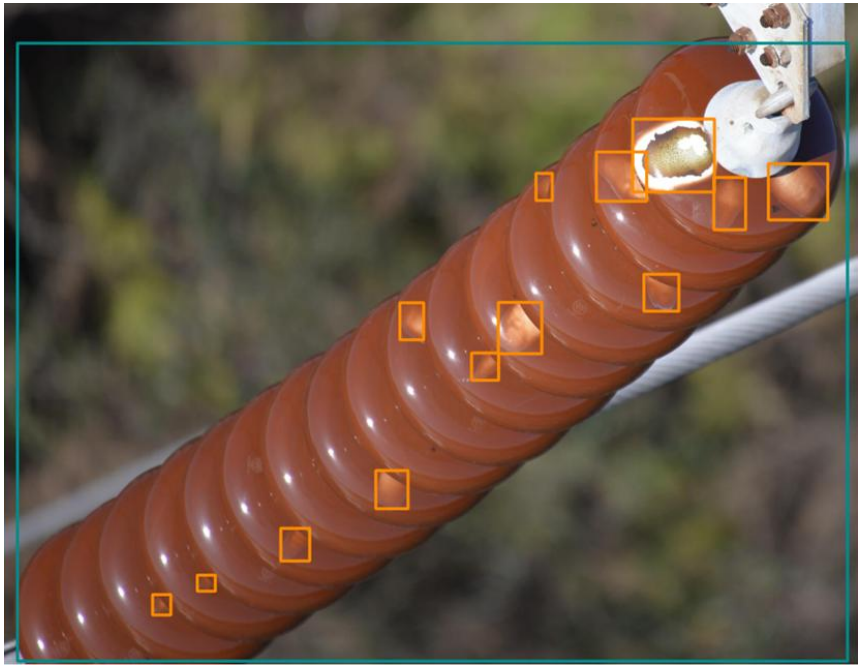
[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

Annotation Consistency

More Samples

Reference [*]	Careful



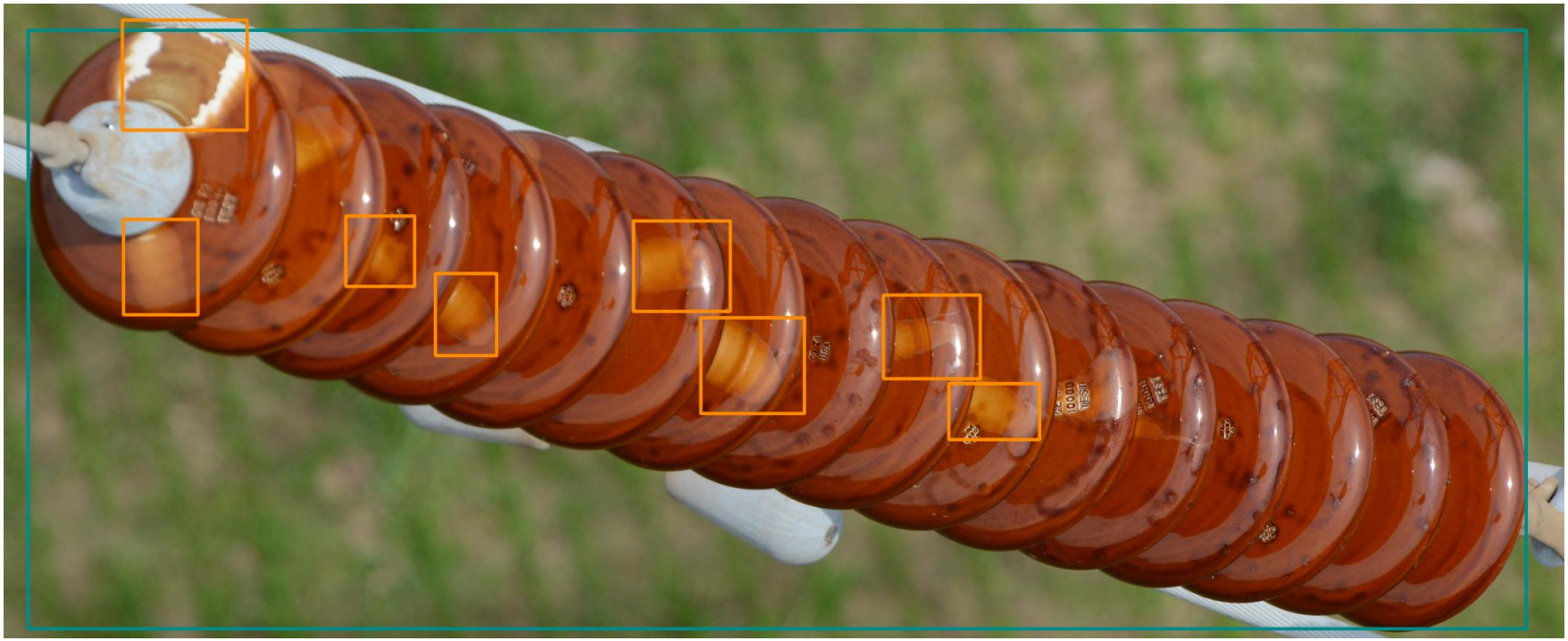


[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

Annotation Consistency

More Samples

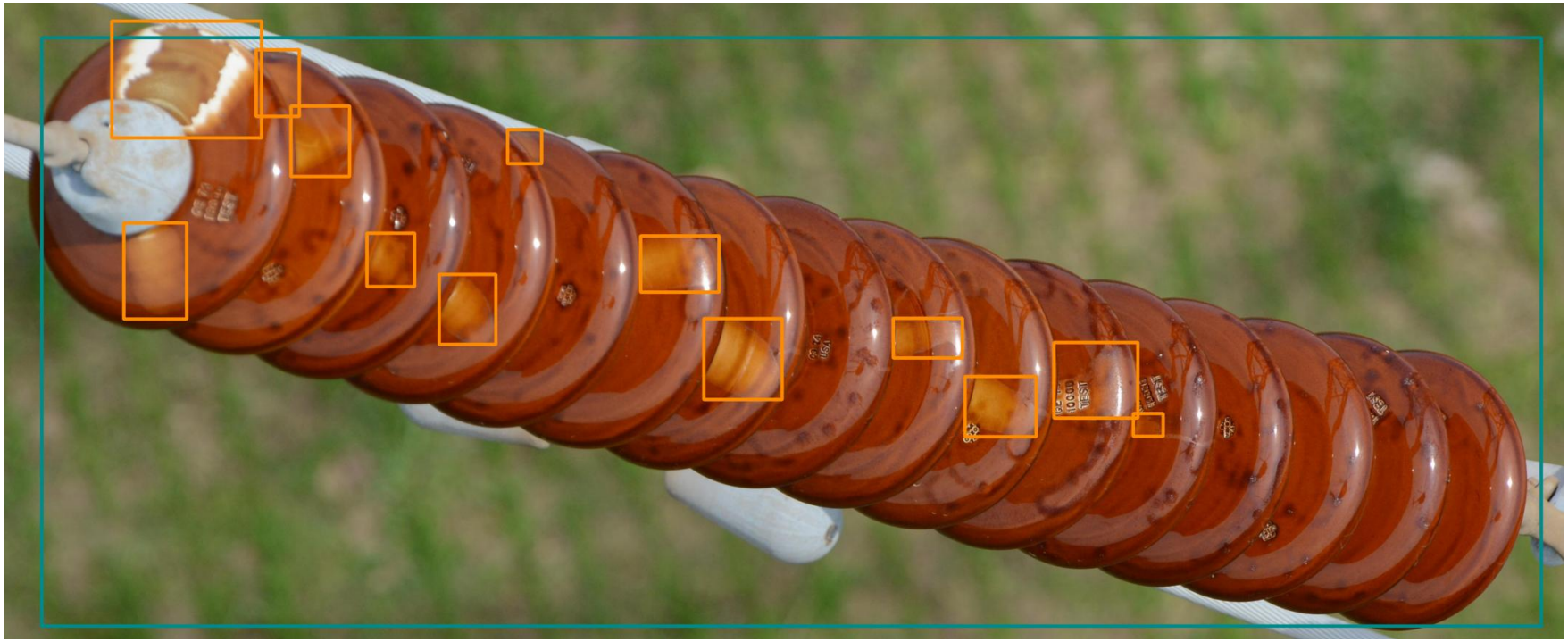
Reference [*]



Annotation Consistency

More Samples

Careful



Annotation Consistency

More Samples

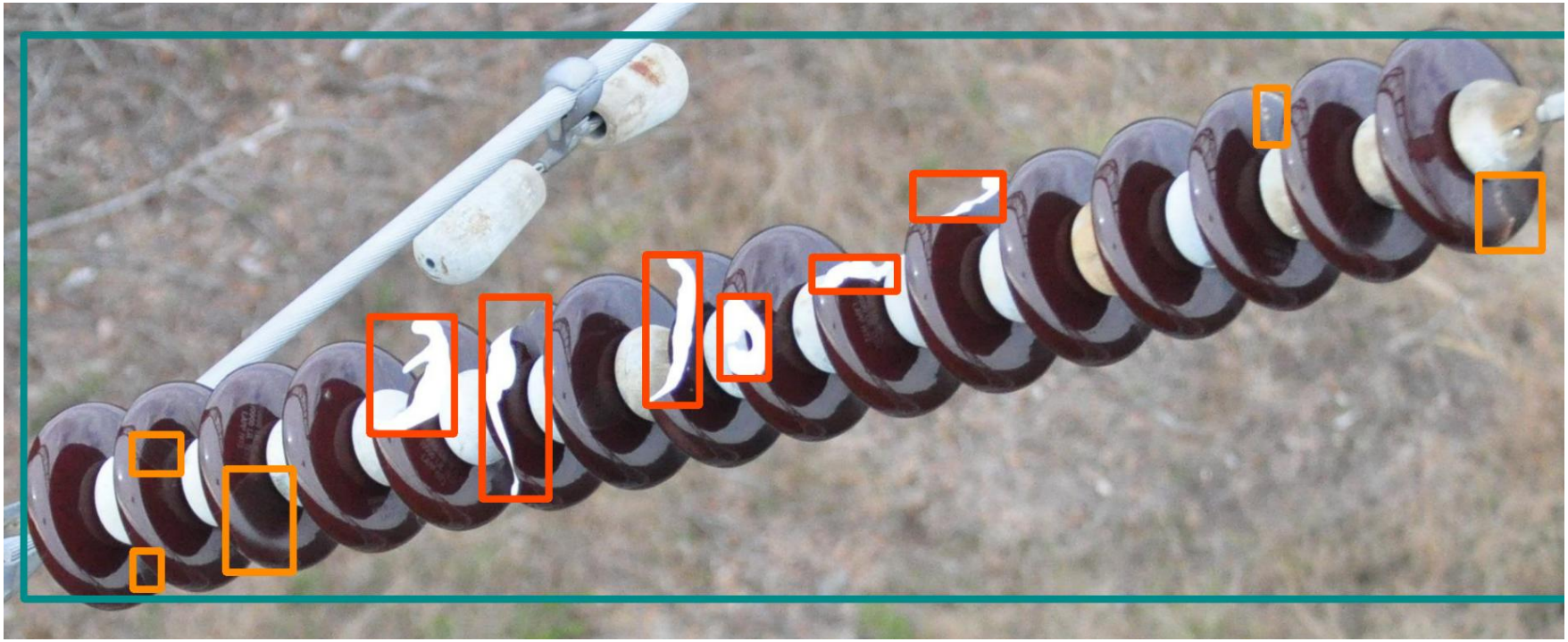
Reference [*]



Annotation Consistency

More Samples

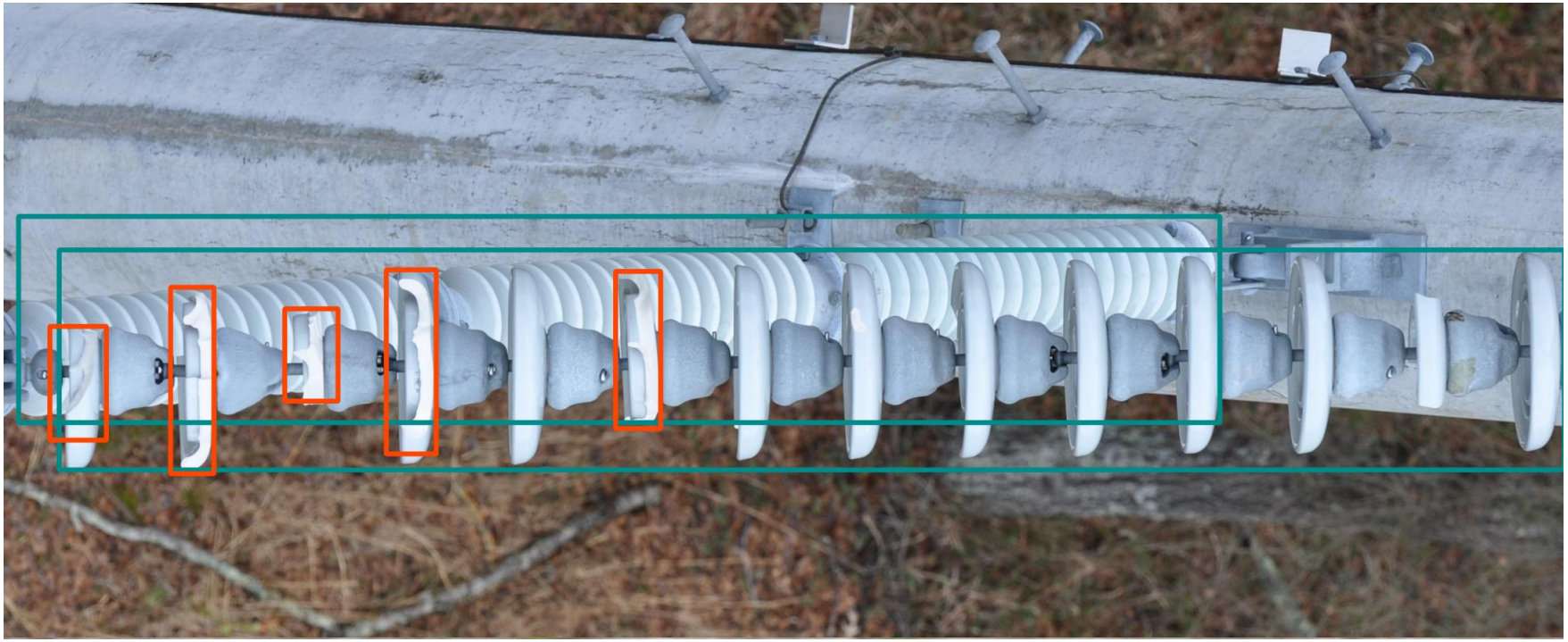
Careful



Annotation Consistency

More Samples

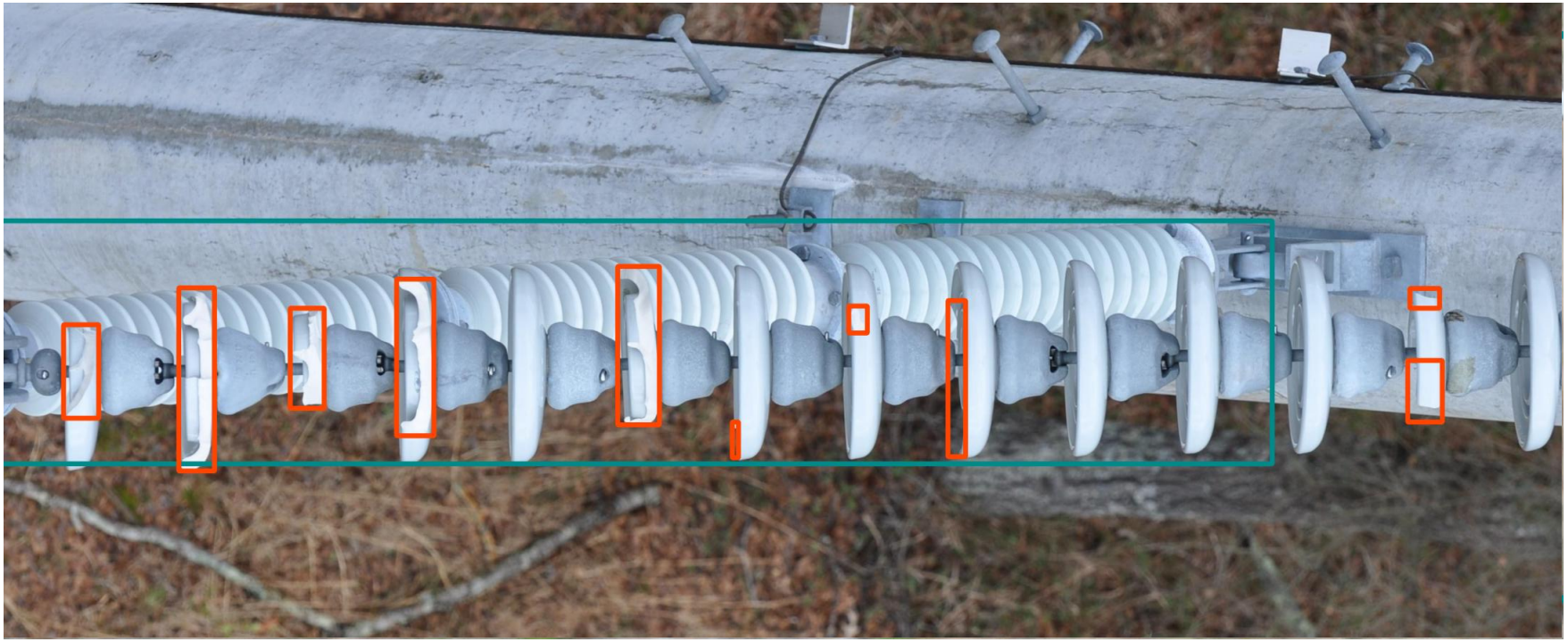
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Annotation Consistency

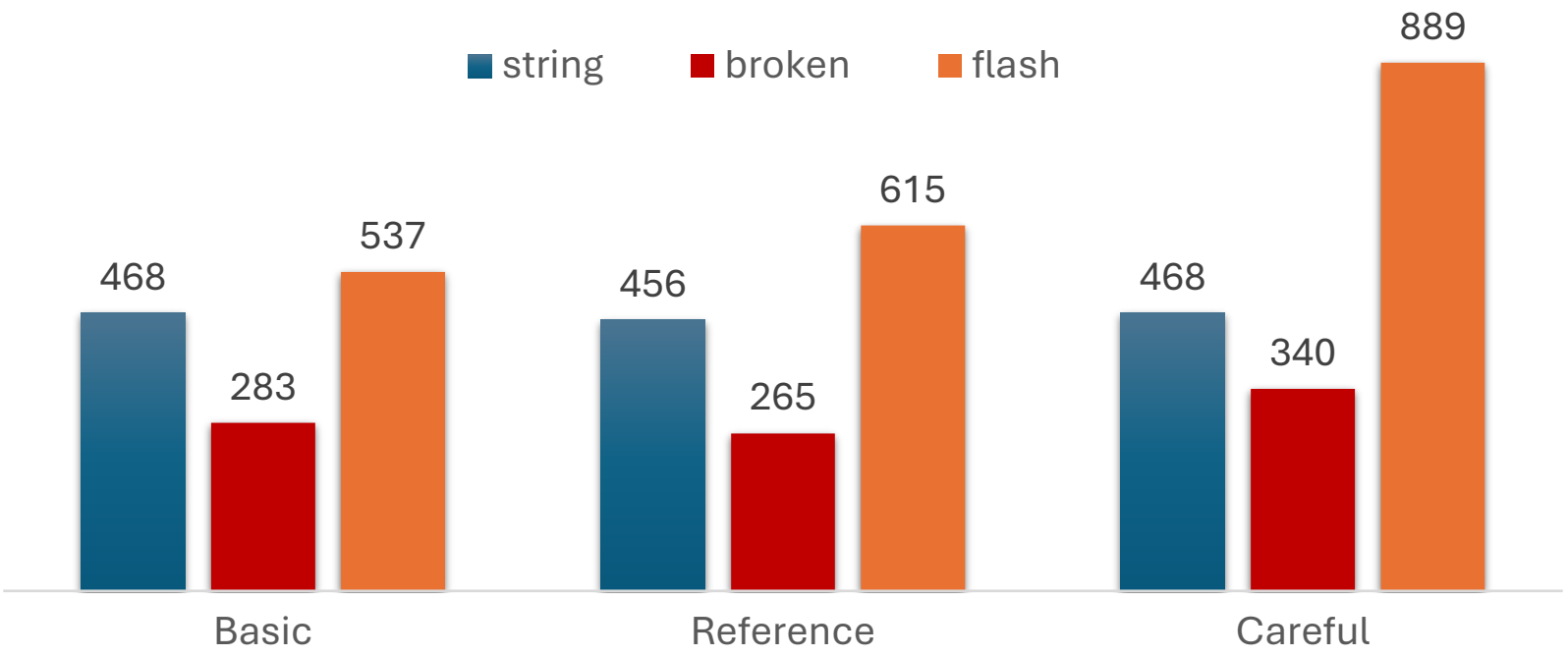
More Samples

Careful



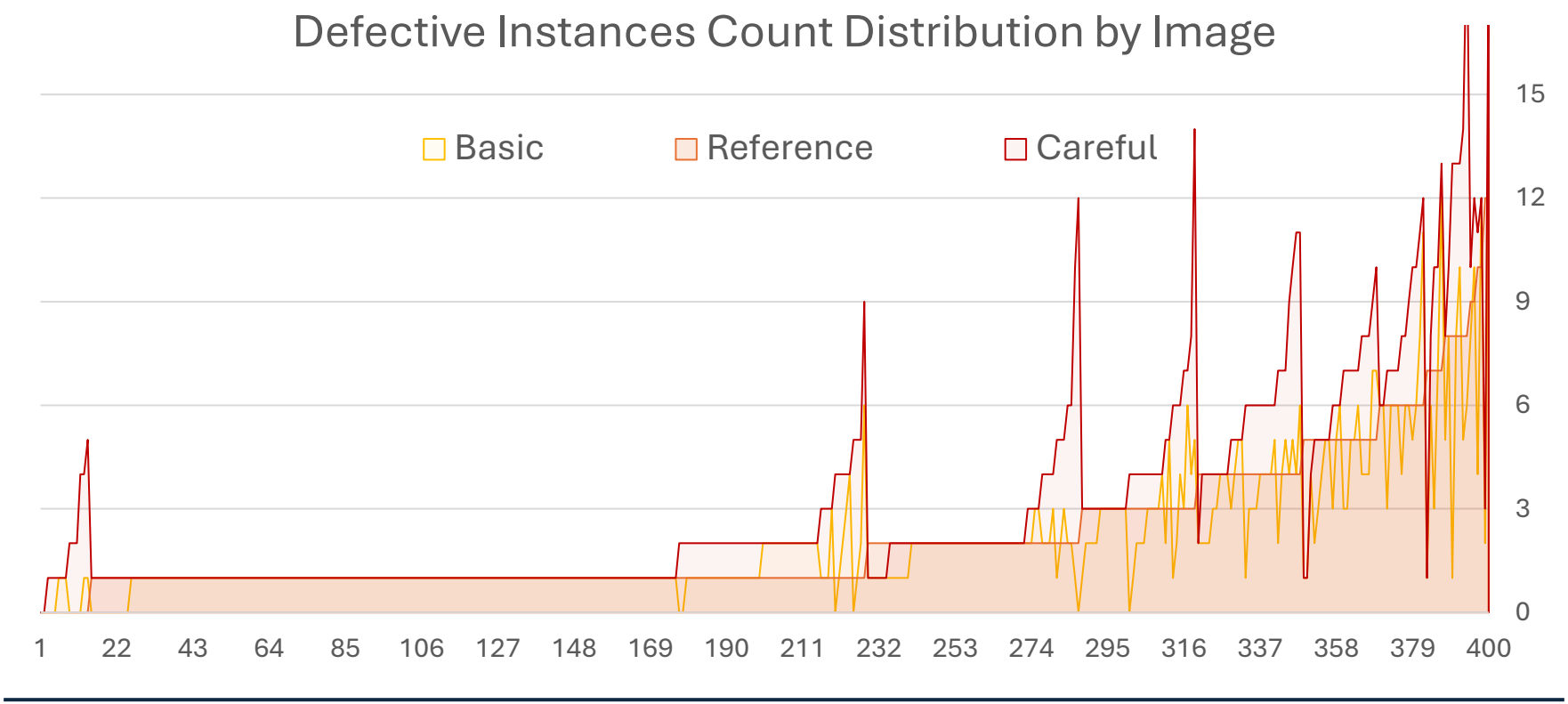
Annotation Consistency

Objects Count



[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

Annotation Consistency By Image

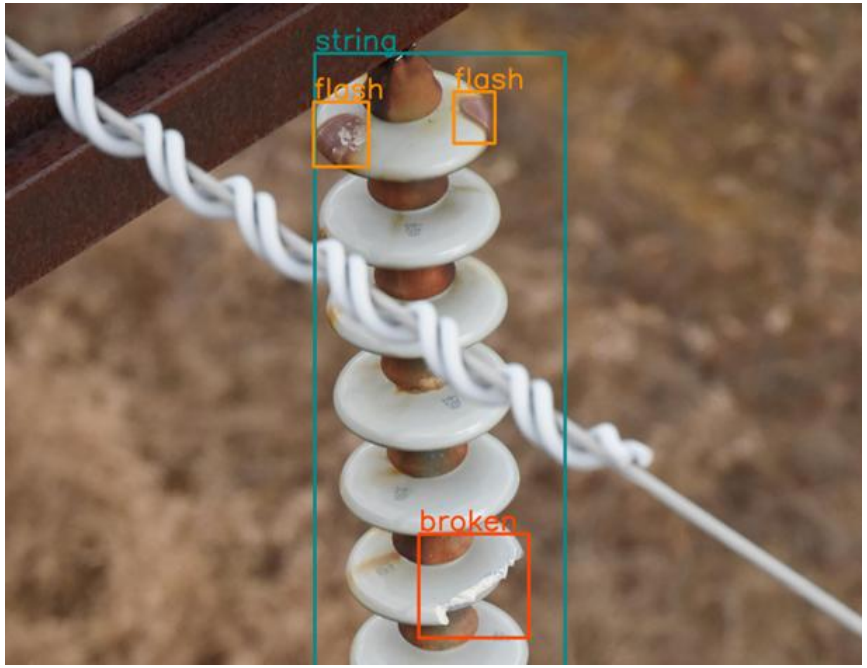


[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

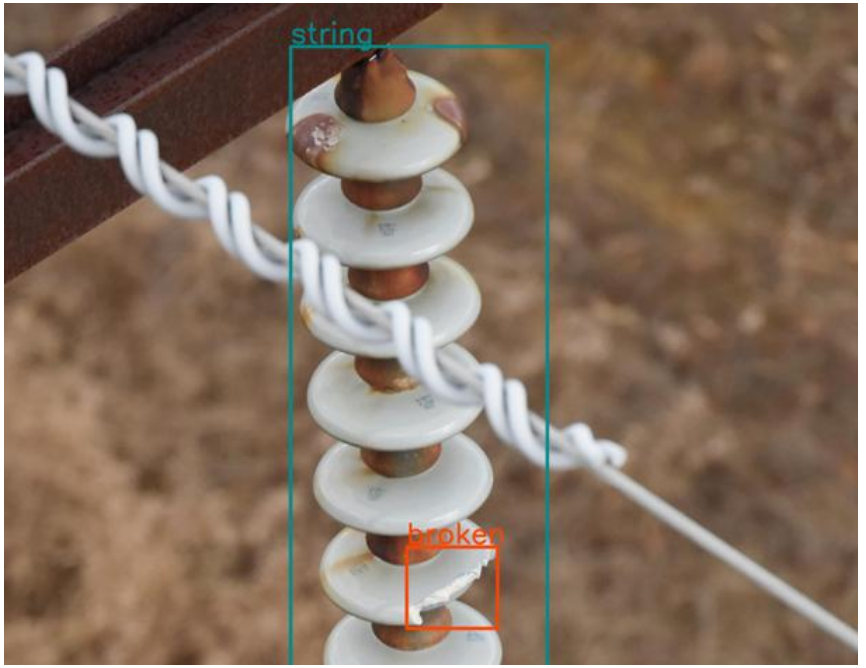
Annotation Consistency

Reference ≥ Careful

Reference [*]



Careful

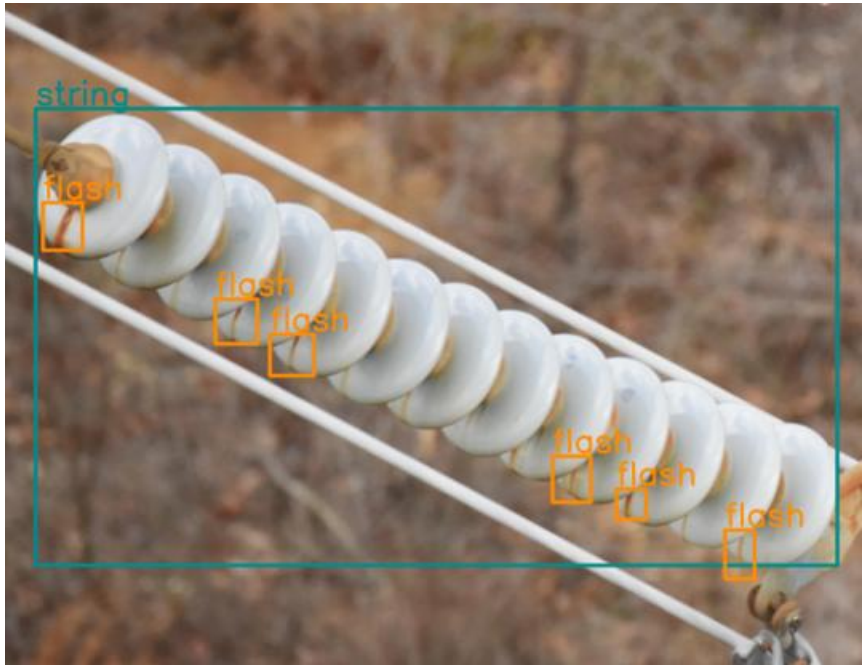


[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

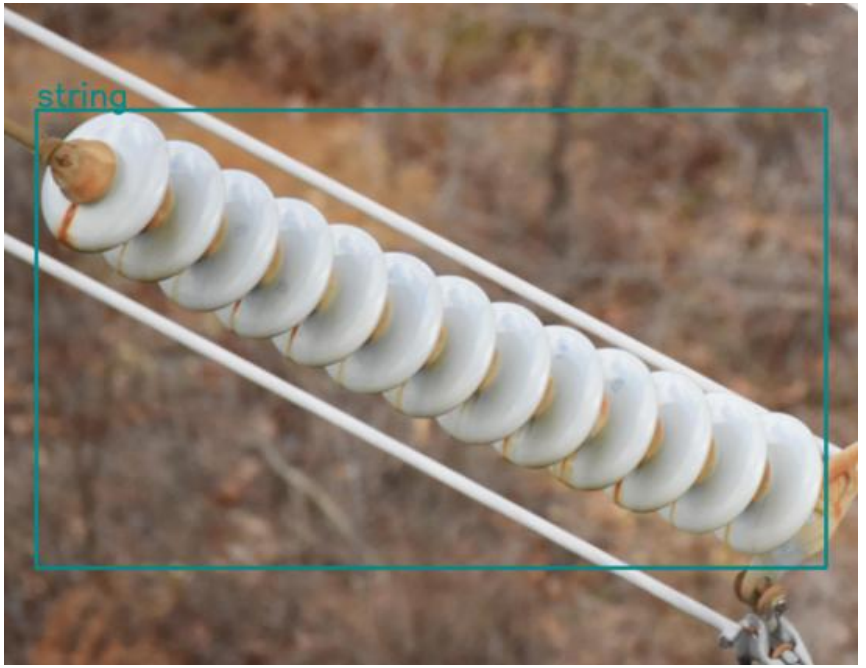
Annotation Consistency

Reference ≥ Careful

Reference [*]



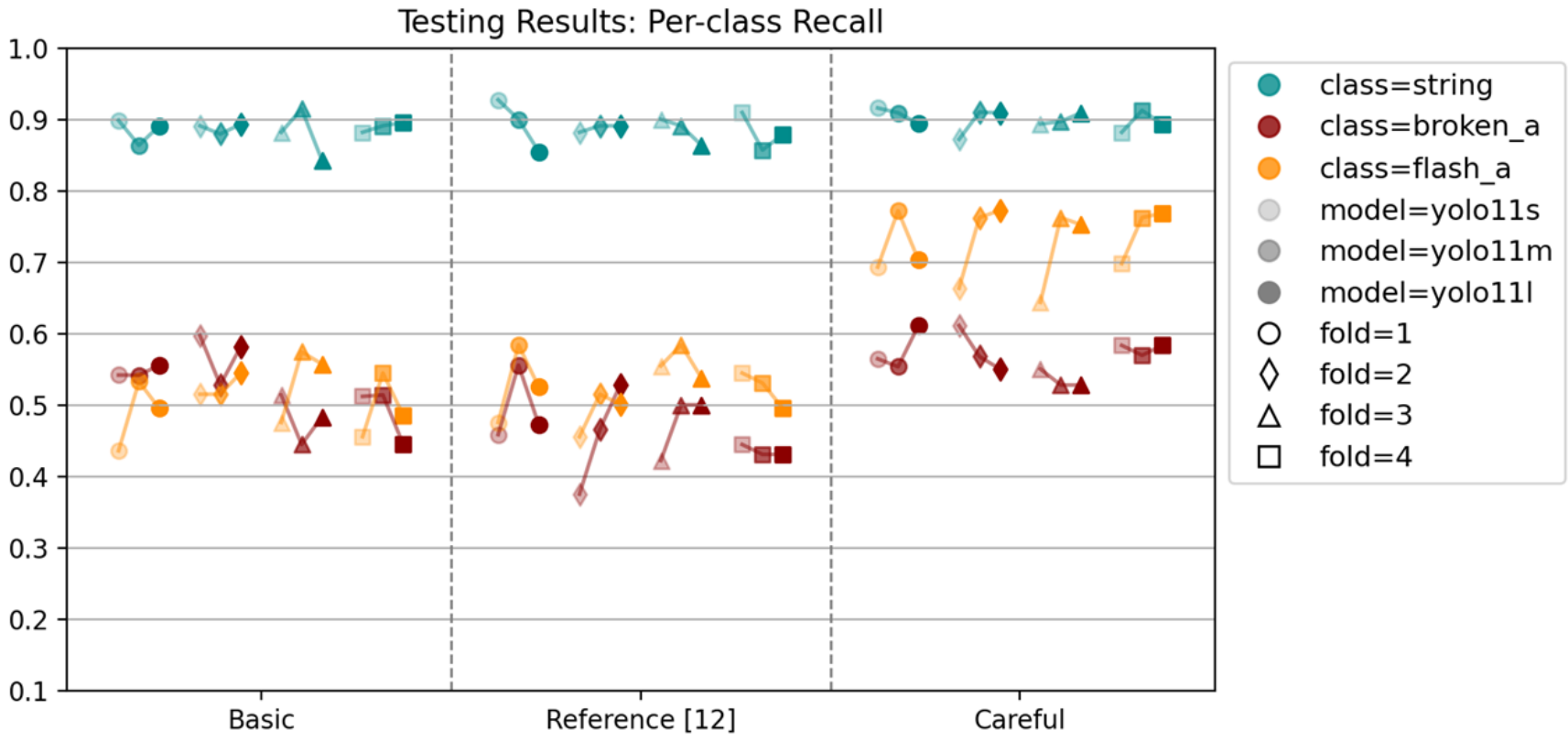
Careful



[*] Zheng, J.; Wu, H.; Zhang, H.; Wang, Z.; Xu, W. Insulator-Defect Detection Algorithm Based on Improved YOLOv7. Sensors 2022, 22, 8801. https://doi.org/10.3390/s22228801

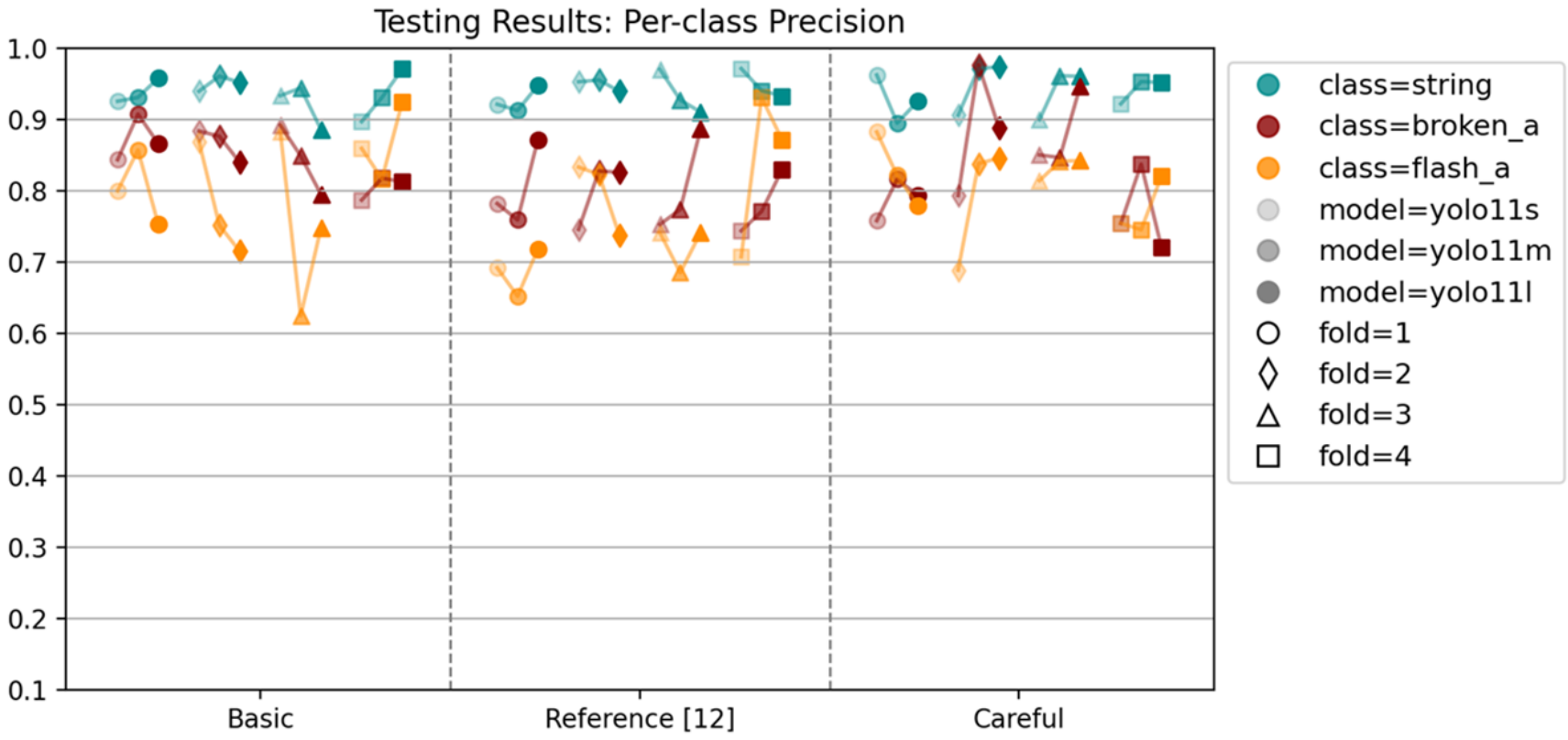
Annotation Consistency

Recall Impact



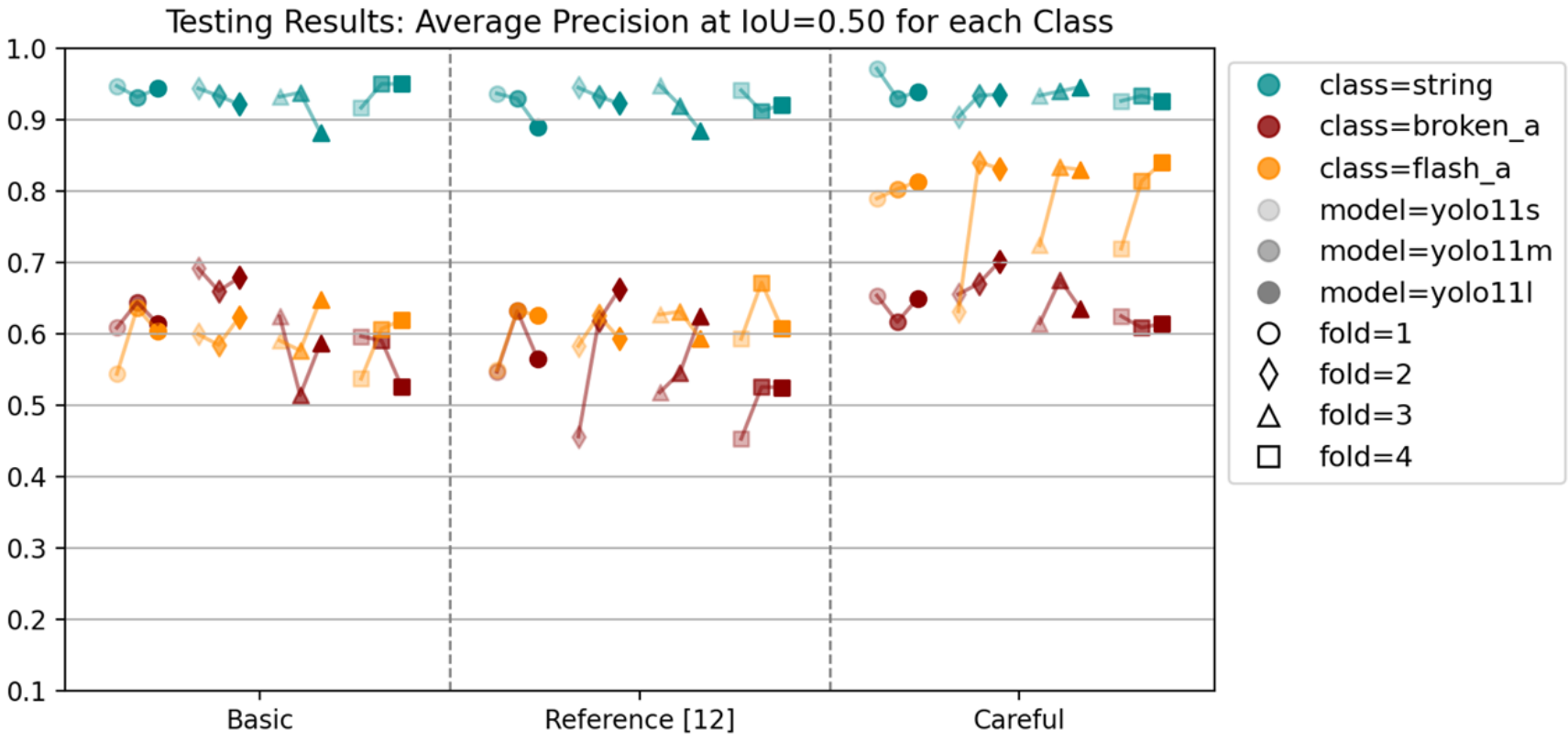
Annotation Consistency

Precision Impact



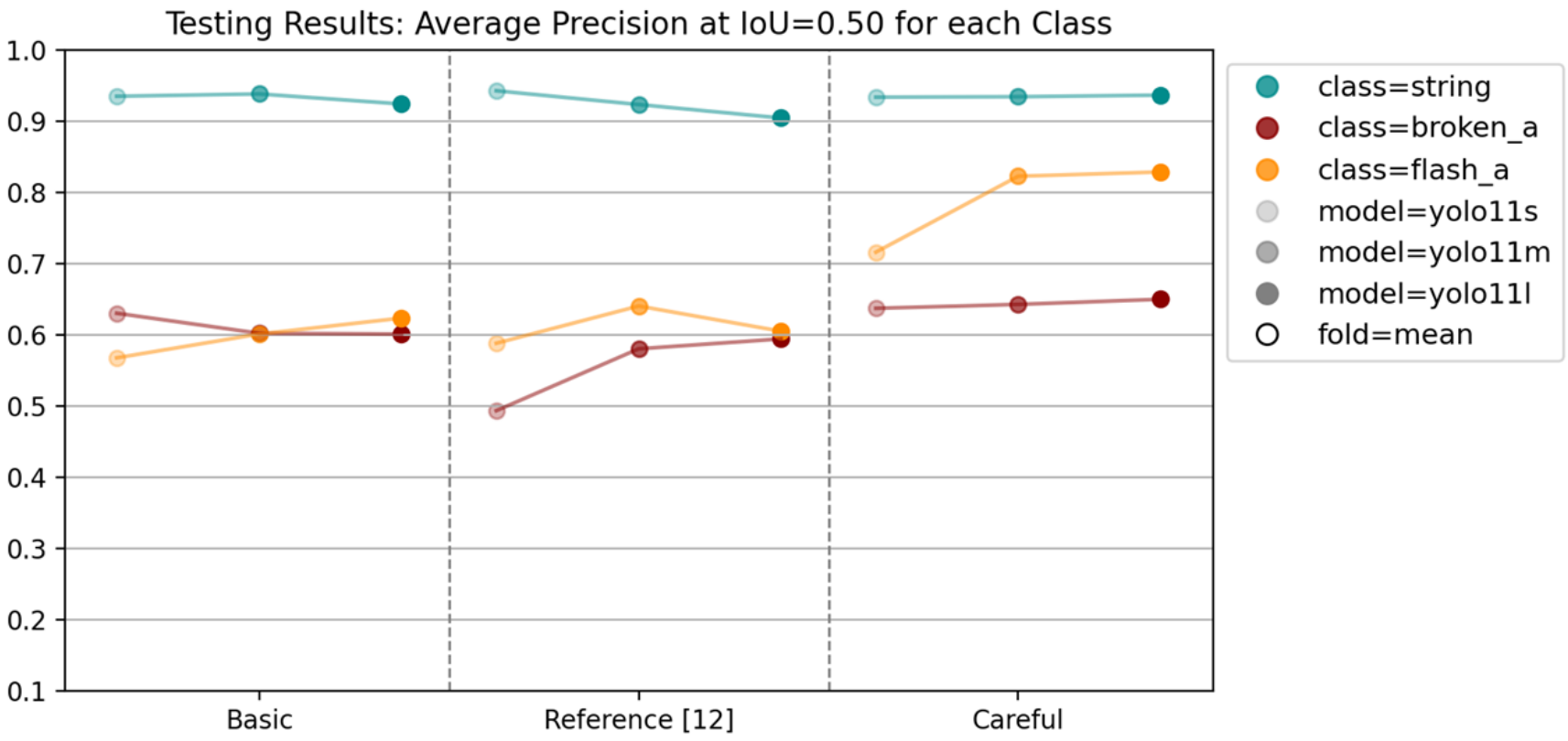
Annotation Consistency

AP50 Impact



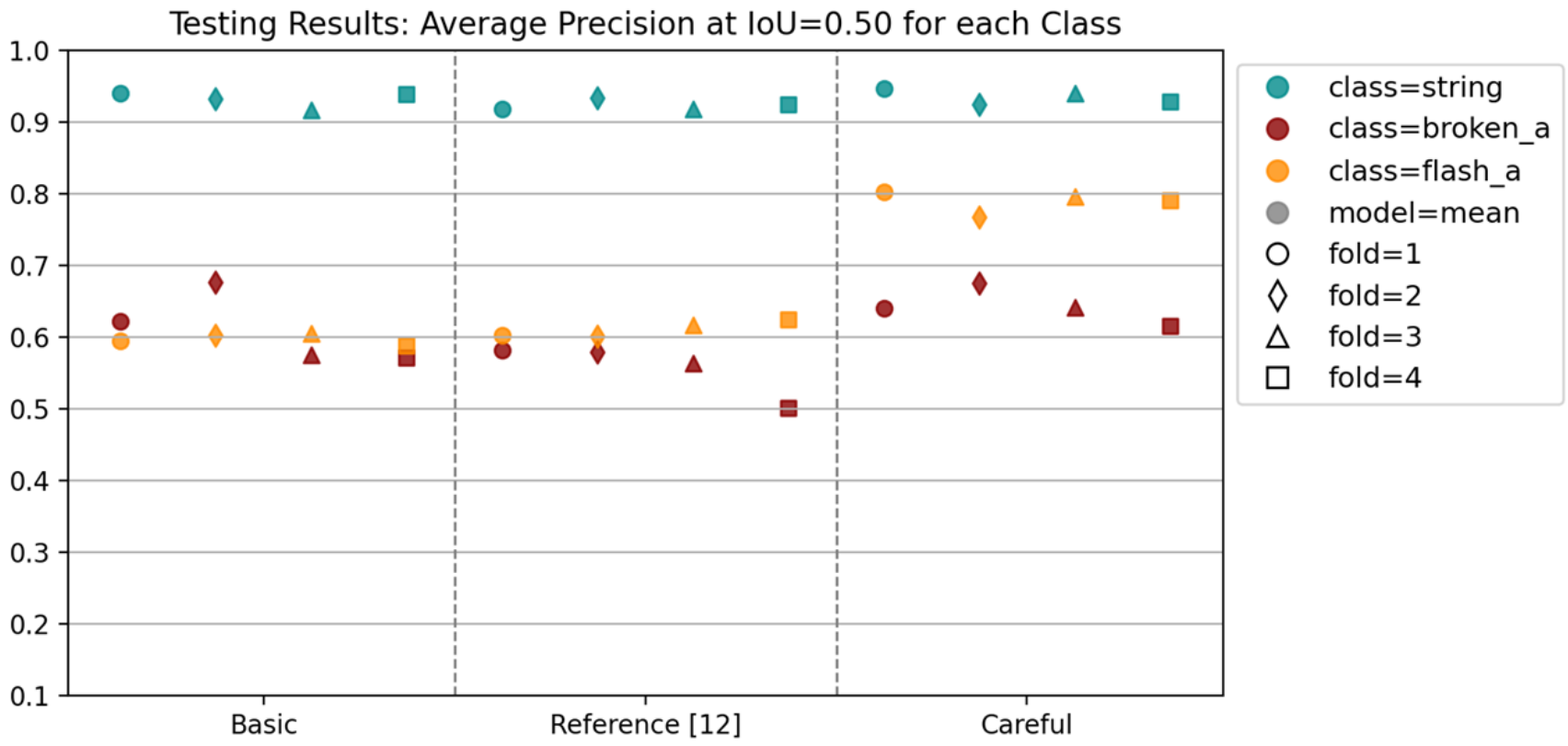
Annotation Consistency

Model Impact



Annotation Consistency

Fold Impact



Annotation Consistency

Key Takeaway

Missing Annotations Penalizes Recall

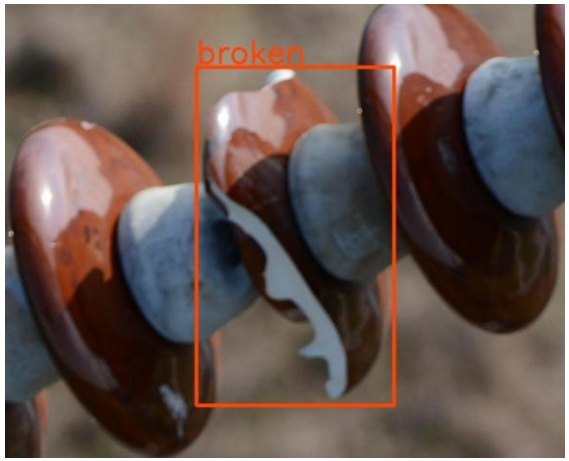
A high recall value indicates that the model is good at capturing all instances of the positive class, minimizing missed positive cases.

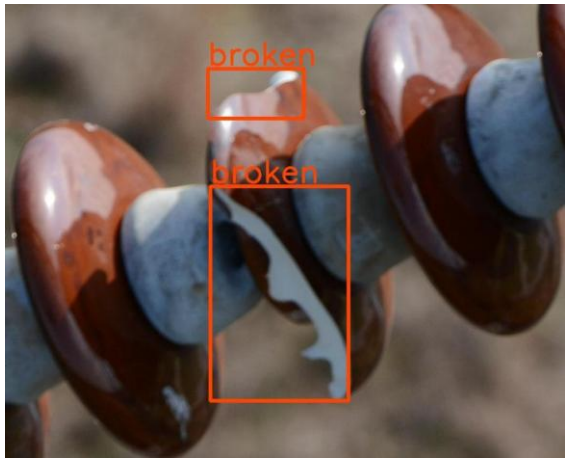
Spatial Granularity

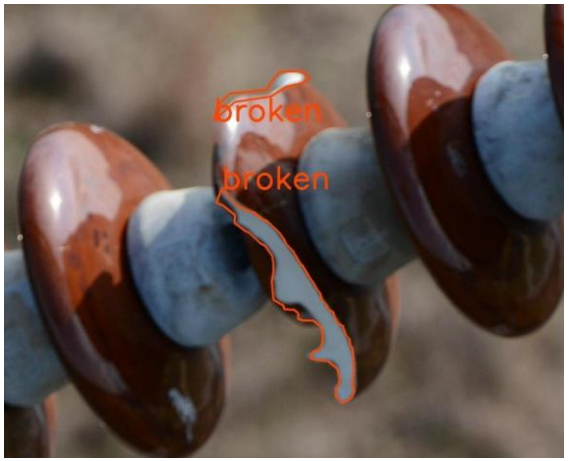
Spatial Granularity

Region Tightness

Box_{Large}	Box_{Small}	Polygon
Use Disc Points IF Defect Overlap	Polygon Max and Min Points	Proposed Annotation

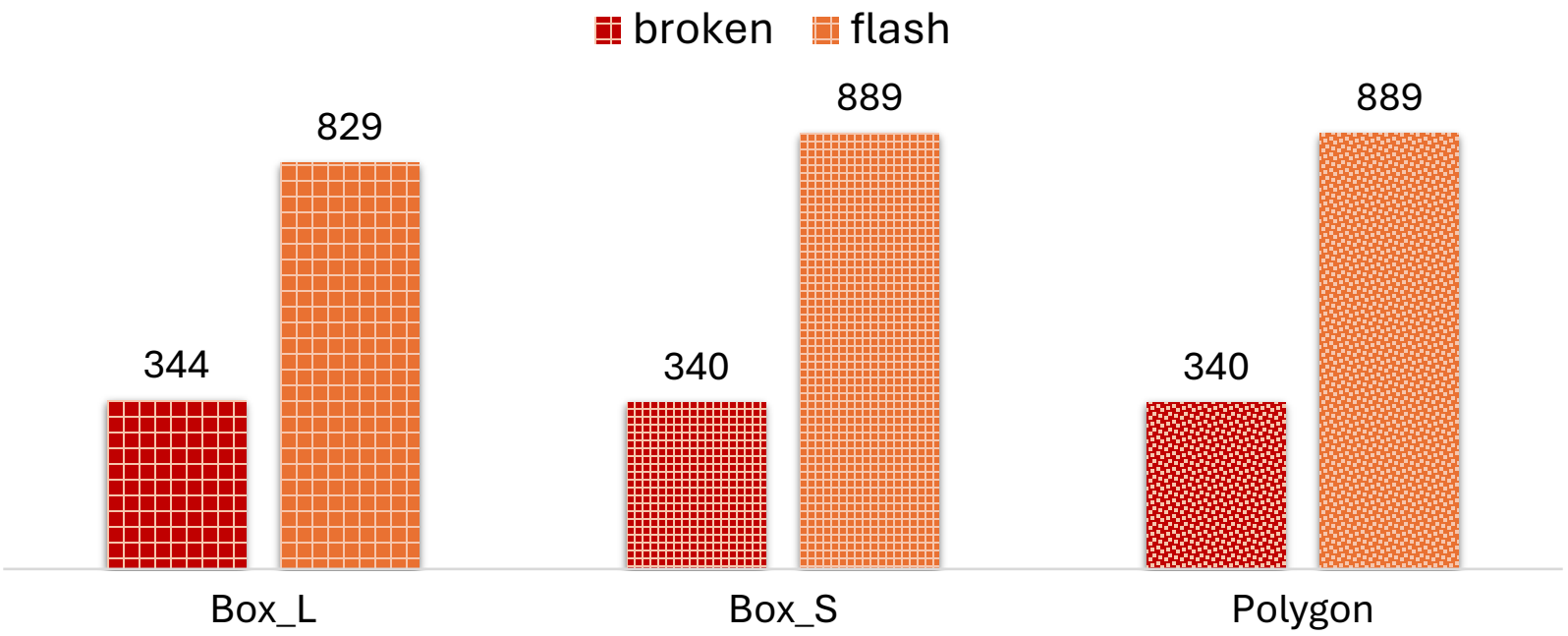






Spatial Granularity

Object Count



Spatial Granularity

$Box_{LARGE} \leq Box_{SMALL}$

Polygon	Box_{SMALL}	Box_{LARGE}

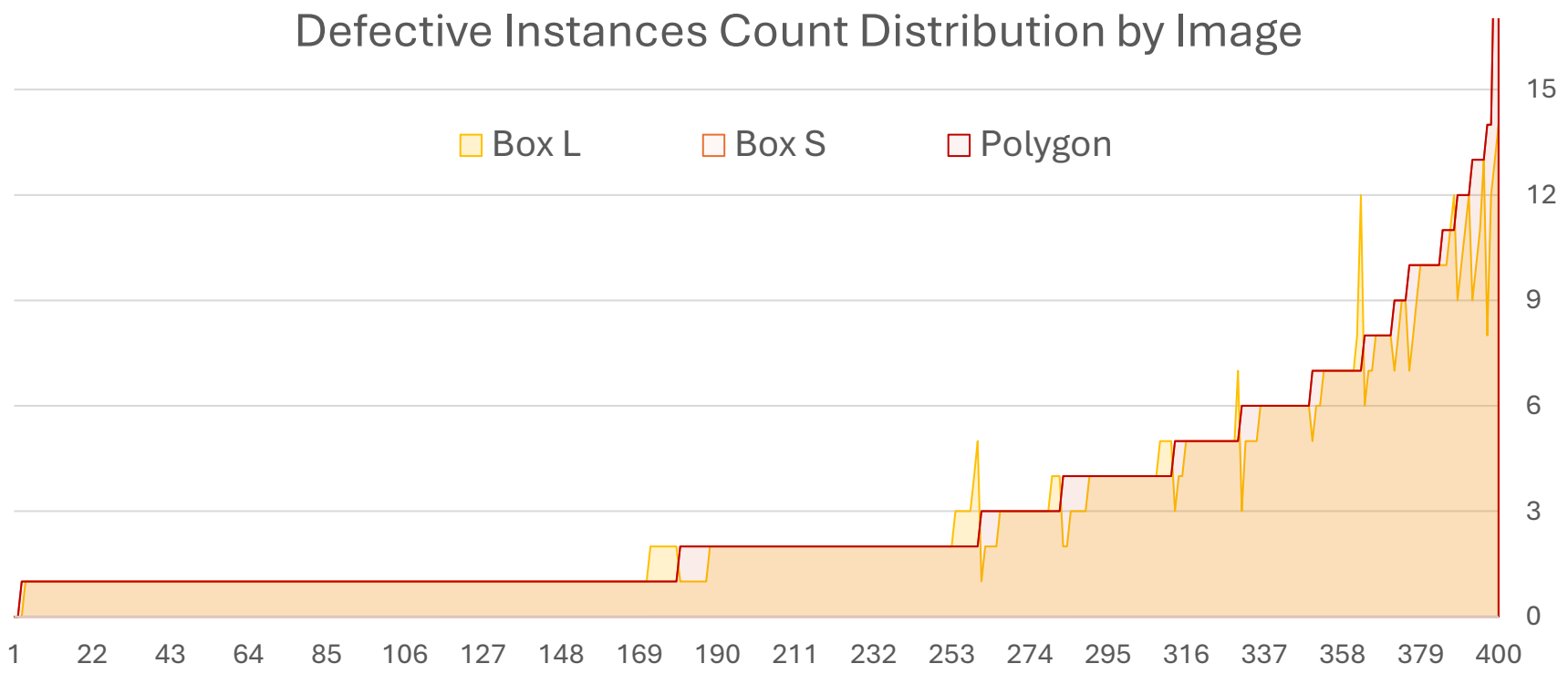






Spatial Granularity

By Image

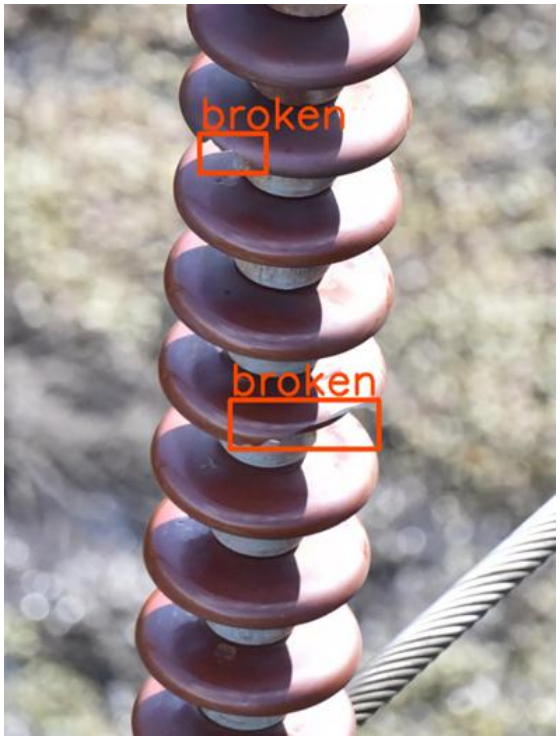


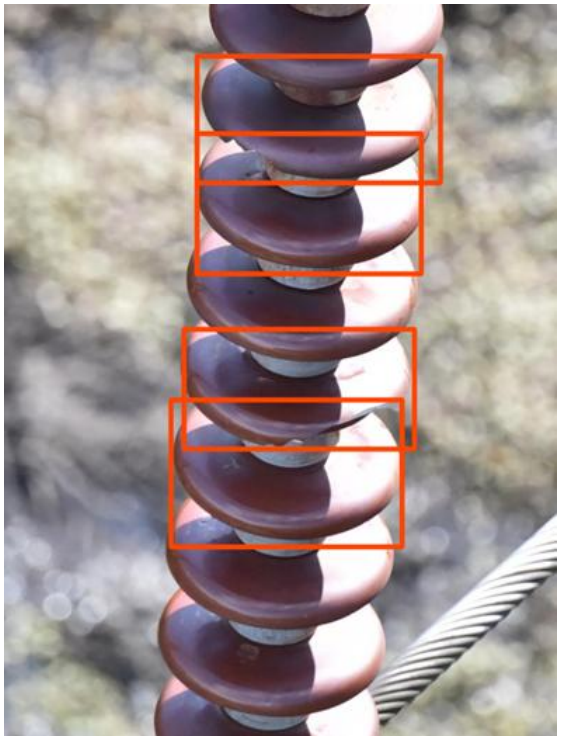
Spatial Granularity

$Box_{LARGE} \geq Box_{SMALL}$

Polygon	Box_{SMALL}	Box_{LARGE}

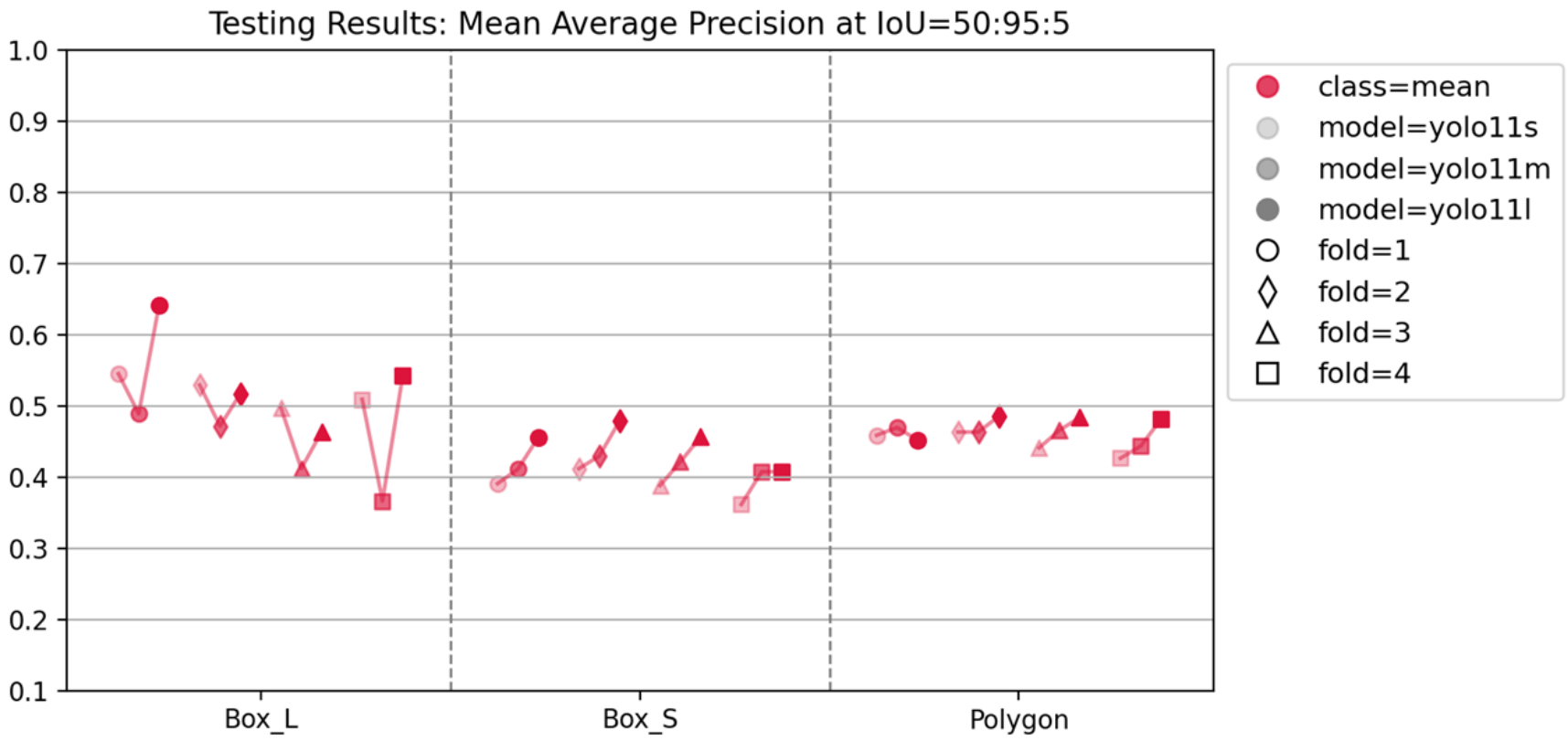






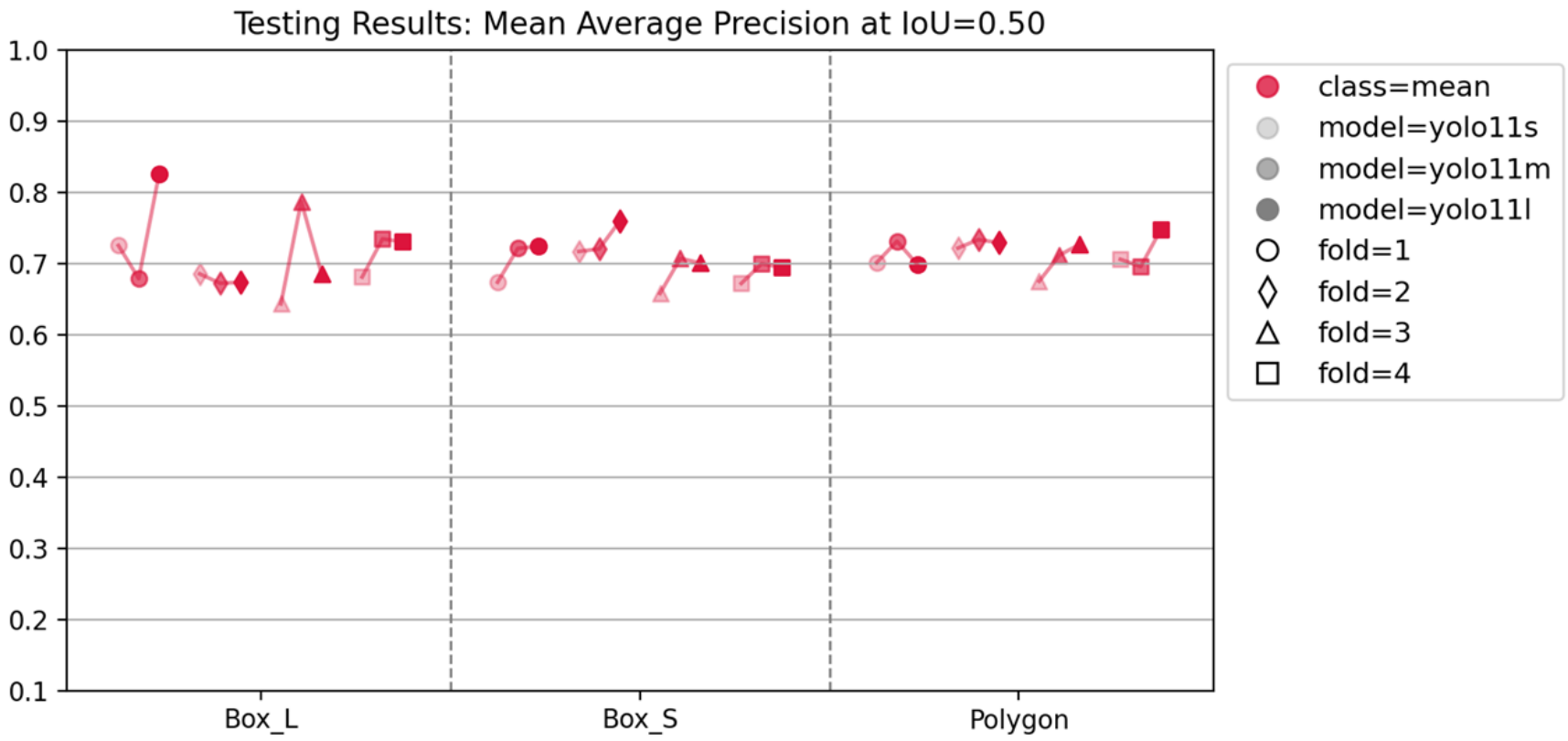
Spatial Granularity

mAP Impact



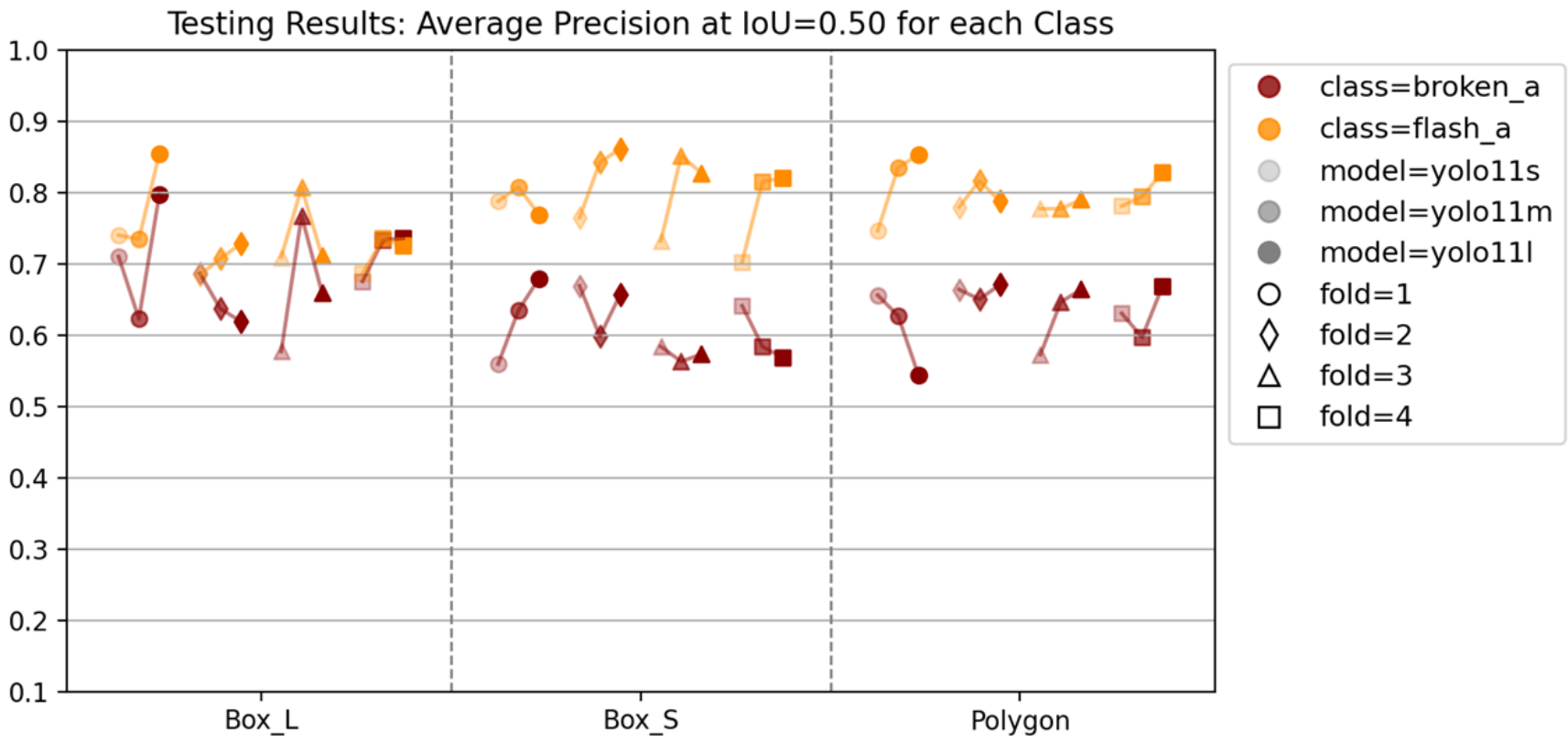
Spatial Granularity

mAP50 Impact



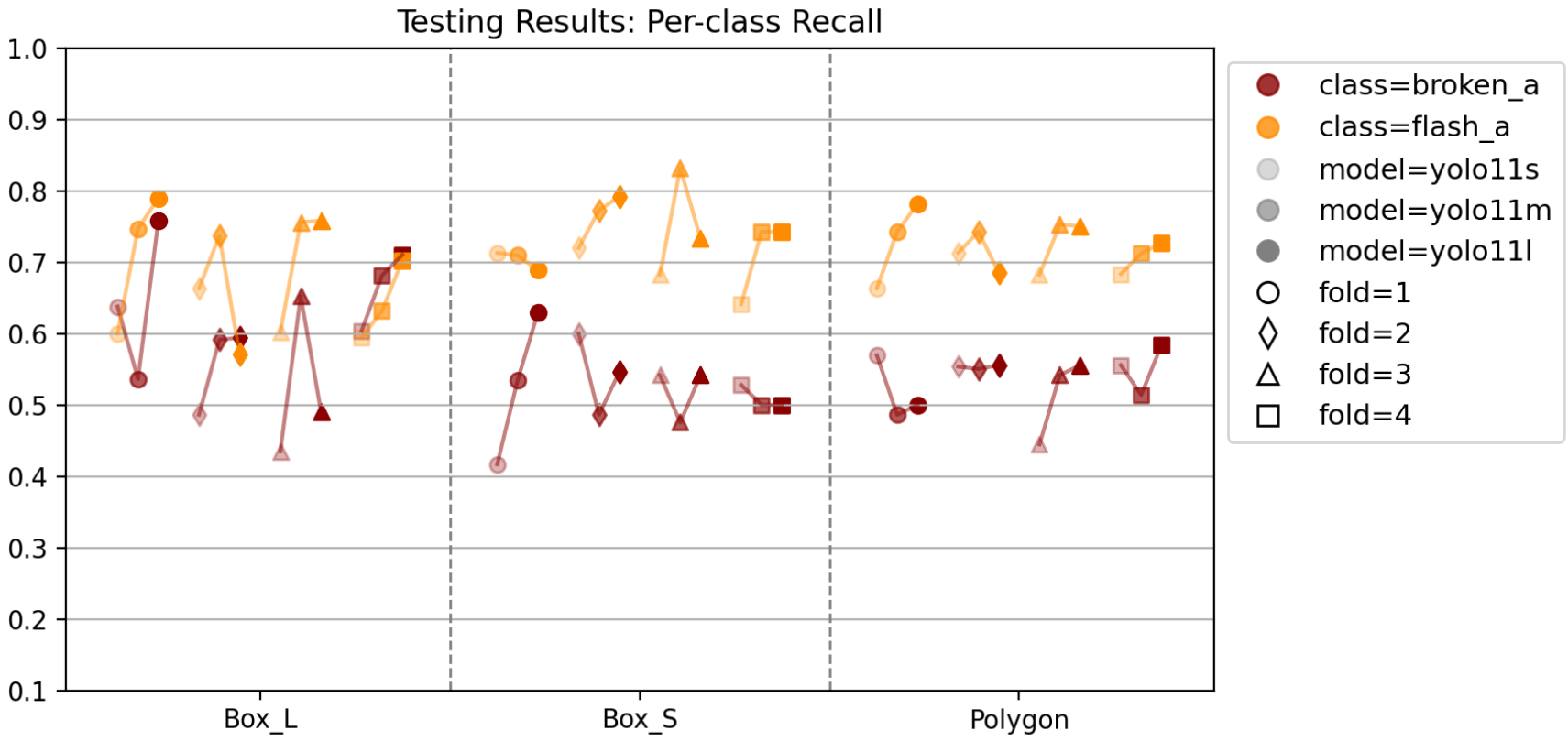
Spatial Granularity

AP50 Impact



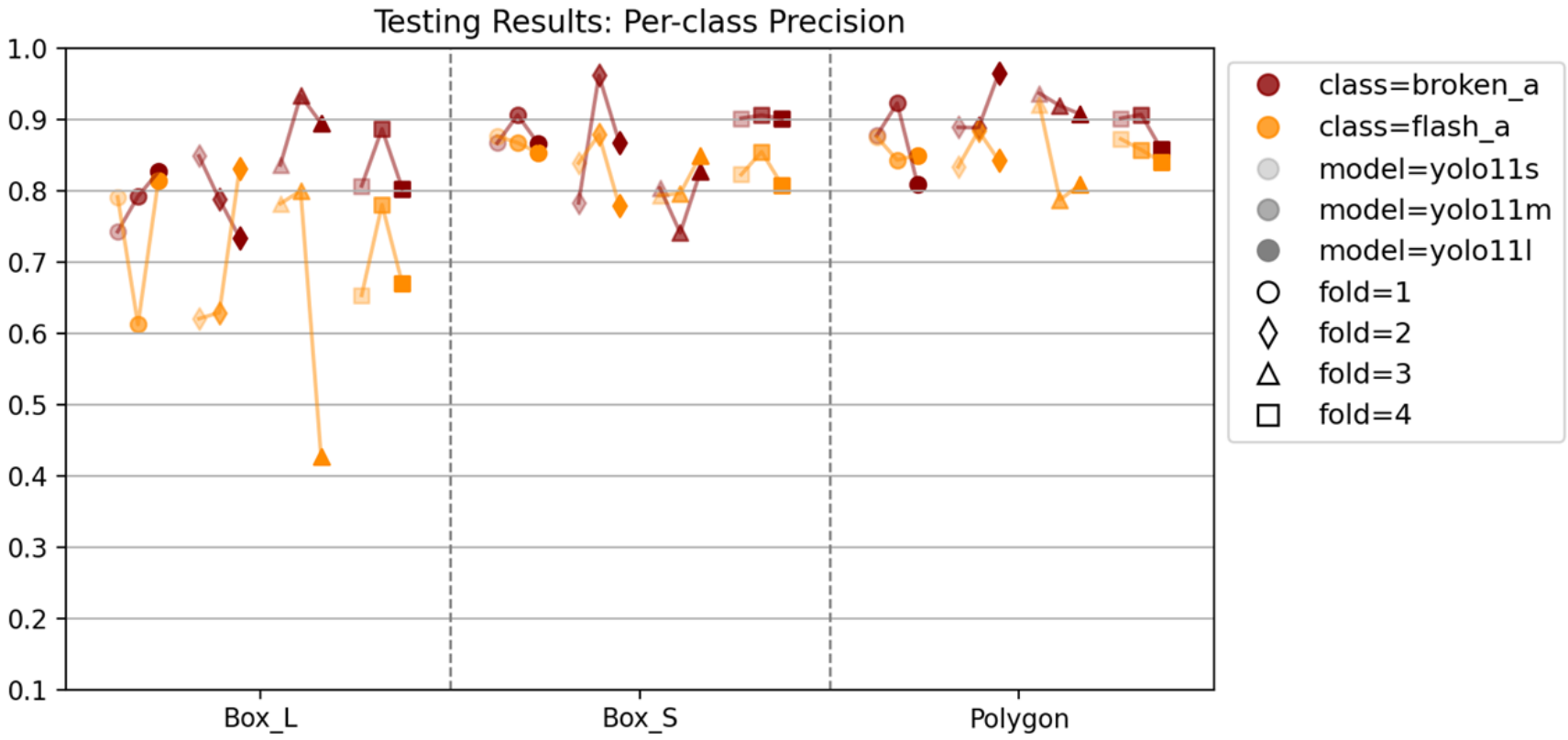
Spatial Granularity

Recall Impact



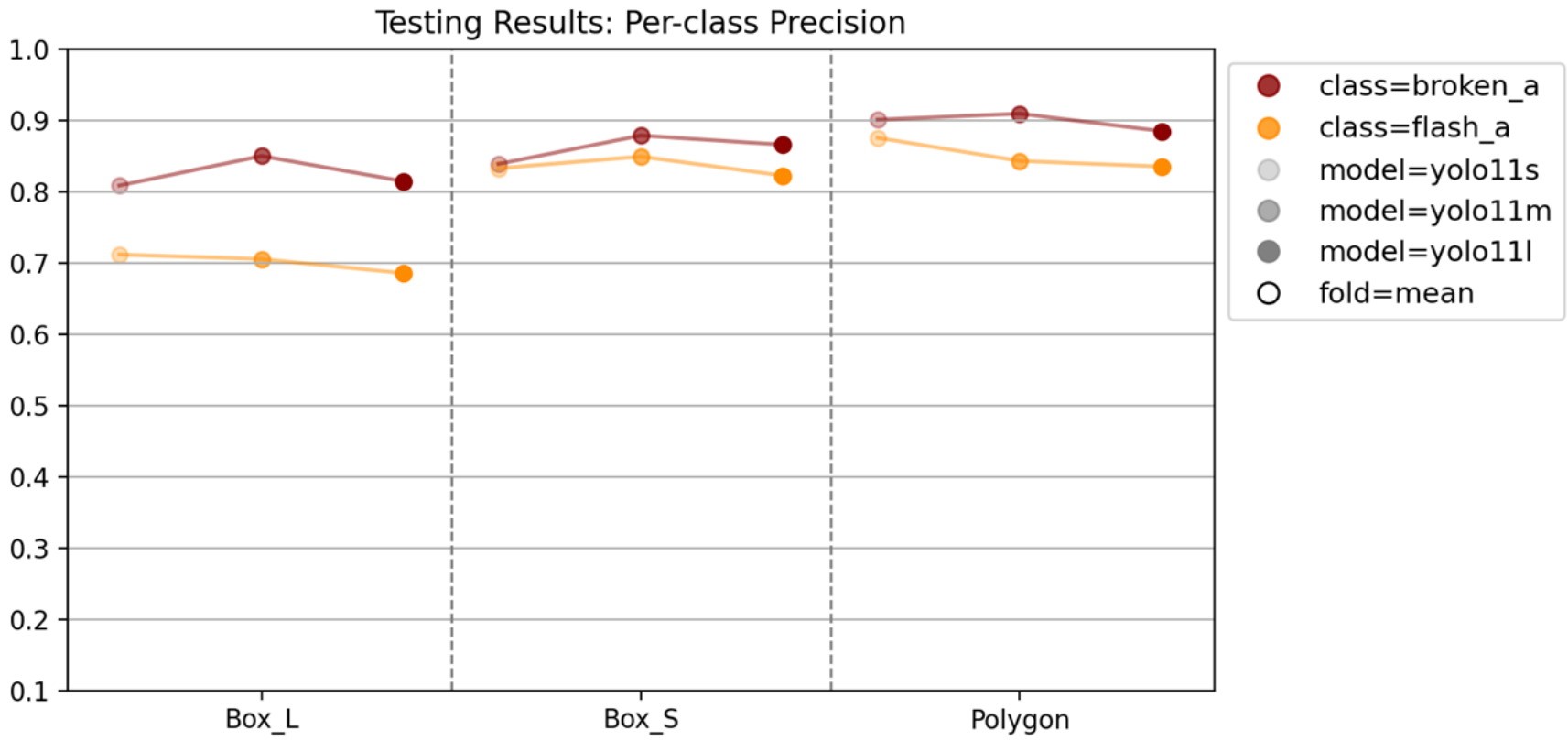
Spatial Granularity

Precision Impact



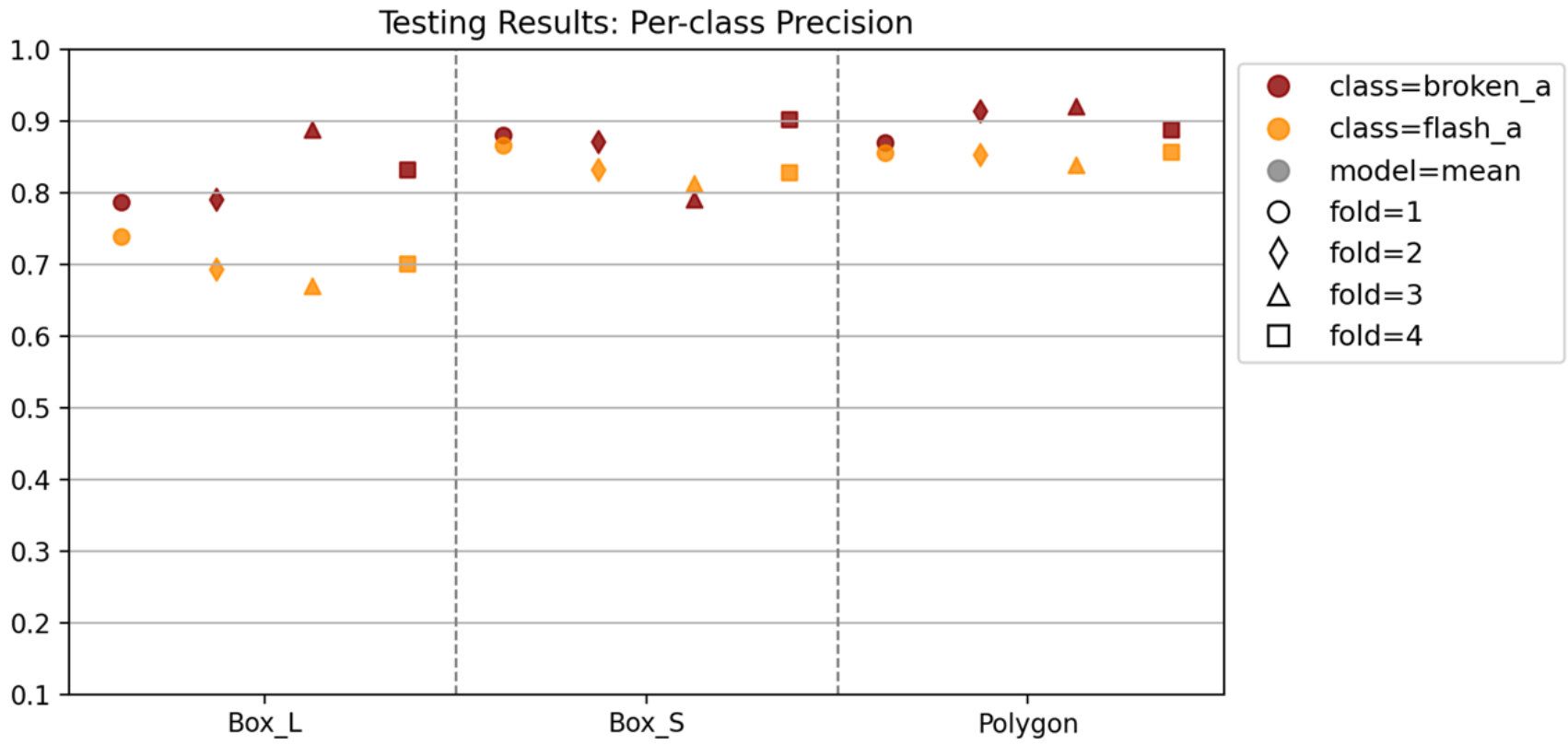
Spatial Granularity

Model Impact



Spatial Granularity

Fold Impact



Spatial Granularity

Key Takeaway

Improving Tightness Favors Precision

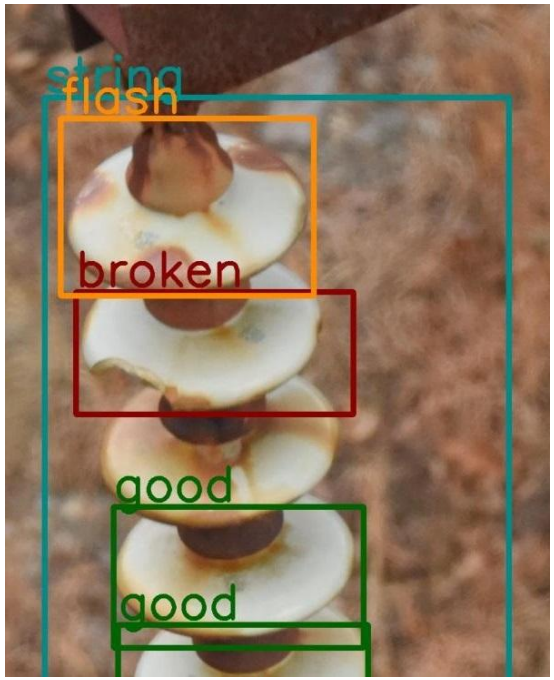
A high precision value indicates that the model is good at correctly identifying existing objects while avoiding false detections on background regions.

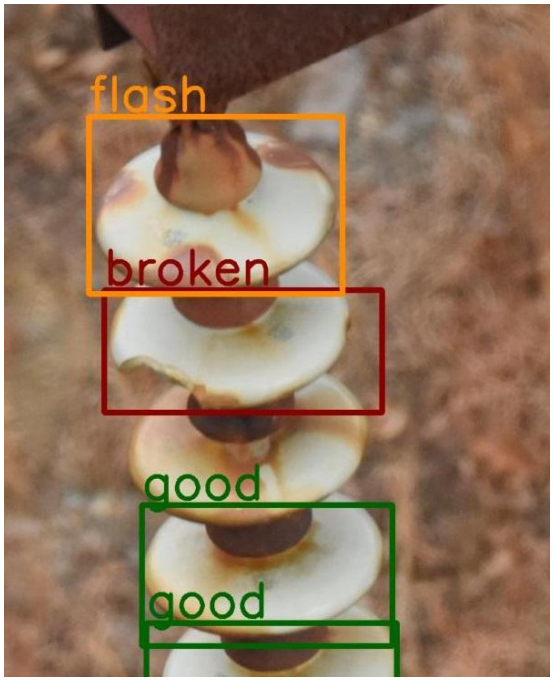
Categorical Granularity

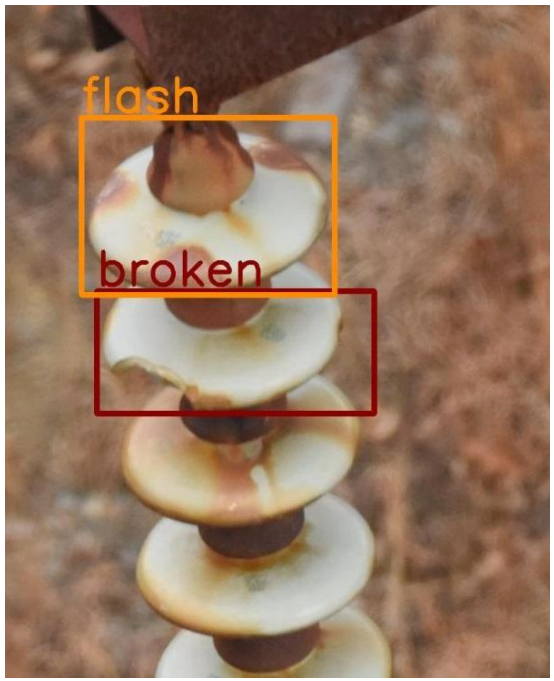
Categorical Granularity

Number of Classes

4	3	2

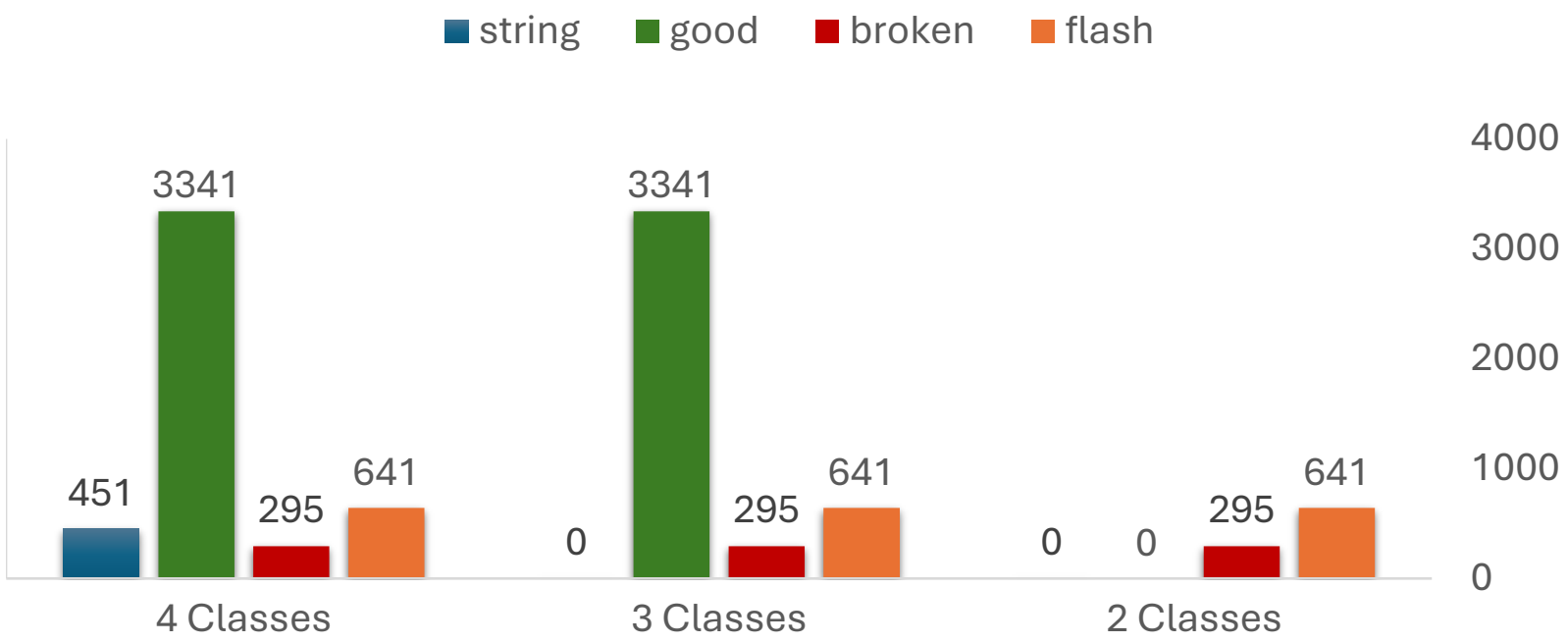






Categorical Granularity

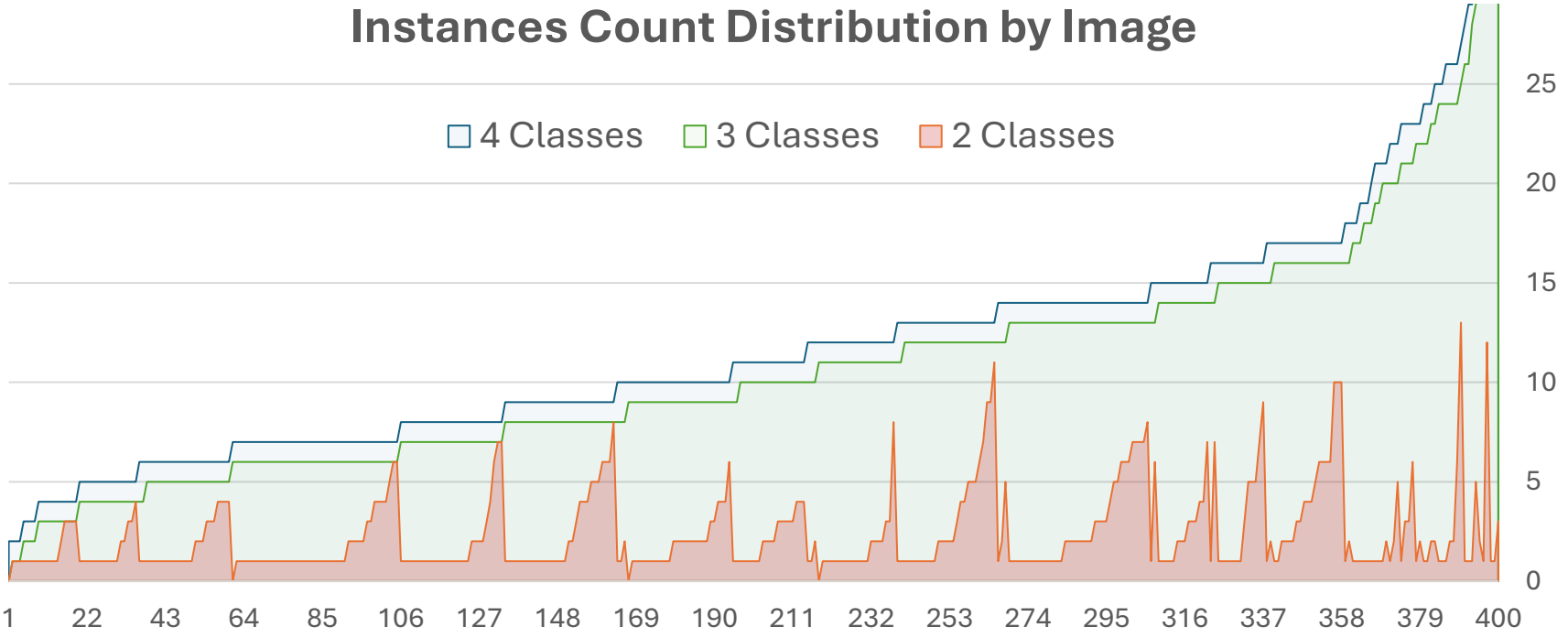
Object Count



D. Lewis and P. Kulkarni, “IDID,” Aug. 2021. [Online]. Available: https://ieee-dataport.org/competitions/insulator-defect-detection

Categorical Granularity

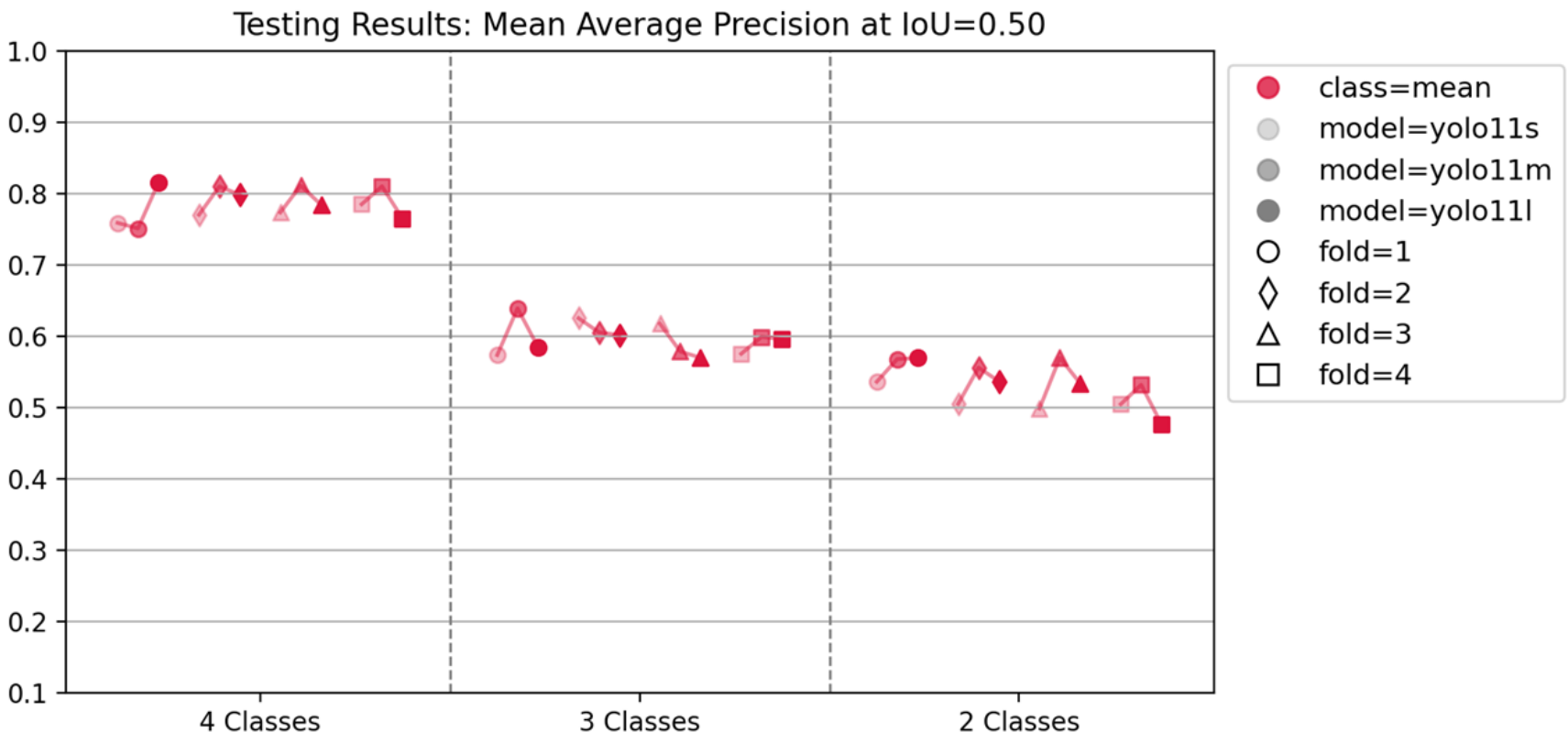
By Image



D. Lewis and P. Kulkarni, “IDID,” Aug. 2021. [Online]. Available: https://ieee-dataport.org/competitions/insulator-defect-detection

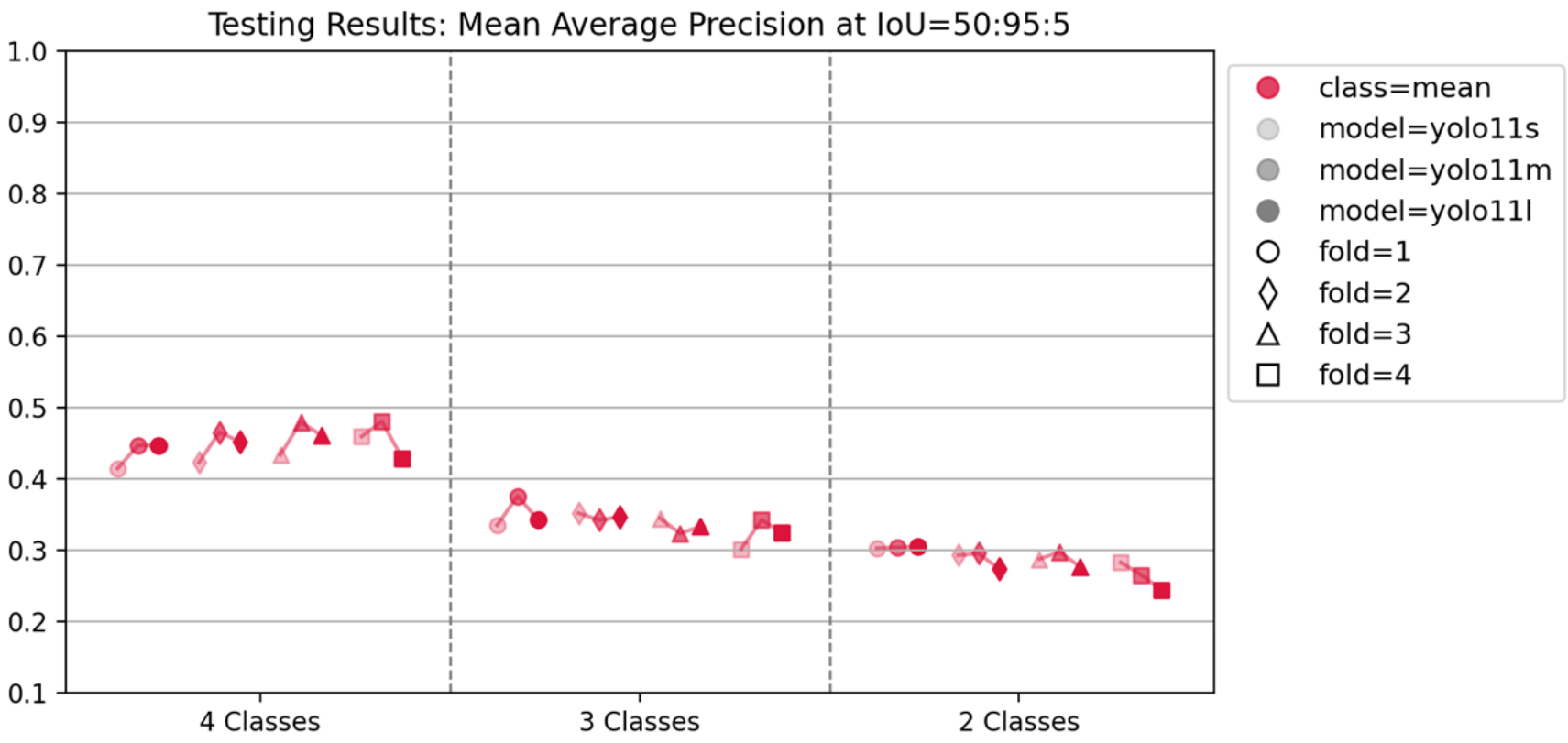
Categorical Granularity

mAP50 Impact



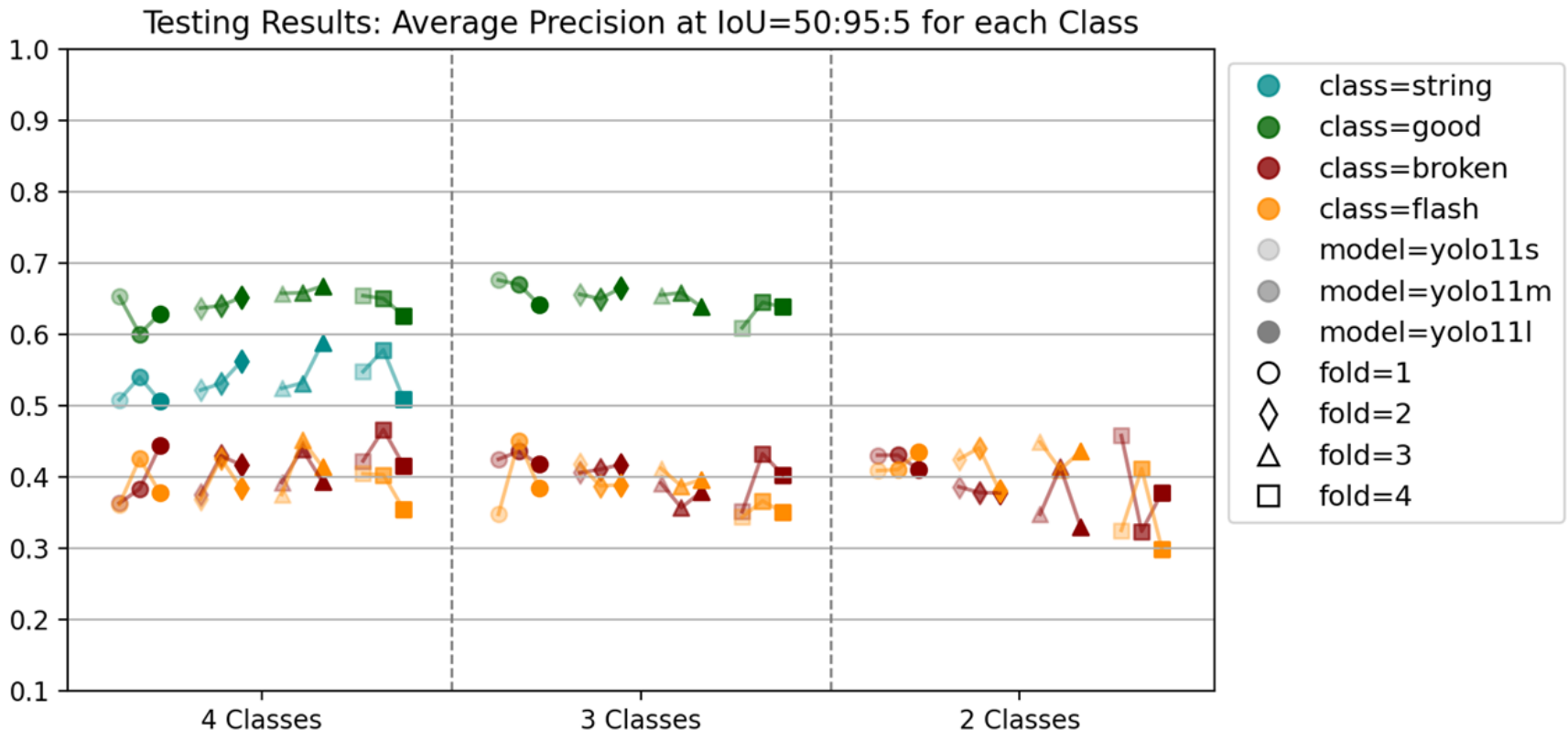
Categorical Granularity

mAP Impact



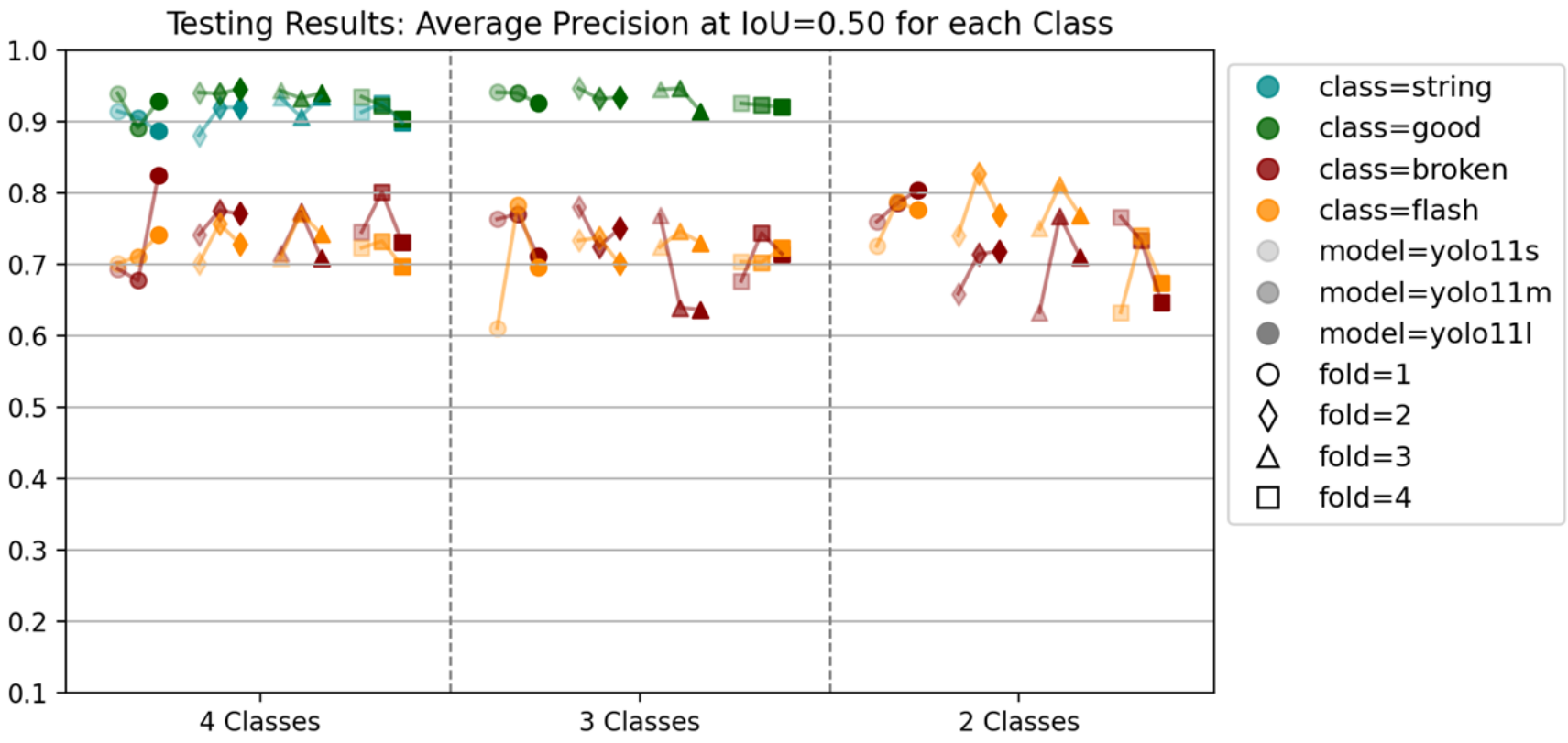
Categorical Granularity

AP Impact



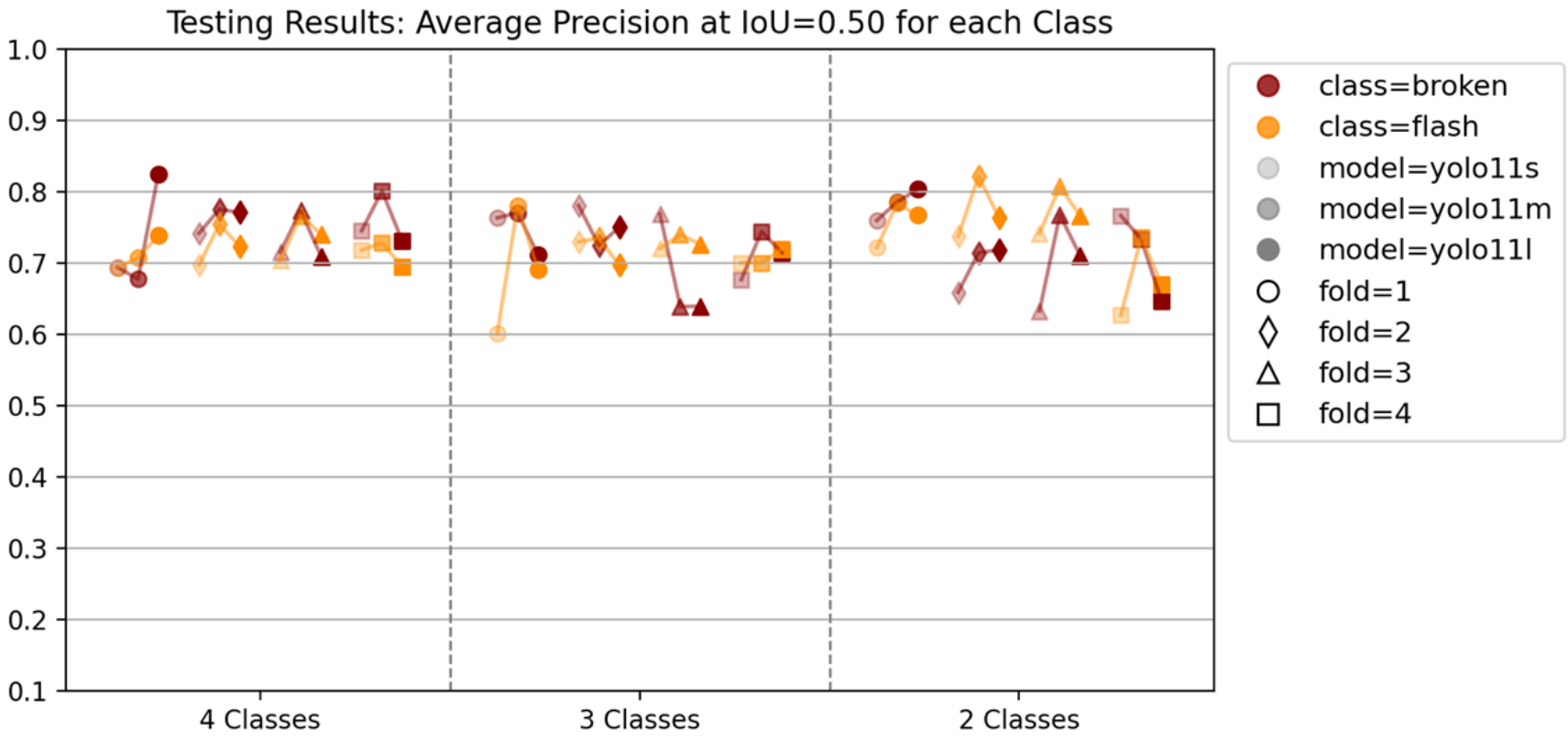
Categorical Granularity

AP50 Impact



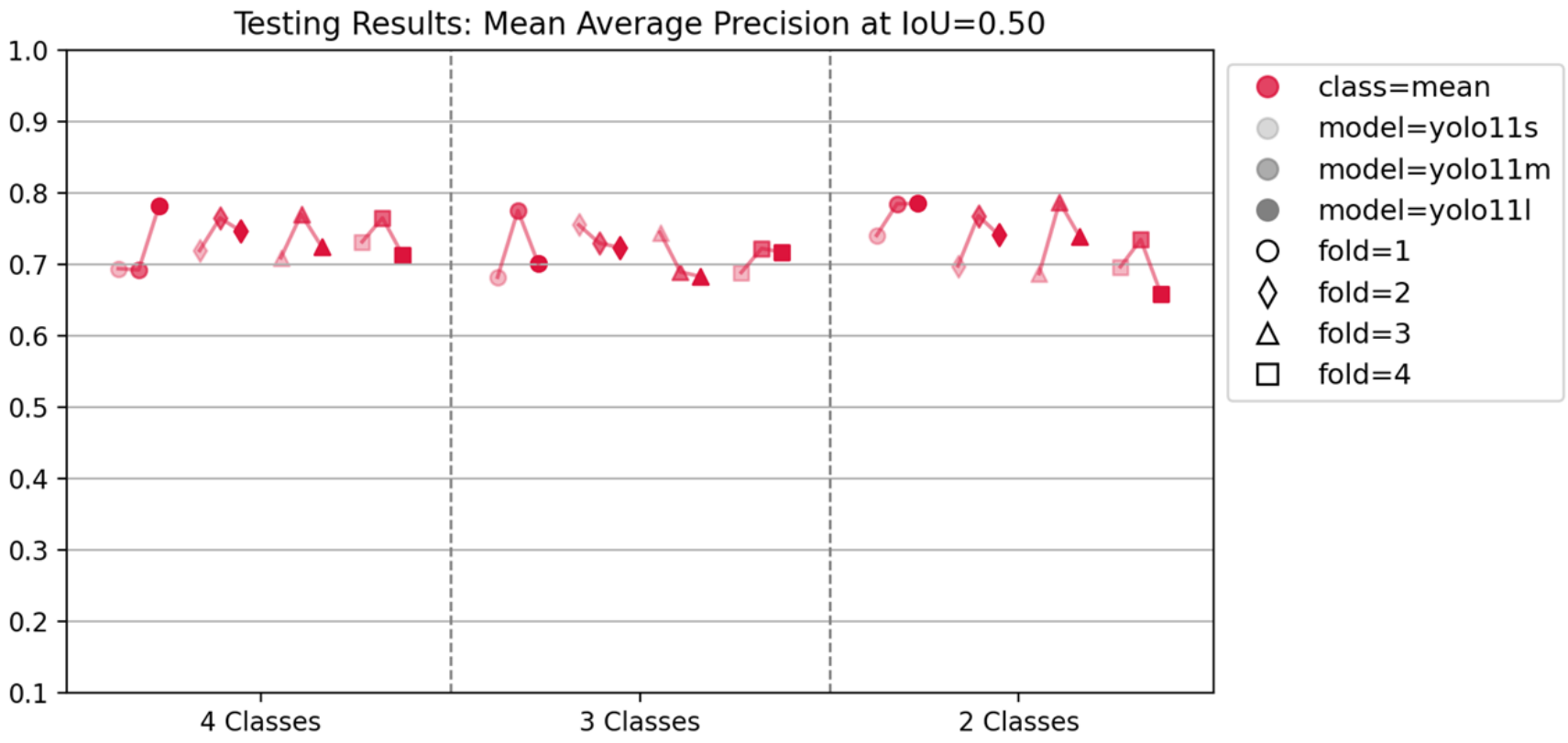
Categorical Granularity

AP50 Impact



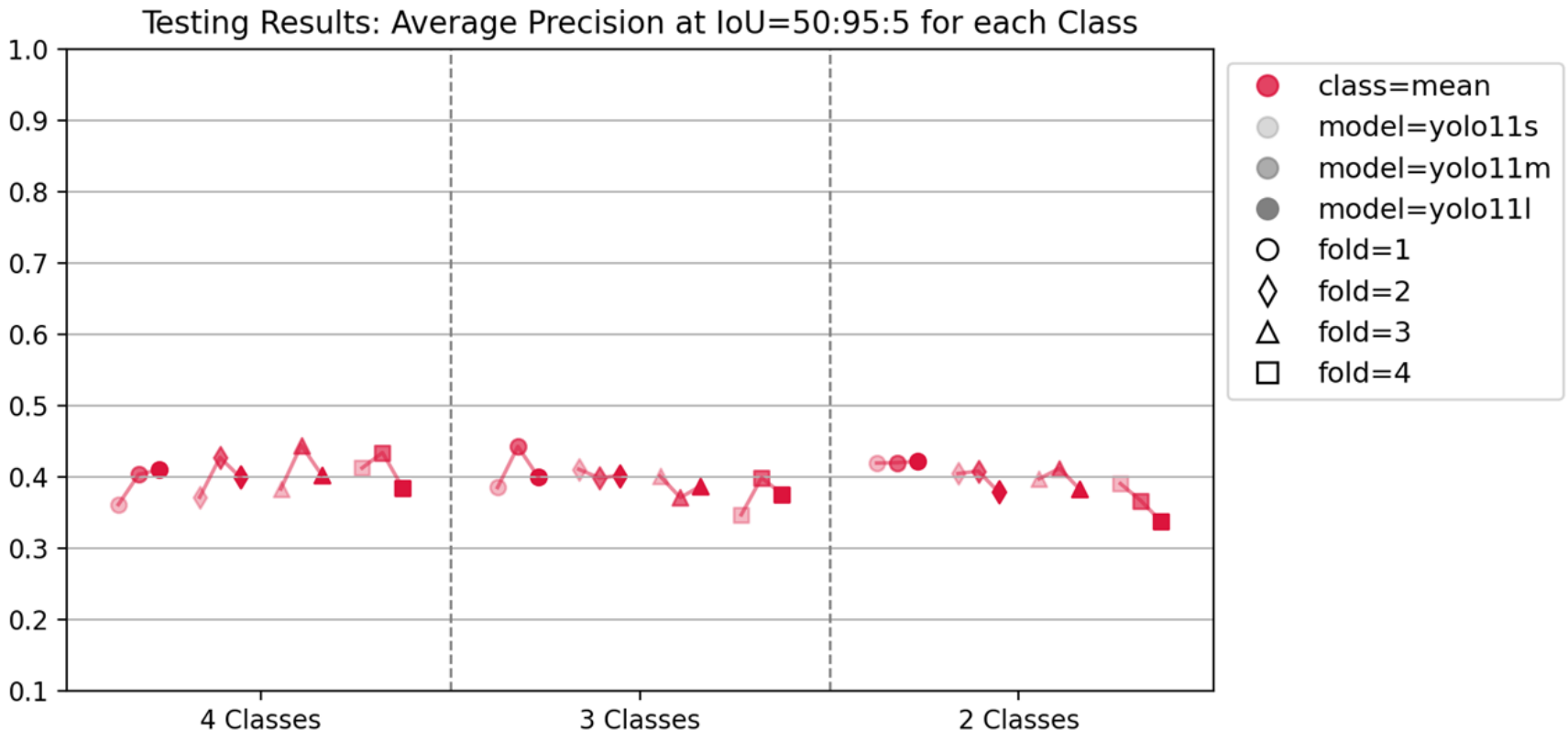
Categorical Granularity

mAP50 Impact



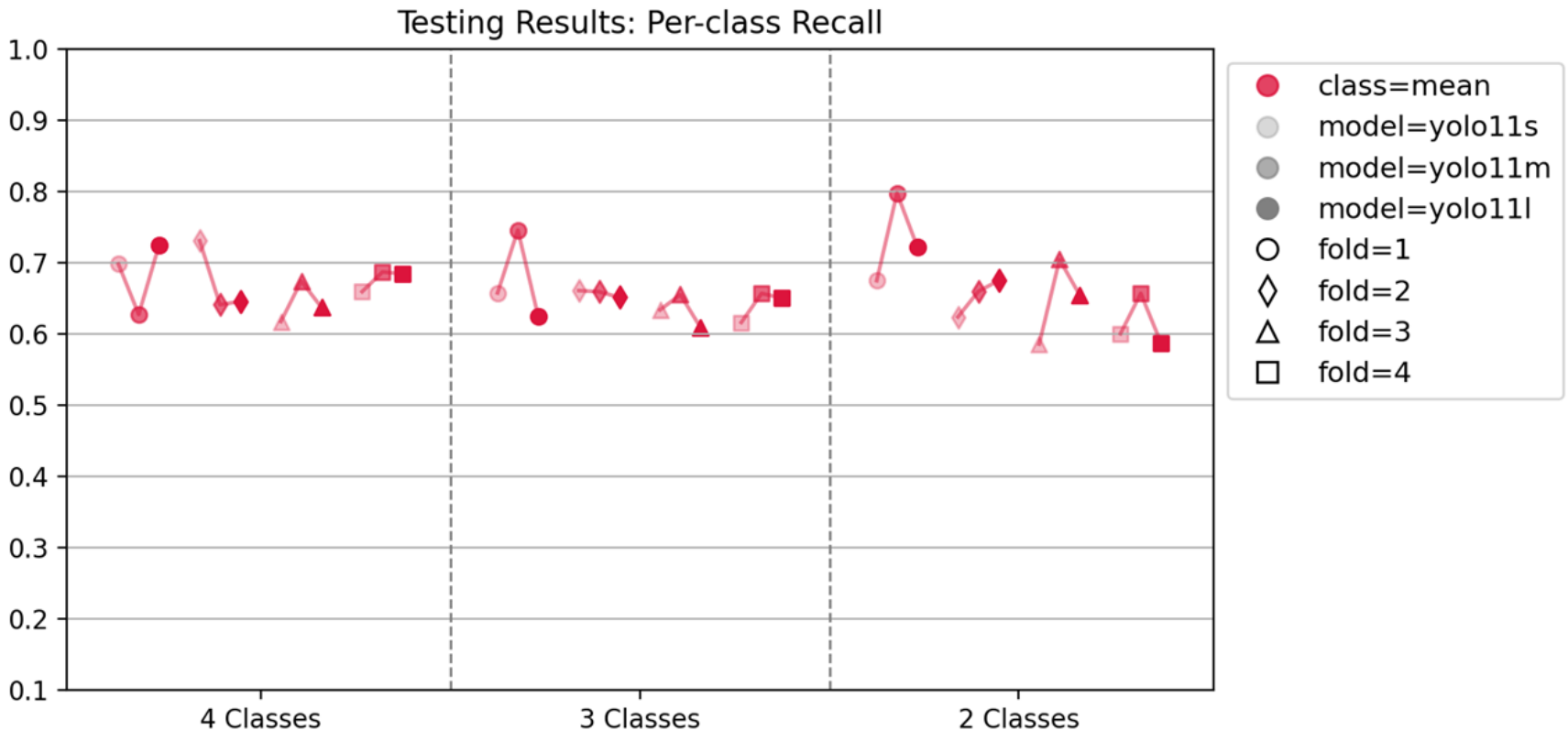
Categorical Granularity

mAP Impact



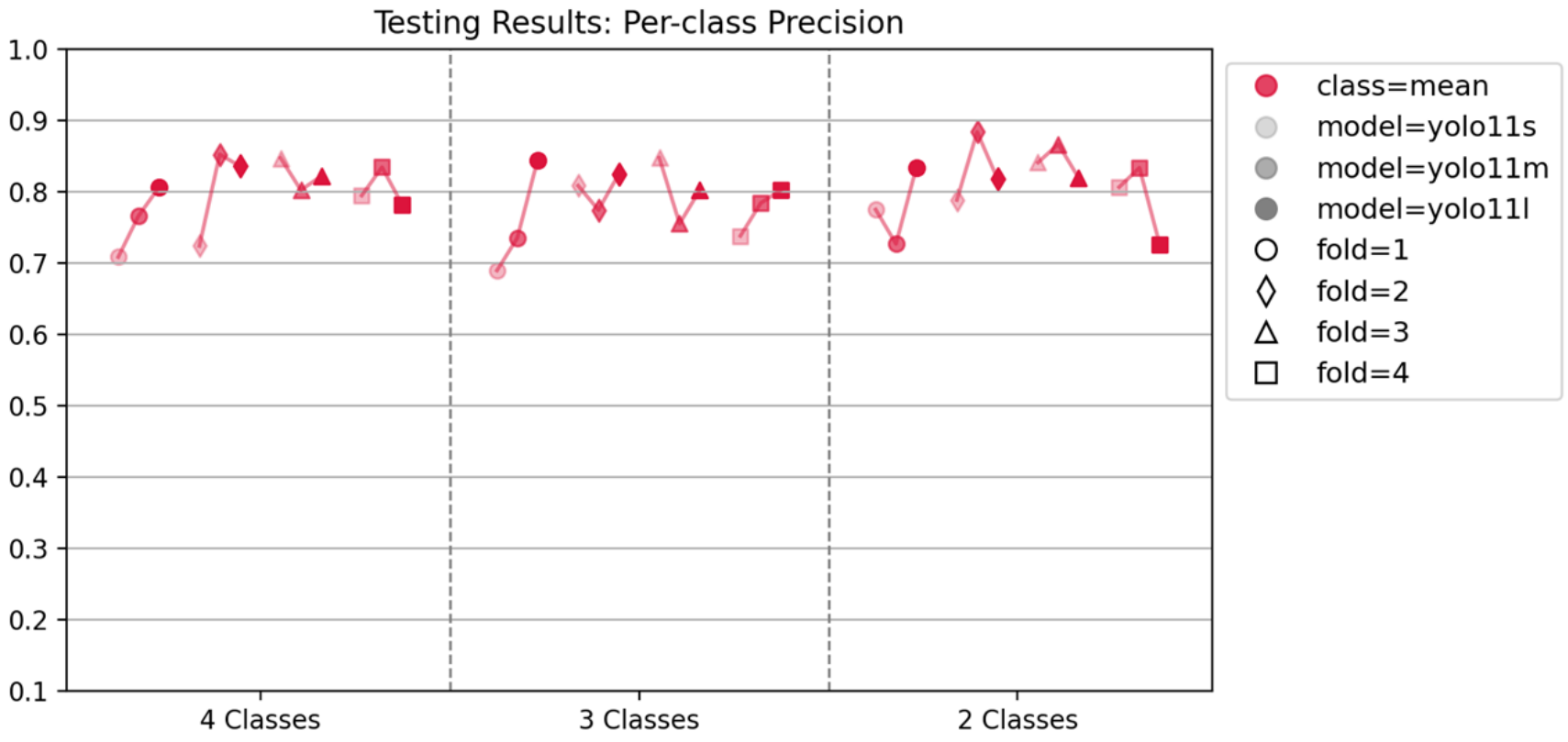
Categorical Granularity

Recall Impact



Categorical Granularity

Precision Impact



Categorical Granularity

Key Takeaway

Irrelevant Labelling Seems Unnecessary Labor

When the goal is anomaly detection, adding non-defective samples may not provide useful information for discriminating defective classes.

Conclusion

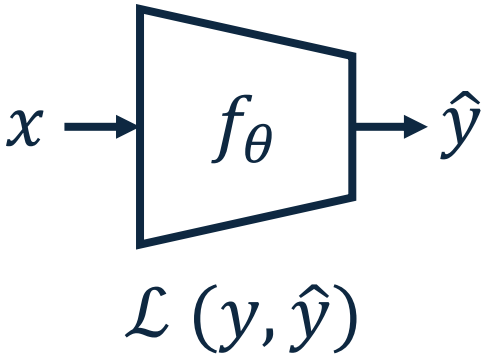
- Summary
- Contributions
- Challenges
- Future Work

SUMMARY



SUMMARY

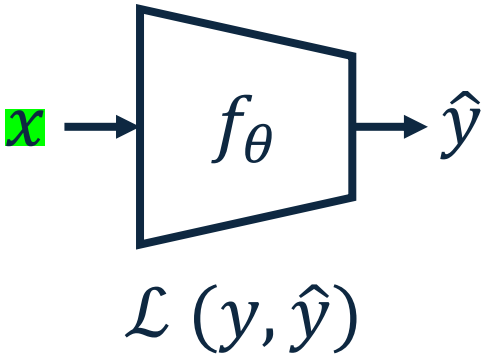
- Performance Impact





SUMMARY

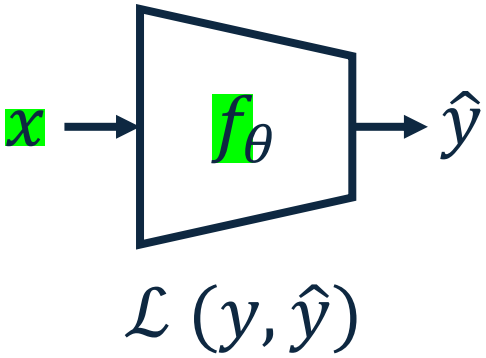
- Performance Impact
  - Input x – Cross Validation





SUMMARY

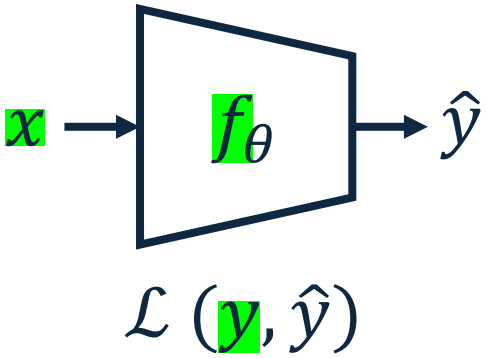
- Performance Impact
  - Input x – Cross Validation
  - Model f – ANN Size





SUMMARY

- Performance Impact
  - Input x – Cross Validation
  - Model f – ANN Size
  - Target y – Detail Level





SUMMARY

- Performance Impact
  - Input x – Cross Validation
  - Model f – ANN Size
  - Target y – Detail Level
- Annotations Dimensions



SUMMARY

- Performance Impact
  - Input x – Cross Validation
  - Model f – ANN Size
  - Target y – Detail Level
- Annotations Dimensions
  - Consistency – Low Noise



SUMMARY

- Performance Impact
  - Input x – Cross Validation
  - Model f – ANN Size
  - Target y – Detail Level
- Annotations Dimensions
  - Consistency – Low Noise
  - Spatial – Tightness



SUMMARY

- Performance Impact
  - Input x – Cross Validation
  - Model f – ANN Size
  - Target y – Detail Level
- Annotations Dimensions
  - Consistency – Low Noise
  - Spatial – Tightness
  - Categorical – Nº Classes



Key Contributions



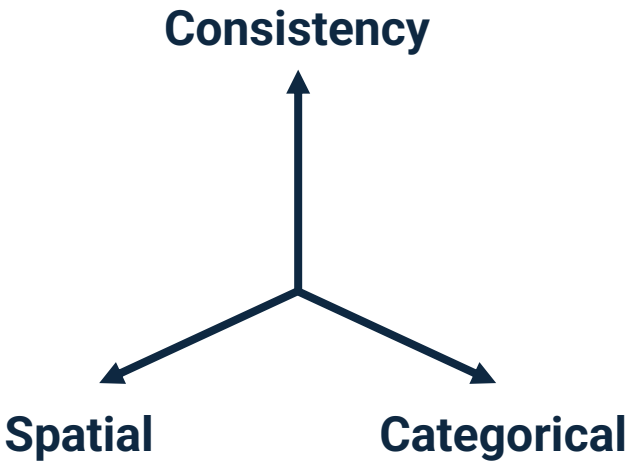
Key Contributions

- Conceptual Definition



Key Contributions

- Conceptual Definition





Key Contributions

- Conceptual Definition
- Compressive Benchmark



Key Contributions

- Conceptual Definition
- Compressive Benchmark
  - 108 Trained Models



Key Contributions

- Conceptual Definition
- Compressive Benchmark
  - 108 Trained Models
  - 4 Dimension Evaluation



Key Contributions

- Conceptual Definition
- Compressive Benchmark
  - 108 Trained Models
  - 4 Dimension Evaluation

Horizontal Axis	XX offset
Color	●●●●
Shade	●●●●
Marker	○ ◊ Δ □



Key Contributions

- Conceptual Definition
- Compressive Benchmark
  - 108 Trained Models
  - 4 Dimension Evaluation
- Takeaways



Key Contributions

- Conceptual Definition
- Compressive Benchmark
  - 108 Trained Models
  - 4 Dimension Evaluation
- Takeaways
  - Consistency – Recall



Key Contributions

- Conceptual Definition
- Compressive Benchmark
  - 108 Trained Models
  - 4 Dimension Evaluation
- Takeaways
  - Consistency – Recall
  - Spatial – Precision

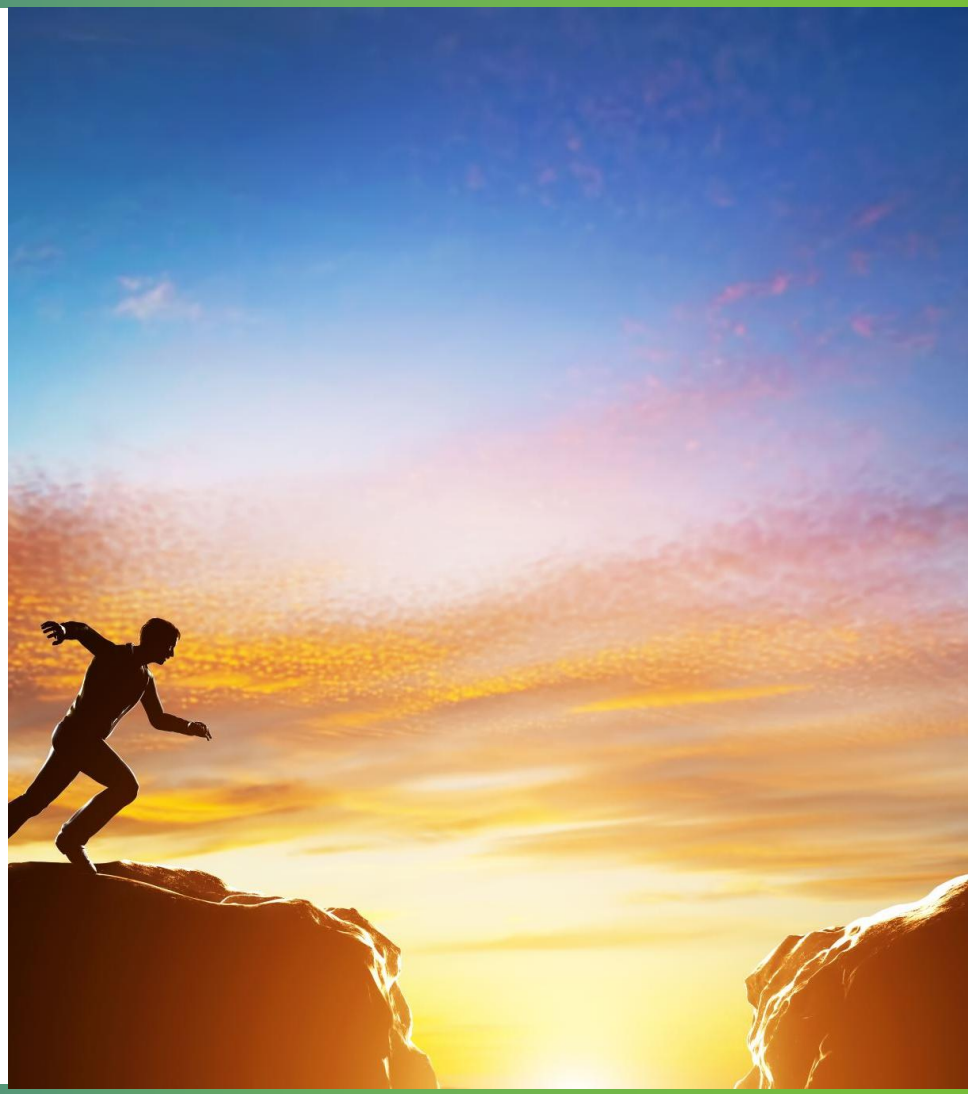


Key Contributions

- Conceptual Definition
- Compressive Benchmark
  - 108 Trained Models
  - 4 Dimension Evaluation
- Takeaways
  - Consistency – Recall
  - Spatial – Precision
  - Categorical – No Impact

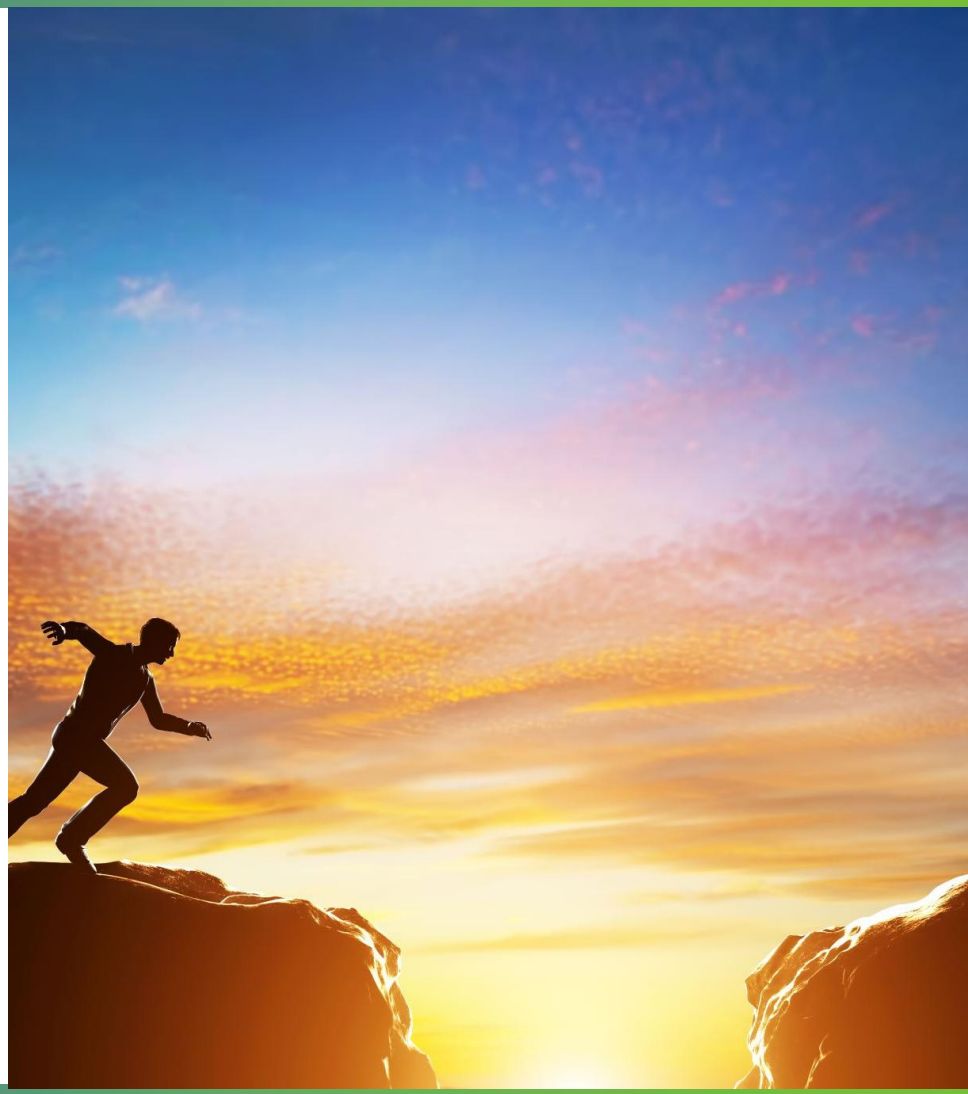


CHALLENGES



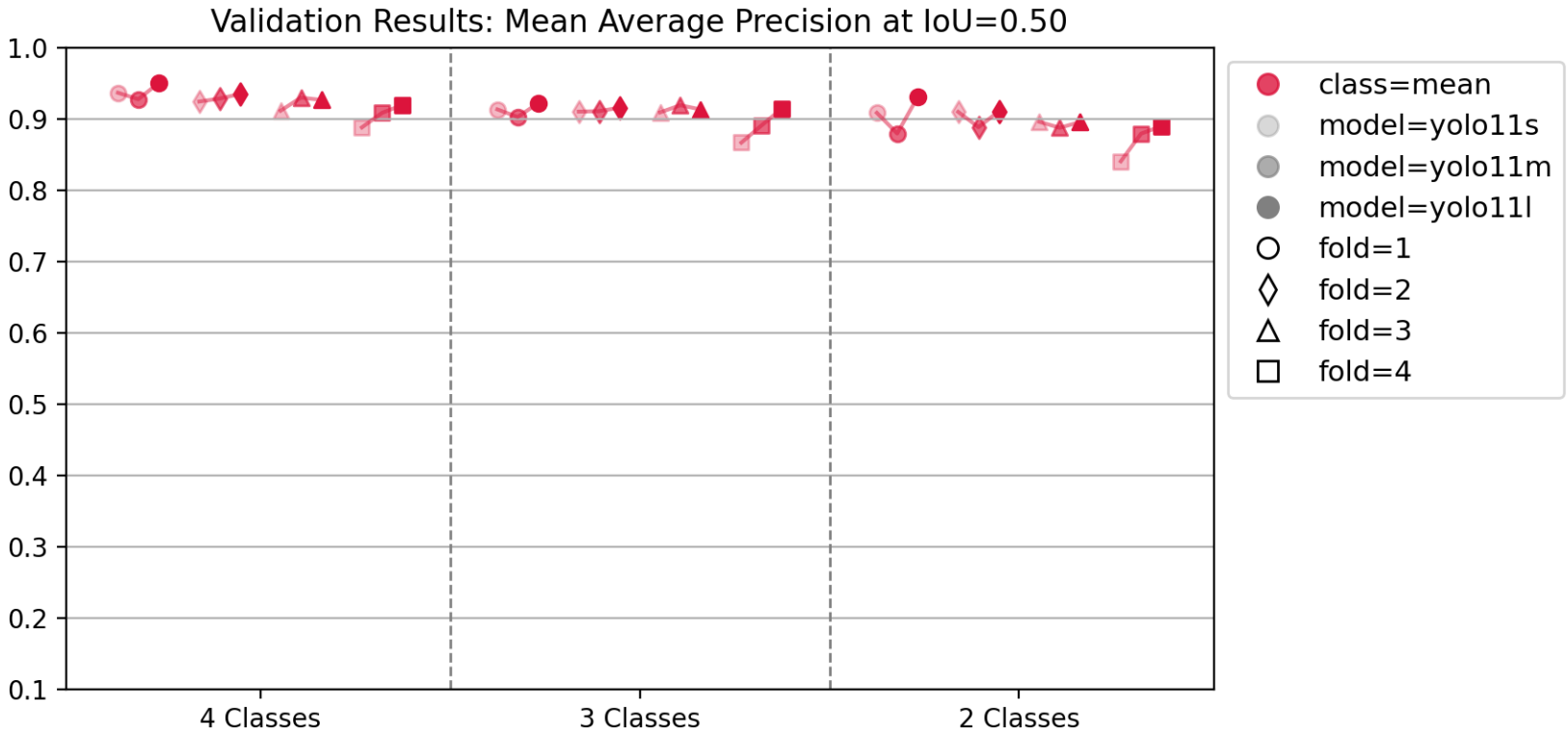
CHALLENGES

- Inflated Performance

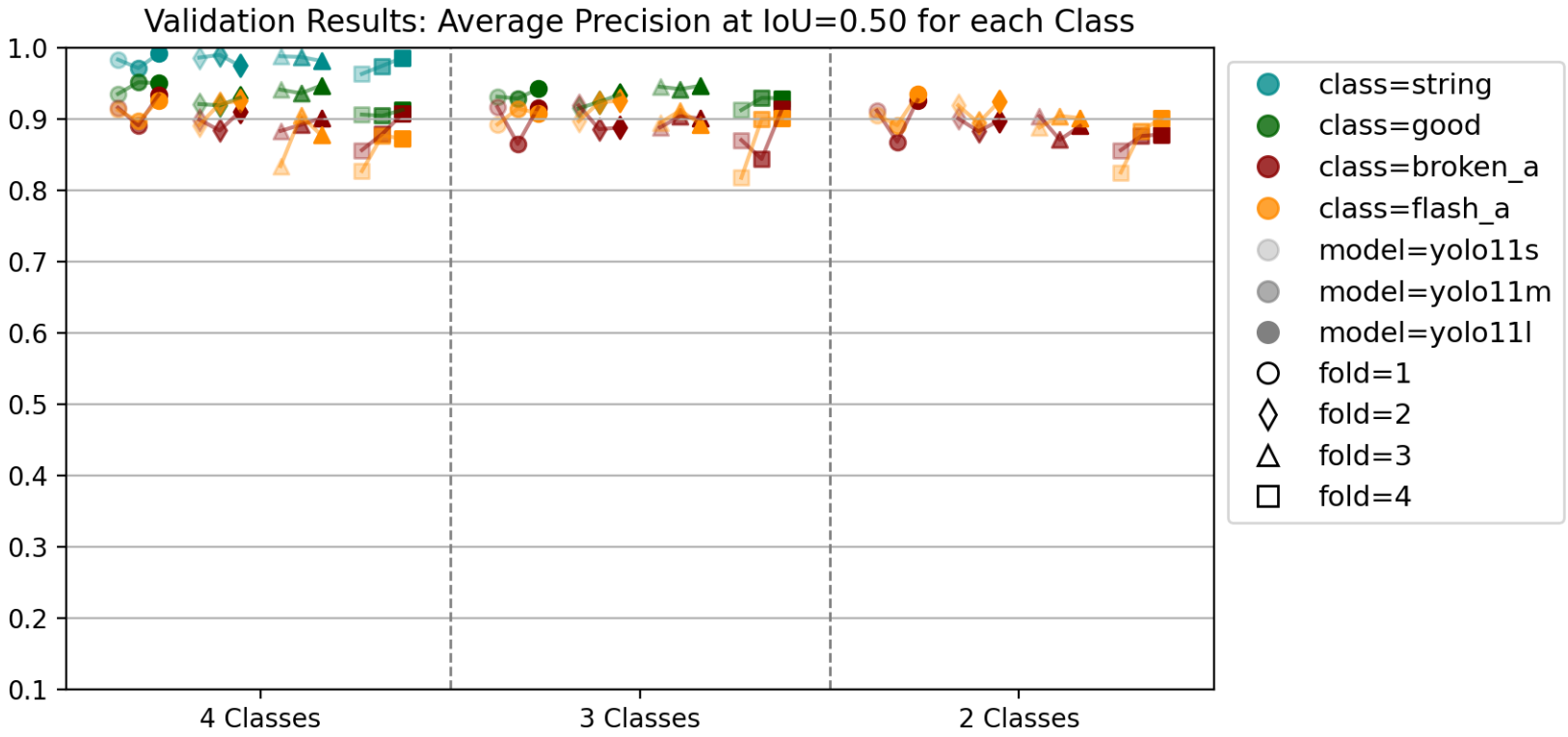


Best Run on Validation Set

94.8% of mAP50

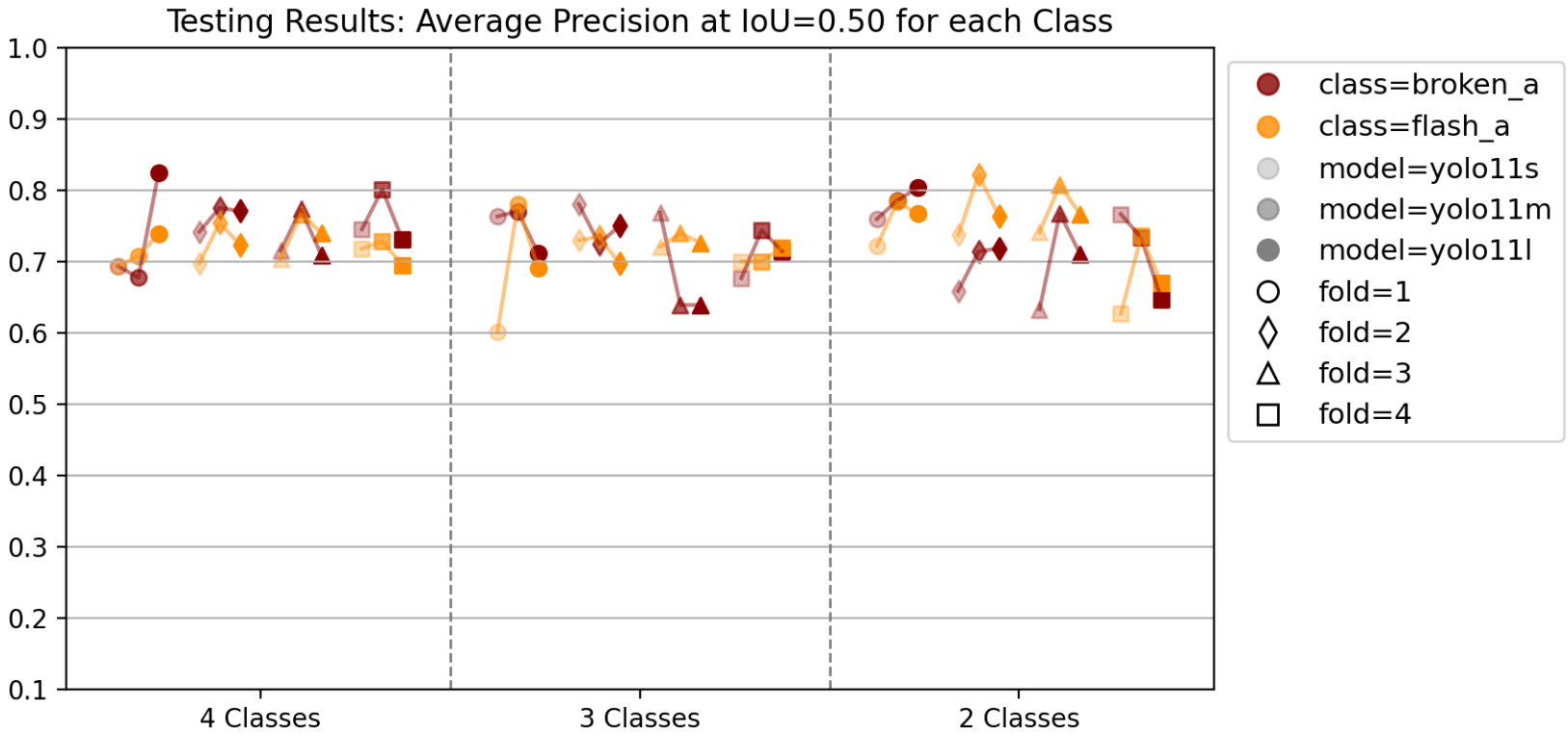


Performance Inflated by Non-Defective Classes



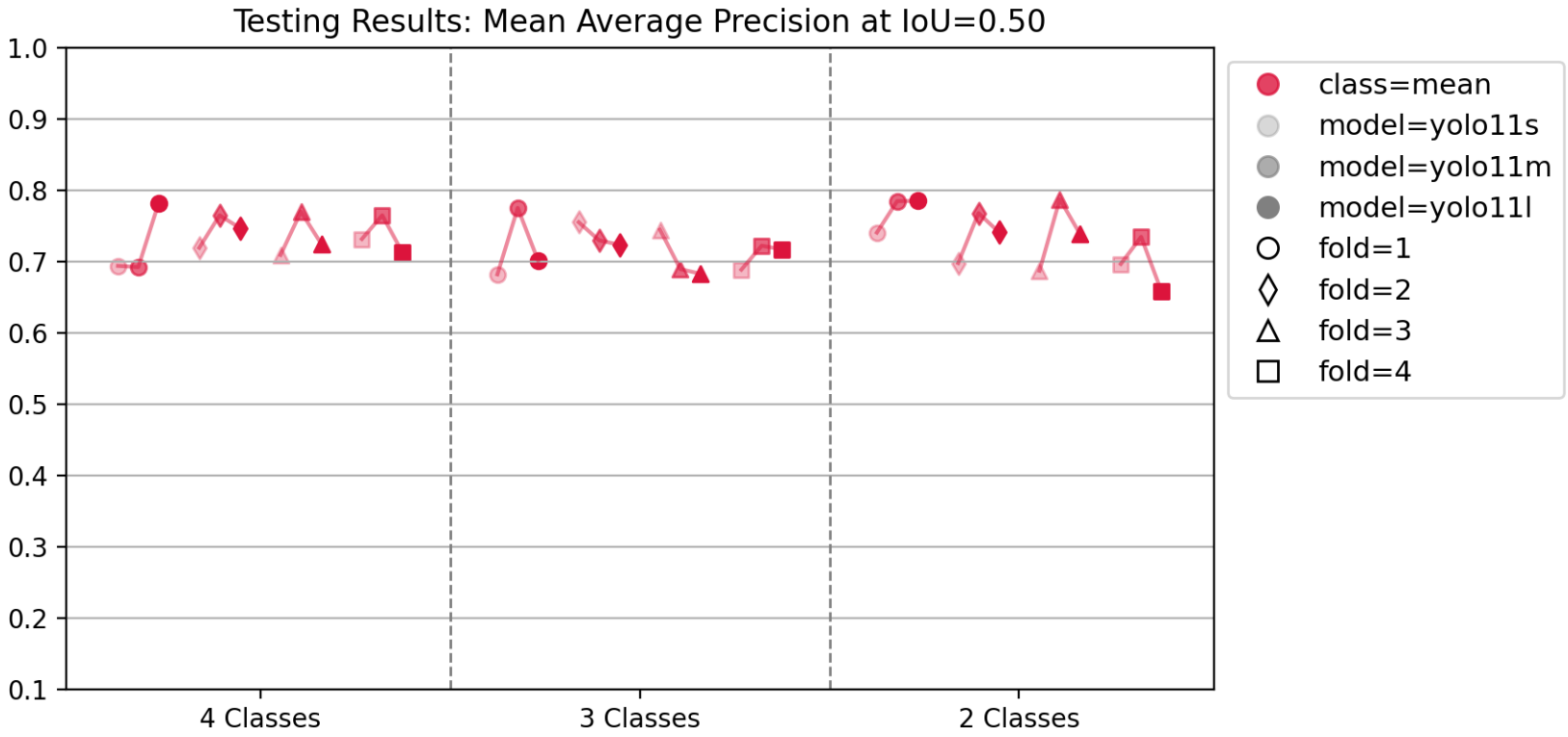
Runing on Test Set

Performance Drop 15%



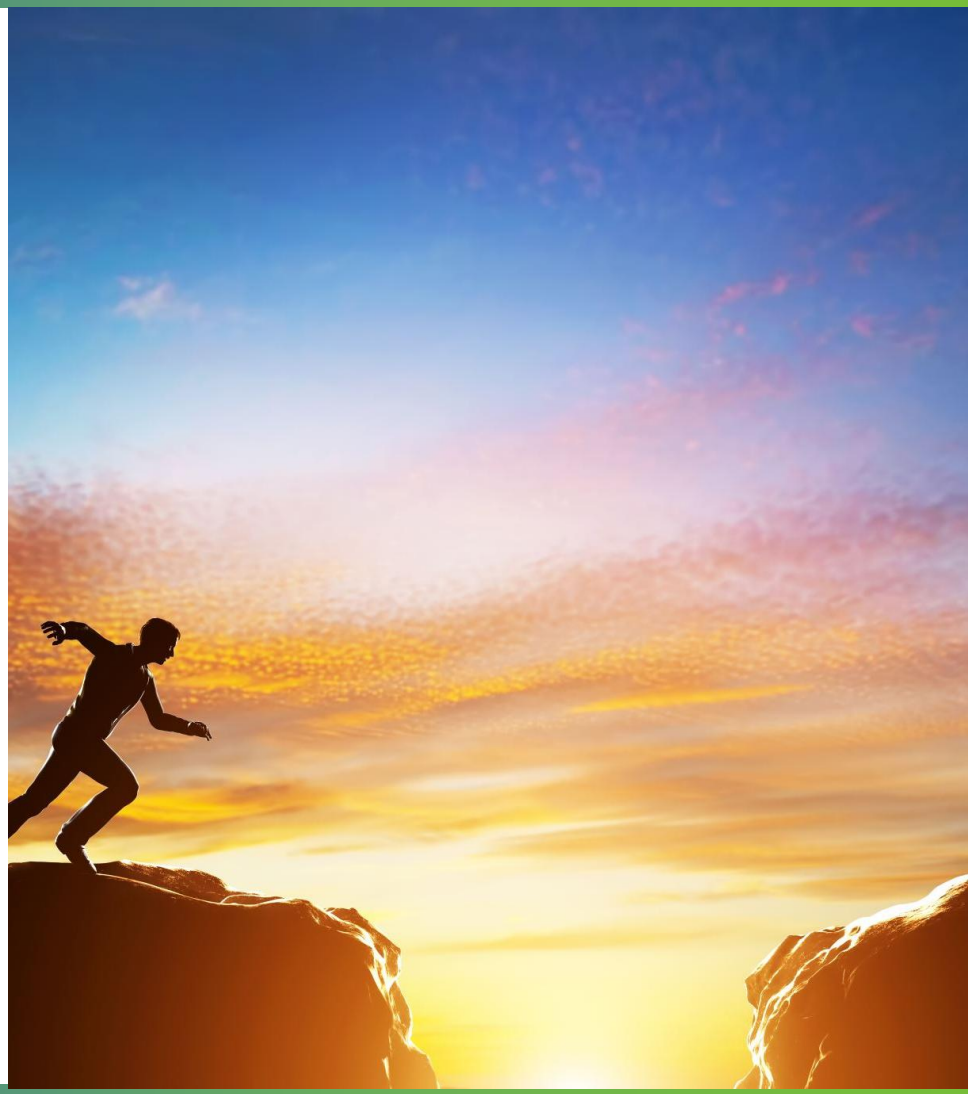
Best Run on Test Set

78.1% of mAP50



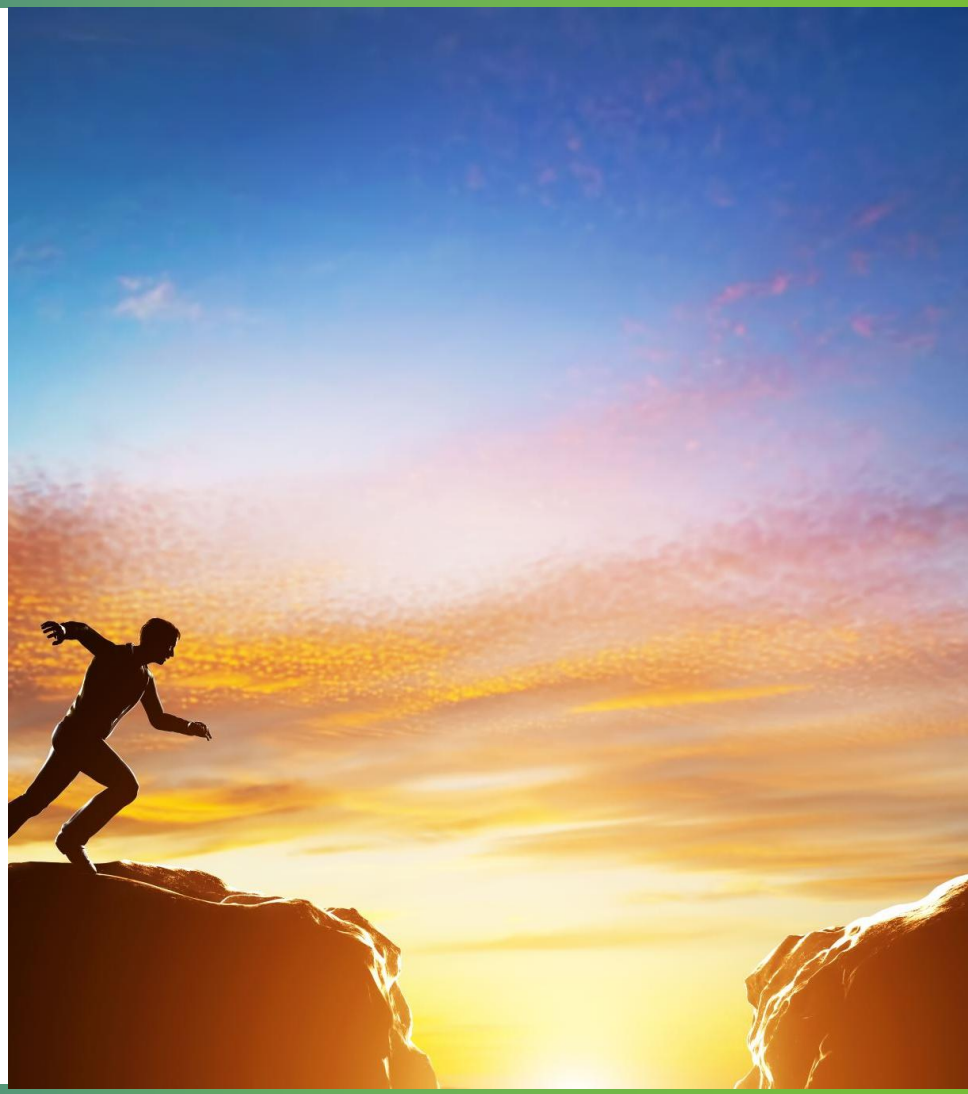
CHALLENGES

- Inflated Performance
  - 94.8% mAP50 Valid – 4 Cls
  - 78.1% mAP50 Test – 2 Cls



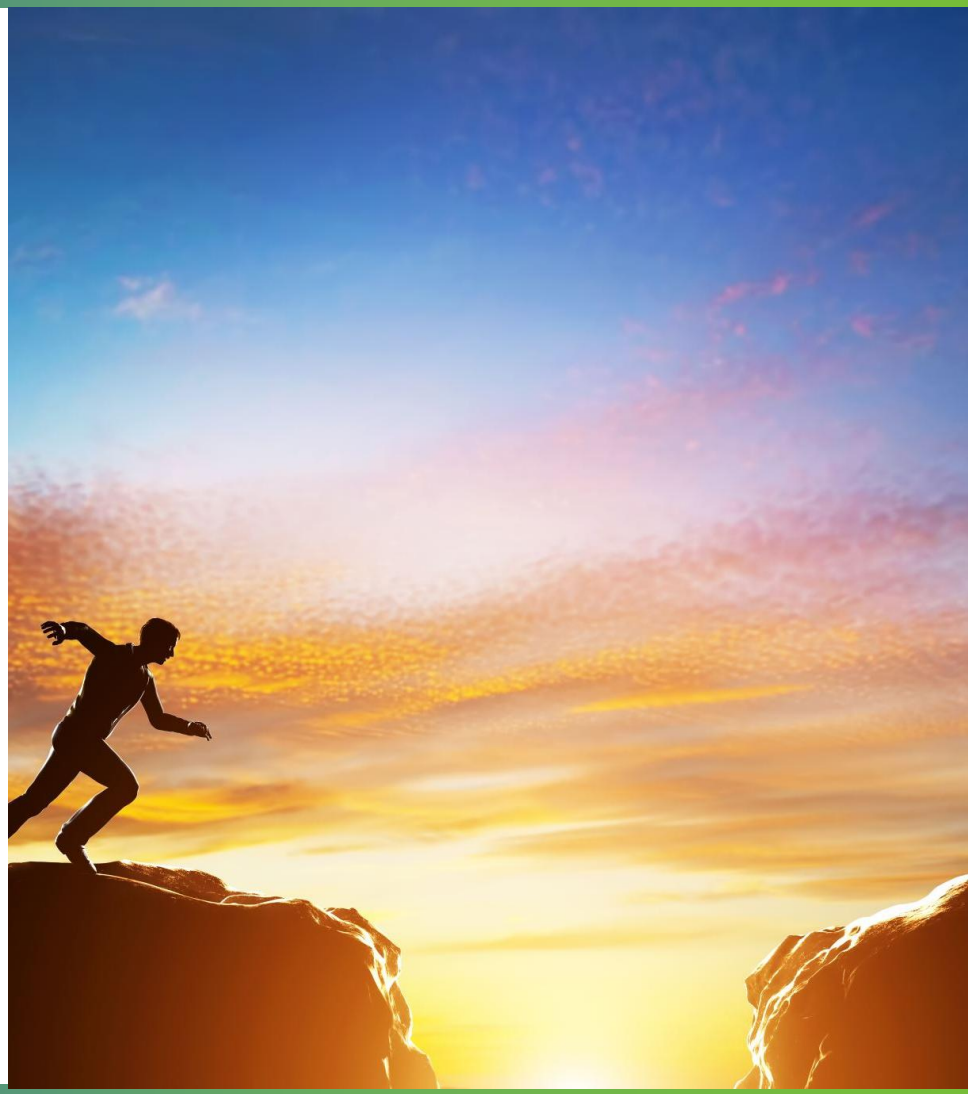
CHALLENGES

- Inflated Performance
  - 94.8% mAP50 Valid – 4 Cls
  - 78.1% mAP50 Test – 2 Cls
- Results Variability



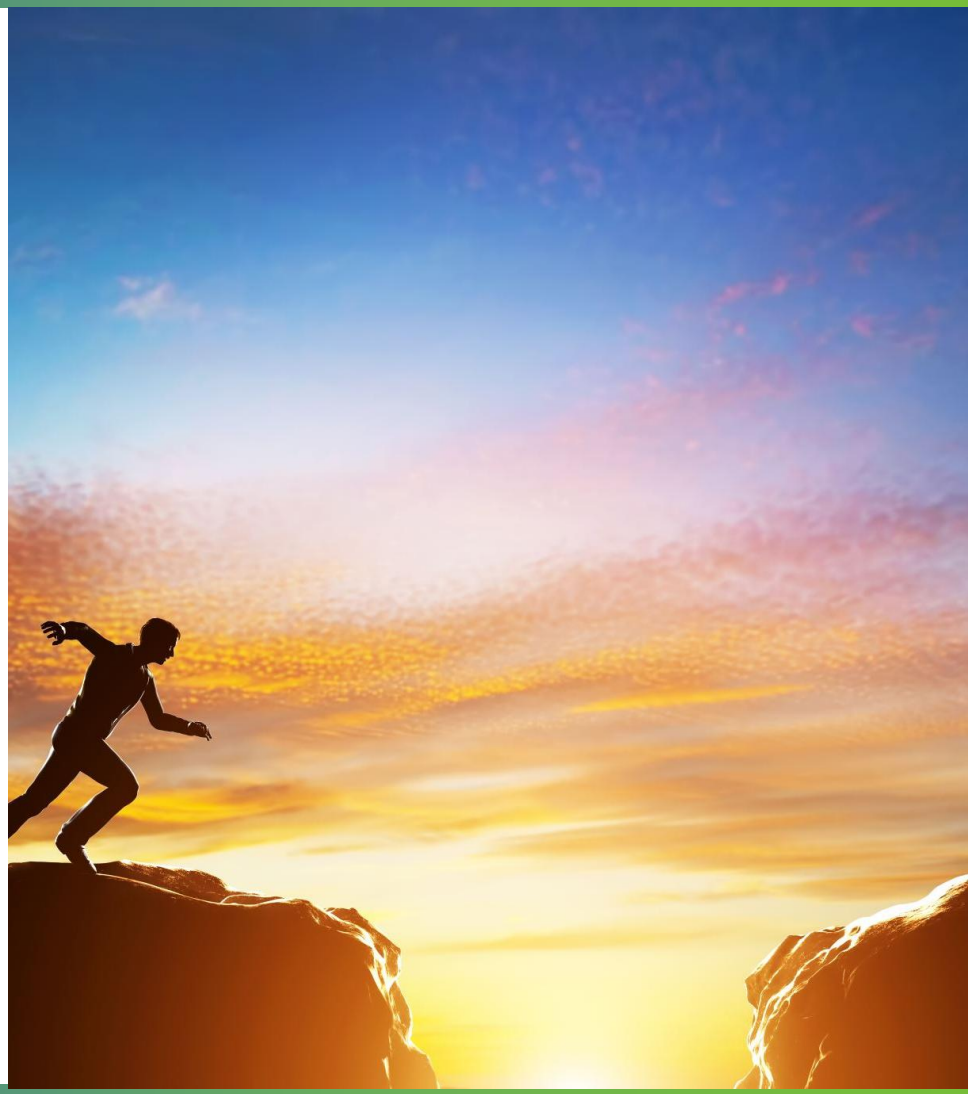
CHALLENGES

- Inflated Performance
  - 94.8% mAP50 Valid – 4 Cls
  - 78.1% mAP50 Test – 2 Cls
- Results Variability
  - Stochastic Training



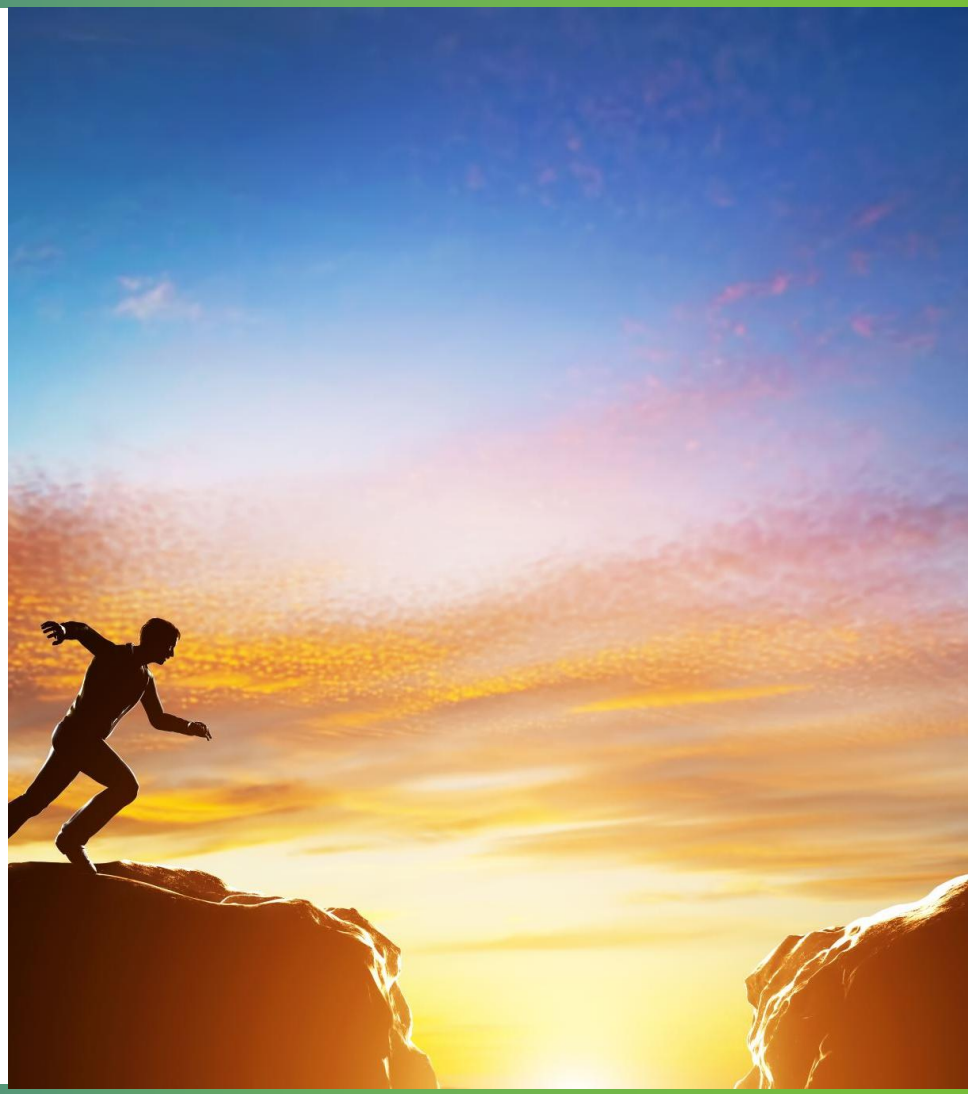
CHALLENGES

- Inflated Performance
  - 94.8% mAP50 Valid – 4 Cls
  - 78.1% mAP50 Test – 2 Cls
- Results Variability
  - Stochastic Training
  - Auto Batch Size



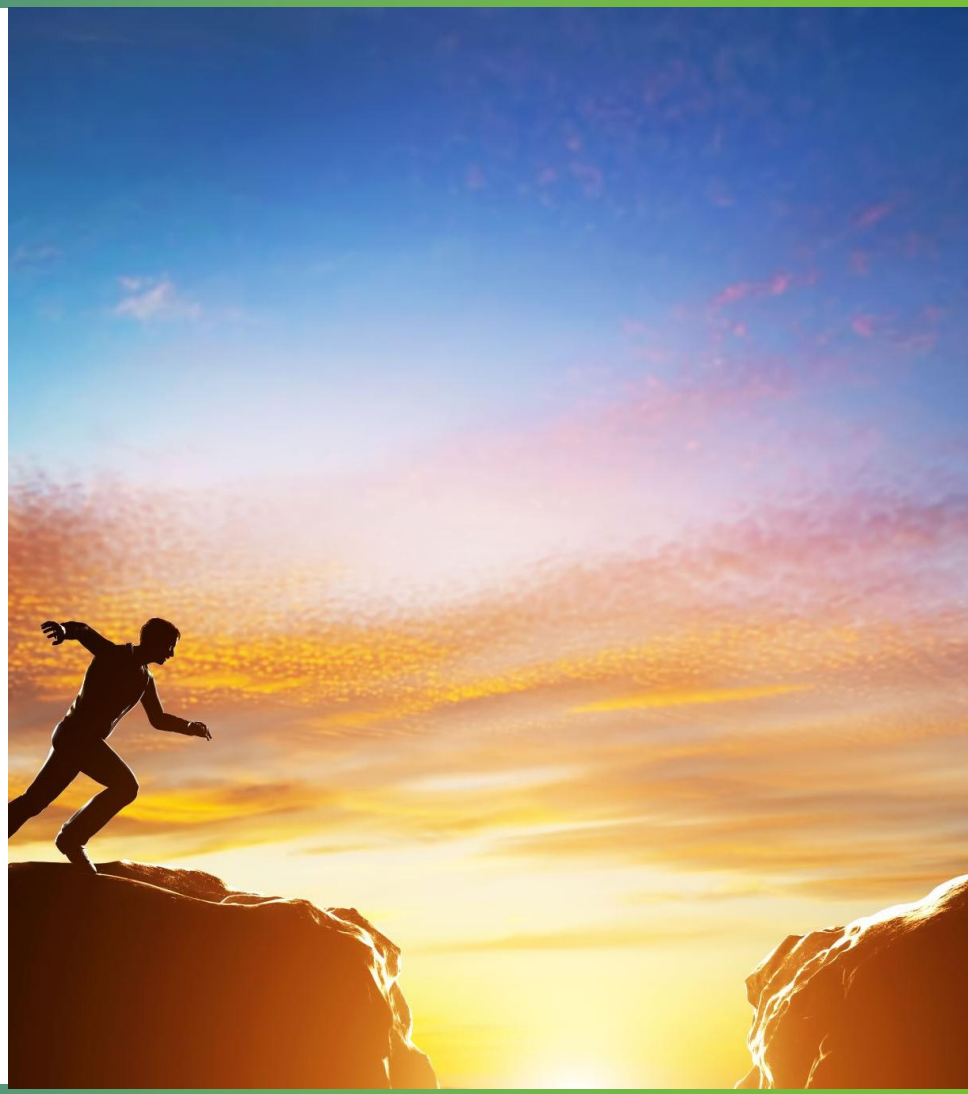
CHALLENGES

- Inflated Performance
  - 94.8% mAP50 Valid – 4 Cls
  - 78.1% mAP50 Test – 2 Cls
- Results Variability
  - Stochastic Training
  - Auto Batch Size
- Labels Noise Correction



CHALLENGES

- Inflated Performance
  - 94.8% mAP50 Valid – 4 Cls
  - 78.1% mAP50 Test – 2 Cls
- Results Variability
  - Stochastic Training
  - Auto Batch Size
- Labels Noise Correction
  - Annotation Pipeline



Future Work



Future Work

- Exhaustive Experiments



Future Work

- Exhaustive Experiments
  - Repeat Runs 10X



Future Work

- Exhaustive Experiments
  - Repeat Runs 10X
  - Average Results

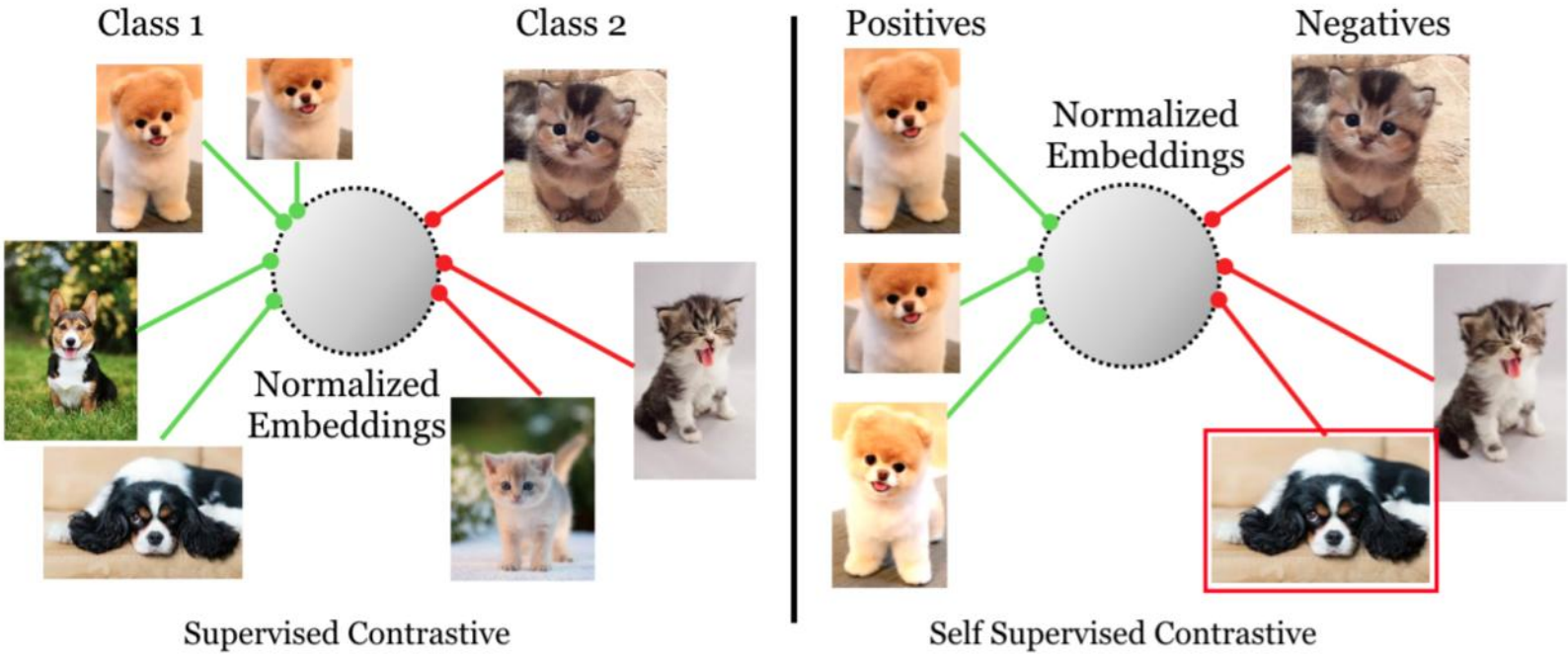


Future Work

- Exhaustive Experiments
  - Repeat Runs 10X
  - Average Results
- Learning Paradigm



Supervised Contrastive Learning



Khosla, Prannay, et al. "Supervised contrastive learning." *Advances in neural information processing systems* 33 (2020): 18661-18673.

Future Work

- Exhaustive Experiments
  - Repeat Runs 10X
  - Average Results
- Learning Paradigm
  - Contrastive Loss



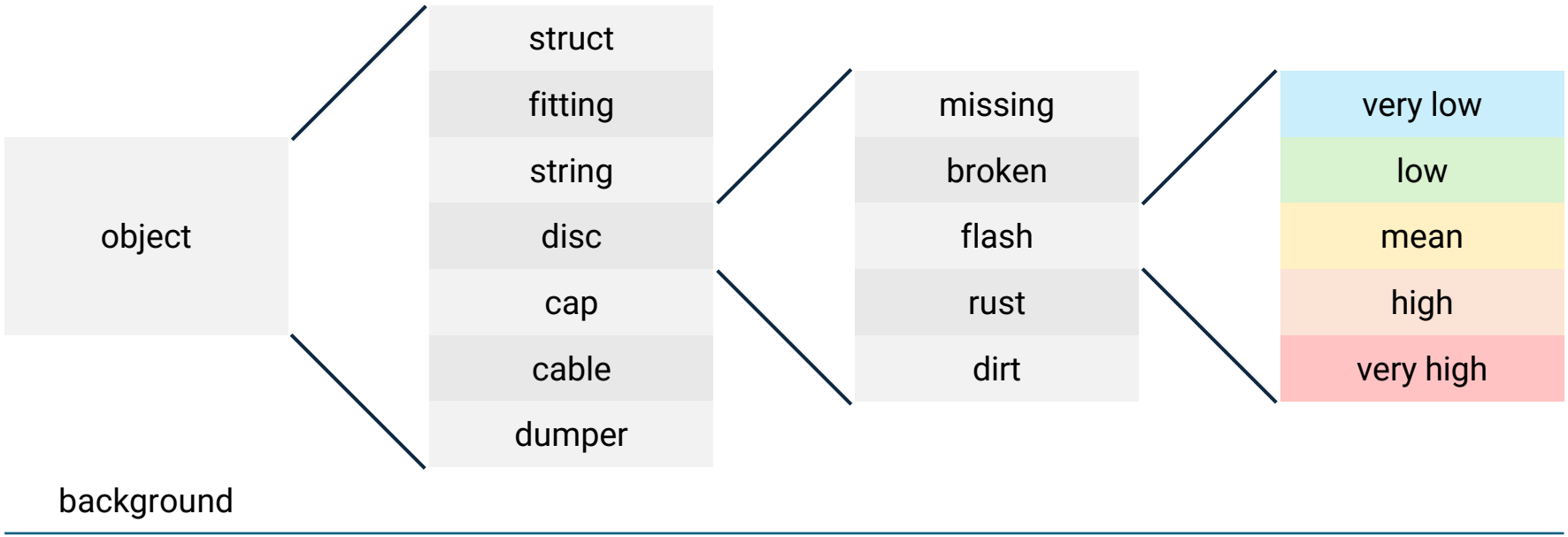
Future Work

- Exhaustive Experiments
  - Repeat Runs 10X
  - Average Results
- Learning Paradigm
  - Contrastive Loss
- Hierarchic Classes



Hierarchic Categories

Semantic Granularity



Future Work

- Exhaustive Experiments
  - Repeat Runs 10X
  - Average Results
- Learning Paradigm
  - Contrastive Loss
- Hierarchic Classes
  - Defect Type
  - Defect Severity



Q&A

Session